

Proposal for Agent/Banca Portal & Back-office Solution

SalesVerse 2.0, configured for IIFT as the IIFT Agent/Banca Portal & Back-office

Technical and commercial proposal in response to IIFT's Request for Proposal



PREPARED FOR

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Commercial-in-Confidence. This proposal contains confidential commercial information of iorta TechNXT and is submitted solely for the evaluation of the above RFP by Insurans Islam Family Takaful Sendirian Berhad. It may not be disclosed to third parties without prior written consent.

Cover Letter

[Day] October 2026

Our ref: [Proposal reference no.]

Your ref: [IIFT RFP / tender reference no.]

The Management

Insurans Islam Family Takaful Sendirian Berhad

[Street address]

Bandar Seri Begawan [Postcode]

Brunei Darussalam

Dear Sir/Madam,

Re: Proposal for Agent/Banca Portal & Back-office Solution

iorta TechNXT is pleased to submit its proposal for the Agent/Banca Portal and Back-office Solution. We offer to configure, integrate, test, deploy and support the solution, with training, documentation, a six-month warranty and five years of maintenance and support.

The solution is SalesVerse 2.0, iorta TechNXT's distribution platform for Takaful and insurance, configured for IIFT as the IIFT Agent/Banca Portal & Back-office. A working version already runs the seven products in Appendix 3 of the RFP, the B\$150,000 referral rule, the seven-day grace-period agency block, maker-checker approvals, AML watch-list screening, e-signature, e-Policy and e-Receipt issuance and the end-of-day FIN interface file. The remaining work (integration with IIFT's systems, IIFT's rates and templates, testing, migration and training) is set out in this proposal at a fixed fee. Our offer in summary:

- Fixed implementation price of **B\$ 166,000** (Agent/Banca Portal B\$ 91,500; Back-office solution B\$ 74,500), inclusive of withholding tax and planned out-of-pocket expenses.
- Licence fee of **B\$0**: a perpetual, royalty-free, enterprise-wide licence to SalesVerse 2.0 for unlimited users, with the source code of the deployed solution.
- Maintenance and support at **B\$ 21,600 a year**, fixed for five years.
- Go-live in **week 24**, then four weeks of hypercare.

The proposal contains the items requested in your invitation, in the order requested:

Item	Content	Where
(a)	Cover letter	This letter
(b)	Corporate profile and relevant experience	Enclosure B, attached separately
(c)	Business registration certificate	Enclosure C, attached separately
(d)	Detailed scope of work and methodology	Sections 5, 6 and 17
(e)	Detailed fee and pricing structure, stating WHT, OPE and third-party charges	Section 19 and the pricing workbook
(f)	Delivery and implementation timeline with key milestones	Section 18
(g)	Quotation validity period	Section 24: 90 days
(h)	Other supporting information	Sections 7 to 11, 16 and 25; technical documents
(i)	Terms and conditions	Section 26

We confirm that this proposal follows the RFP and would welcome the opportunity to demonstrate the working application. For clarification, please contact [Name of contact person], [Designation], at [e-mail address] or [telephone number].

Yours faithfully,

[Name of authorised signatory]

[Designation]

For and on behalf of **iorta TechNXT**

[Company stamp]

Document Control

Version history

Version	Date	Author	Description
0.9	[Date]	iorta TechNXT	Internal review draft
1.0	October 2026	iorta TechNXT	Submission version

Review and approval

Role	Name	Signature / date
Prepared by (Bid Manager)	[Name]	
Reviewed by (Solution Architect)	[Name]	
Approved by (Authorised Signatory)	[Name, designation]	

Distribution

Recipient	Organisation	Format
Tender / Evaluation Committee	Insurans Islam Family Takaful Sendirian Berhad	Signed PDF and editable DOCX; pricing workbook (XLSX)
Bid file	iorta TechNXT	Archive copy

Companion documents

- IIFT-Commercial-Pricing-iorta-TechNXT.xlsx: the commercial workbook (summary in the format requested by IIFT, RFP section 10 breakdown, integration per interface, five-year maintenance, rate card, payment milestones, third-party charges and assumptions).
- Enclosure B: corporate profile and relevant experience (attached separately).
- Enclosure C: business registration certificate (attached separately).
- Technical document pack in docs/technical: solution architecture, data dictionary, code standards and quality report, security assessment report, production support handover.

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How this Proposal Responds to the Invitation

The invitation asked for items (a) to (i). The table shows where each is answered and lists the additional chapters we included.

Invitation item	Where addressed
(a) Cover letter	Cover Letter, directly after the cover page
(b) Corporate profile and relevant experience	Enclosure B, attached separately (Section 3)
(c) Business registration certificate	Enclosure C, attached separately (Section 4)
(d) Detailed scope of work and methodology	Sections 5 (scope of work), 6 (fitment) and 17 (methodology)
(e) Detailed fee and pricing structure (WHT, OPE, third-party charges)	Section 19: Commercial Proposal; pricing workbook
(f) Delivery and implementation timeline with key milestones	Section 18
(g) Quotation validity period	Section 24
(h) Other supporting information	Section 25, plus personas, journeys, screens and the technical document pack
(i) Terms and conditions	Section 26
Additional chapters	
Personas and user journeys	Sections 7 and 8
Functional rules, products and screen catalogue	Sections 9, 10 and 11
Architecture, security, deployment, infrastructure	Sections 12 to 15
Technical document pack	Section 16
Five-year maintenance and support, SLA	Section 20
Team, assumptions and risks	Sections 21 to 23
Compliance and fitment matrix (every RFP requirement ID)	Annex A

Abbreviations

Takaful and technical terms are explained in the glossary in Annex D.

Abbreviation	Meaning
AD / LDAP	Active Directory / Lightweight Directory Access Protocol, used for single sign-on
AMBD	Autoriti Monetari Brunei Darussalam, the central bank and financial regulator of Brunei Darussalam
AML / KYC	Anti-Money Laundering / Know Your Customer screening and due diligence
API	Application Programming Interface
ASVS	OWASP Application Security Verification Standard
BRR	The issuance register / report referenced in RFP Appendix 3 against which daily issuance and FIN postings are tallied
BRS / FRS	Business Requirement Specification / Functional Requirement Specification
B\$ / BND	Brunei Dollar
CI/CD	Continuous Integration / Continuous Delivery
CR	Change Request
DR	Disaster Recovery
EOD	End-of-Day processing
FIN	IIFT financial / accounting system referenced in the RFP
FFR	Functional Flow and Requirements (RFP Appendix 3)
HA	High Availability
IC	Identity Card (Brunei national identity card)
IIFT	Insurans Islam Family Takaful Sendirian Berhad
IIGT	Insurans Islam General Takaful, the sister company that operates the IIGT Agent Portal
IITH	Insurans Islam TAIB Holding, the group holding company
MFA / OTP / TOTP	Multi-Factor Authentication / One-Time Password / Time-based One-Time Password
OPE	Out-of-Pocket Expenses (travel, accommodation, subsistence)
OWASP	Open Worldwide Application Security Project
PDPO	Personal Data Protection Order 2025 (Brunei Darussalam)
PITR	Point-in-Time Recovery
PMO	Project Management Office
QC	Quality Check
RACI	Responsible, Accountable, Consulted, Informed
RBAC	Role-Based Access Control
RPO / RTO	Recovery Point Objective (maximum data loss) / Recovery Time Objective (maximum downtime)
RTM	Requirements Traceability Matrix
SAD / TDD	Solution Architecture Document / Technical Design Document
SAST / SCA	Static Application Security Testing / Software Composition Analysis
SIT / UAT	System Integration Testing / User Acceptance Testing
SLA	Service Level Agreement
SSO	Single Sign-On
TCO	Total Cost of Ownership
VAPT	Vulnerability Assessment and Penetration Testing
WAF	Web Application Firewall
WHT	Withholding Tax

1. Executive Summary

IIFT will receive SalesVerse 2.0, configured as the IIFT Agent/Banca Portal & Back-office, live in 24 weeks for a fixed B\$ 166,000. Banca submissions, approvals, payments and reporting move from paper and e-mail to one controlled system.

Solution	SalesVerse 2.0, configured as the IIFT Agent/Banca Portal & Back-office: a portal for agents and bank officers and a back-office for IIFT and IITH staff.
Scope	All 29 in-scope components of RFP section 3.1, the seven Appendix 3 products, and integration with IIFT's core system, FIN, AML provider, Active Directory, SMS gateway and e-mail.
Timeline	Go-live in week 24. Hypercare in weeks 25 to 28. Warranty for six months from go-live.
One-time price	B\$ 166,000 (USD 127,692), fixed. WHT inclusive; planned OPE included.
Support	B\$ 21,600 a year, flat for five years (B\$ 108,000 in total).
Licence	B\$0. Perpetual, royalty-free, enterprise-wide, unlimited users.
Source code	Source code of the deployed solution at go-live and with every release; escrow optional.
Five-year cost	B\$ 274,000 (USD 210,769) with on-premise hosting.
Compliance	234 of 234 RFP requirement IDs addressed (Annex A); fitment in Section 6.

Table 1: What IIFT receives

1.1 Why iorta TechNXT

The platform already works. IIFT's seven products, the B\$150,000 referral, the seven-day agency block, maker-checker with amount thresholds, watch-list screening, e-signature, e-Policy and e-Receipt PDFs, the EOD report and FIN interface file all run today. Discovery workshops start from working screens, which is why requirements and design take five weeks.

The price follows IIFT's scale. The RFP describes 26 named users and about 600 policies a year. There is no per-user or per-policy fee, so more bank branches, agents or other IITH companies do not change the licence cost.

Controls sit in the data layer. The API enforces maker-checker, a database trigger makes the audit table append-only, and IC and passport numbers are encrypted field by field with AES-256-GCM. These controls cannot be bypassed from a screen.

Delivery risk is shared. The price is fixed, 95% of it is paid against signed milestones, and maintenance carries no escalation for five years.

1.2 About iorta TechNXT and SalesVerse 2.0

iorta TechNXT builds and supports software for insurance and Takaful distribution. SalesVerse 2.0 is its platform for agency and bancassurance portals, back-office administration and the integrations that connect them to core, finance and compliance systems. The corporate profile and relevant experience are in Enclosure B.

2. Understanding of IIFT's Requirements

2.1 Background

IIFT sells Mortgage Takaful and Annual (Individual) Family Takaful plans through agents and bank partners. Banca work today relies on paper or e-mailed submissions and manual follow-up with the back-office. Status is hard to track and management has little real-time visibility.

The IIGT Agent Portal went live on 17 July 2026 for Motor Takaful. It covers agent onboarding, e-quotation, e-Cover Notes, AML screening, documents, e-signature, payment proof verification, audit and reporting. IIFT needs the same kind of capability for Family Takaful products and the Banca channel, plus a back-office in which IIFT staff review, approve and report on every Banca transaction.

2.2 Project objectives and our response

#	RFP objective	How the solution meets it
1	Digitalise Agent/Banca operations	Agent/Banca Portal for onboarding, quotation, issuance, servicing, payments, claims notification and issues.
2	Improve operational efficiency	Data captured once at source; contribution calculated by the system; e-Policy, e-Receipt and EOD files produced automatically.
3	Strengthen Agent/Banca management	Agency and bank hierarchy, main/sub-agent links, five agent statuses and document expiry tracking.
4	Strengthen governance and controls	Role-based access with data scoping, maker-checker with configurable levels and thresholds, append-only audit trail.
5	AML/KYC management	Watch-list screening with fuzzy name matching at onboarding and on change; Compliance review queue.
6	Improve data quality	Field validation, controlled master data, duplicate checks on IC and registration numbers, daily reconciliation.
7	Improve integration	Transactional outbox with retry and dead-letter; adapters for Core, FIN, AML, Active Directory, SMS and e-mail.
8	Improve reporting	Dashboards and standard reports with Excel, CSV and PDF export and scheduled distribution.
9	Improve user experience	Step-by-step quotation wizards, one design system themed to Insurans Islam TAIB colours, WCAG 2.1 AA target.
10	Strengthen security	Controls aligned with OWASP ASVS Level 2 and an independent VAPT before go-live.
11	Ensure scalability	Sized for 100 concurrent and 500 named users, more than ten times the projected policy volume.
12	Long-term sustainability	Five years of support at a flat fee, maintained documentation and source code of the deployed solution.

2.3 Users and volumes

Appendix 1 lists 26 named users and Appendix 2 projects 628 policies a year within five years. These volumes are small for a relational database and two application servers, so we size for 100 concurrent and 500 named users and ten times the projected volume (NFR-02, NFR-03, NFR-06, NFR-07).

Company	Application	Expected users
Bank	Portal	12
IIFT	Back-office (Admin)	5
IIFT	Portal	5
Agent	Portal	4
Total		26

Table 2: Expected users (RFP Appendix 1)

Line of business	Product	2025	2026 (to July)	Projected in 5 years
Mortgage Takaful	Financing Takaful Plan – Hire Purchaser	59	16	75
Mortgage Takaful	Financing Takaful Plan – Non-Participating	158	65	205
Mortgage Takaful	Property Financing Takaful Plan	88	60	115
Annual (Individual)	Personal Home Assistant Takaful Plan	102	41	132
Annual (Individual)	Professional Takaful Plan	40	19	52
Annual (Individual)	Khairat Takaful Plan	31	16	40
Annual (Individual)	Overseas Student Assist Takaful Plan	7	2	9
Total		485	219	628

Table 3: Policy volumes (RFP Appendix 2)

2.4 Business rules that shape the design

- Financing above **B\$150,000** is the high-risk limit. The agent or banker cannot complete it; the case is referred to IIFT Sales (FFR01, AP-60).
- A **seven-day payment grace period** applies to issued policies. If payment is not submitted in time, every agent in that agency is blocked from issuing until it is settled (AP-42).
- **Maker-checker** covers agent registration and profile changes, participant updates, endorsements, cancellations, payment verification and AML review (BO-10, BO-17).
- **End-of-day processing** produces the daily issuance report and FIN entries, and issuance is reconciled with BRR for every product (FFR01 to FFR05).
- Product eligibility differs by plan: Professional is limited to Occupational Class I, Khairat has no renewal, and Overseas Student Assist requires a Brunei citizen registered as a student and aged 65 or under.

3. Enclosure B – Corporate Profile and Relevant Experience

Attached separately. iorta TechNXT's latest corporate profile and relevant experience (invitation item b) are submitted as a separate document with this proposal.

Document	[File name of the corporate profile]
Contents	Company overview, leadership, delivery capability and relevant reference projects
Contact	[Name, designation, e-mail, telephone]

Table 4: Enclosure B

4. Enclosure C – Business Registration Certificate

Attached separately. A certified copy of iorta TechNXT's business registration certificate (invitation item c) is submitted with this proposal.

Document	[File name of the certificate]
Business registration no.	[Business registration no.]

Table 5: Enclosure C

5. Scope of Work

We will configure, integrate, test, deploy and support SalesVerse 2.0 as the IIFT Agent/Banca Portal & Back-office, with training, documentation, a six-month warranty and five years of maintenance. This chapter follows the RFP structure. For each item it states what we deliver, how, and the evidence IIFT uses to accept it. Fitment against the working platform is in Section 6; the per-ID matrix is in Annex A.

5.1 In-scope components (RFP 3.1)

#	Component	What we deliver	How	Acceptance evidence
1	Agent/Banca Portal	Web portal (/portal) for agents, main/sub-agents and bank officers	Configure the SalesVerse 2.0 portal for IIFT roles, products and branding	UAT sign-off of portal scenarios
2	Back-office (Admin)	Internal application (/backoffice) for IIFT and IITH staff	Configure back-office roles, queues and dashboards; publish only on the internal network	UAT sign-off; network test showing no internet exposure
3	Agent/Banca management	Registration, profile, status, agent code, hierarchy	Load agent master data; set numbering pattern and status rules	SIT cases; migrated agent reconciliation
4	Agency/Banca management	Agencies, banks, branches and their relationships	Load agency and bank structure; map hierarchy to data scopes	Hierarchy review signed by Banca/Sales
5	Agent/Banca onboarding	Online application, KYC data, documents, AML, approval	Configure document checklist and approval levels	Journey 1 executed in UAT
6	Participant management	Shared participant record, search, updates with approval, AML	Set search criteria, duplicate rules and update workflow	SIT and UAT cases for AP-11 to AP-16
7	Policy management	Seven products, quotation, drafts, issuance, statuses, servicing	Load IIFT rates, questionnaires and templates for all seven products	Rate verification sheet signed by IIFT actuarial; UAT
8	Billing management	Payment dashboard, single and bulk payment, verification, e-Receipt, 7-day block	Configure payment rules, thresholds and receipt numbering	Journey 4 in UAT; grace-period test evidence
9	Claims management	Claim notification with policy validation and documents	Configure claim types and document lists	UAT cases for AP-43
10	AML/KYC	Watch-list screening with fuzzy matching, Compliance review	Connect to IIFT's AML provider or lists; set match threshold	Screening test pack agreed with Compliance
11	Workflow management	Configurable workflows for all approval transactions	Load IIFT approval matrix as workflow definitions	Approval matrix sign-off; SIT cases
12	Maker-checker	Segregation of duties on every configured transaction	Assign maker and checker permissions per role	Negative tests: maker cannot approve own request
13	Document management	Encrypted document store; e-Policy, schedule and e-Receipt PDFs	Load IIFT templates and wording; deploy ClamAV in each environment	Template sign-off; upload security tests
14	Dashboard	Agent, bank, back-office and management dashboards	Agree KPI definitions and targets	Dashboard review in UAT
15	Reporting	Report catalogue with filters, schedules, Excel/CSV/PDF export	Configure the 12 existing reports; build the remaining catalogue items	Report sign-off against sample data
16	Issue management	Issue reporting, assignment, priorities, SLA timers	Set categories, assignment rules and SLA targets	Journey 7 in UAT
17	Notification	E-mail, SMS and in-portal notifications	Connect SMTP and SMS gateway; agree templates	Notification test log
18	Audit trail	Append-only audit with search and export	Set retention; optional SIEM export	Audit tamper test (UPDATE/DELETE rejected)
19	Administration	Users, roles, permissions, master data, parameters	Load IIFT roles and parameter values	Role matrix sign-off
20	Integration	Core, FIN, AML, AD, SMS, e-mail adapters with monitoring and reconciliation	Build adapters against agreed interface specifications	End-to-end interface tests with each system owner
21	Security	Controls in Section 13; MFA; independent VAPT	Apply IITH policy values; remediate VAPT findings	VAPT and re-test report (DEL-17)
22	Data migration	Agent, agency, bank, participant and reference data	Scripted extract, cleanse, load, reconcile; two rehearsals	Signed reconciliation report
23	Testing	Unit, API, SIT, regression, performance, security; UAT support	Automated suites in CI; SIT and UAT plans	SIT and UAT results (DEL-14, DEL-16)

#	Component	What we deliver	How	Acceptance evidence
24	Deployment	DEV, SIT, UAT, PROD and DR environments; CI/CD; cut-over	Container deployment on IIFT infrastructure	Deployment plan and go-live checklist signed
25	Training	Agents, bank officers, back-office, administrators, IT	Train-the-trainer, hands-on sessions, manuals	Attendance records and feedback
26	Documentation	User, administrator, technical and operations manuals	Update the SalesVerse 2.0 document pack for IIFT	Document acceptance by IIFT
27	Warranty	Six months of defect correction from go-live	Fixes under the SLA at no charge	Monthly service report
28	Maintenance	Five years of support, patches and upgrades	Service desk, monthly patch cycle, releases	Monthly service report; quarterly review
29	Transition	Knowledge transfer, handover and exit support	Year 5 exit plan and shadow support	Signed transition acceptance

Table 6: In-scope components 1 to 29

5.2 Agent/Banca Portal functions (RFP 4.1)

RFP area and IDs	What we deliver	How	Acceptance evidence
Authentication (AP-01–04)	Secure login, password change and reset, session control, lockout	Server-side sessions; policy values set to IITH standards	Security test cases; VAPT
Agent/Banca profile (AP-05–06)	Profile view and change requests with approval	Profile change workflow with before and after values	UAT case; audit record of approved change
Agent/Banca management (AP-07–12)	Online registration, agent code, hierarchy, agency information, shared participants	Registration journey; data scoping by agency and bank	Journey 1 in UAT
Participant management (AP-13–16)	Individual and corporate registration, search, approved updates, AML screening	Duplicate checks on encrypted IC numbers via blind index	SIT cases; AML test pack
Quotation (AP-17–20)	Authorised product list, product wizards, automatic contribution, drafts	Seven configured products with IIFT rate tables	Rate verification sheet; UAT
Issuance and status (AP-21–25, AP-32–36)	Policy list, search, details, issuance, validation, routing, restrictions, notifications, audit	Server-side validation and workflow routing	Journeys 2 and 3 in UAT
Policy servicing (AP-26–31)	Renewal, endorsement, cancellation, documents, payment status, history	Servicing requests through maker-checker	Journey 5 in UAT
Billing (AP-37–42)	Payment dashboard, bulk payment per policy, proof upload, verification, e-Receipt, 7-day block	Billing module and nightly grace-period job	Journey 4; grace-period test evidence
Claims (AP-43)	Claim notification with policy validation and documents	Claims module	Journey 6 in UAT
Documents (AP-44–47)	Policy schedule, e-Policy and e-Receipt PDFs; download, e-mail, upload, status and expiry	IIFT templates; encrypted store; ClamAV scan	Template sign-off; upload tests
AML information (AP-48)	AML/KYC data captured at onboarding	Configurable KYC fields	Compliance review of form
Workflow (AP-49–51)	Save, submit, track; status list; rejection remarks and resubmission	My requests screen over the workflow engine	Journey 9 in UAT
Dashboard (AP-52–53)	Agent dashboard and pending actions	Role-based dashboard widgets	UAT review
Notifications (AP-54)	Alerts on status changes, approvals, rejections, actions	Notification templates per event	Notification test log
Issue management (AP-55–57)	Report issues with reference number, tracking and attachments	Issues module with SLA timers	Journey 7 in UAT
Reporting (AP-58)	Role-scoped reports for agency, agent and bank	Report catalogue with data scoping	Report sign-off
Security and approvals (AP-59–61)	Role-based access, authority limits, approval history	Permissions and data scopes; amount thresholds	Role matrix sign-off; negative tests
Digital signature (AP-62)	On-screen signature, single-use e-mail link, upload of signed documents	e-Signature module with evidence record	UAT case; evidence sample reviewed by IIFT Legal

5.3 Back-office functions (RFP 4.2)

RFP area and IDs	What we deliver	How	Acceptance evidence
Authentication and access (BO-01–02)	Internal login, MFA, Active Directory sign-on, role-based access	MFA and AD adapter built during implementation	Security tests; VAPT
User management (BO-03–04)	User administration, roles and permissions	Role builder; IIFT role matrix loaded	Role matrix sign-off
Agent management (BO-05–08, BO-10)	Agent records, search, status lifecycle, hierarchy, approval of registrations and changes	Agents module with maker-checker	SIT cases; Journey 1
Agency management (BO-09)	Agency and bank records with linked agents	Agencies and banks screens	Hierarchy review
Documents (BO-11–12)	Validate, approve, reject documents; completeness and expiry	Document checks queue	UAT case
AML/KYC (BO-13–15)	Screening, results with dates, Compliance review	AML cases and watch-list screens	AML test pack
Workflow (BO-16–19)	Configurable approvals, maker-checker, status tracking, rejection remarks	Workflow definitions for the IIFT matrix	Journey 9; negative tests
Dashboards (BO-20–21)	Management KPIs and pending actions	Back-office dashboard	UAT review
Reporting (BO-22–25)	Standard reports, filters, Excel/CSV/PDF export, schedules	12 existing reports plus the agreed catalogue	Report sign-off
Audit (BO-26–28)	Audit of activity and changes with before and after values; search	Append-only audit table protected by trigger	Tamper test; audit search demo
Issue management (BO-29–31)	Assignment, priorities, SLA monitoring	Issue rules and SLA targets	Journey 7
Administration (BO-32–33)	Master data and configurable rules, statuses, levels, parameters	Parameters and master data screens	Configuration workbook signed

5.4 Integration (RFP 4.3)

RFP area and IDs	What we deliver	How	Acceptance evidence
Core system (INT-01–03)	Agent, agency and bank master synchronisation; agent information exchange	Adapter over the outbox; API or file per IITH capability	End-to-end test with core owner
Financial system (INT-04–05)	EOD FIN interface file, receipts, commission and referral fee data	FIN file format agreed; SFTP or API delivery	Journey 8; FIN reconciliation sign-off
Identity (INT-06)	Active Directory single sign-on for back-office	LDAPS adapter	Login test with IITH accounts
Notifications (INT-07–08)	E-mail through IIFT SMTP; SMS through IIFT gateway	SMTP and SMS adapters	Delivery test log
APIs (INT-09–11)	Secure REST APIs, authentication, monitoring	OpenAPI 3; mTLS or OAuth2; integration monitor	API security tests; monitor demo
Data (INT-12–15)	Validation, error handling and retry, logging, reconciliation	Schema validation; outbox retry and dead-letter; daily reconciliation	Failure-injection tests; reconciliation report

5.5 Shared platform (RFP 4.4)

RFP area and IDs	What we deliver	How	Acceptance evidence
Security (COM-01, 02, 10)	Protection of sensitive data, access control, encryption	Controls in Section 13	VAPT; encryption check
Audit (COM-03)	Common audit service	Append-only audit	Tamper test
Workflow (COM-04)	Configurable workflows by type, role, level, rule	Workflow engine	Approval matrix tests
Notification (COM-05)	E-mail, SMS, portal notifications	Notification service and templates	Notification test log
Documents (COM-06)	Secure document repository	Encrypted storage; access through API only	Access tests
Search (COM-07)	Global search across records	Search with configurable criteria	UAT case
Reporting (COM-08)	Common reporting framework	Report definitions, exports, schedules	Report sign-off

RFP area and IDs	What we deliver	How	Acceptance evidence
Configuration (COM-09)	Parameters maintained without code changes	Parameters and master data screens	Configuration workbook signed

5.6 Non-functional requirements (RFP 5)

RFP area and IDs	What we deliver	How	Acceptance evidence
Performance (NFR-01–03)	Agreed response times for 100 concurrent users and projected volume	k6 load, stress and soak tests on production-like SIT	Performance test report (DEL-18)
Availability (NFR-04–05)	99.5% overall, 99.9% in business hours; no single point of failure	Two app nodes, database standby, load balancer pair	Monthly availability report
Scalability (NFR-06–07)	Growth in users and volume without redesign	Stateless API nodes; indexed schema	Capacity statement in SAD
Security (NFR-08–14)	Encryption, authentication, authorisation, audit, vulnerability management, VAPT, secure SDLC	Section 13	VAPT report; scan results
Reliability (NFR-15–16)	Controlled errors; transaction integrity	Generic messages with correlation id; ACID transactions	Negative tests
Backup and DR (NFR-17–19)	Backups, DR to RPO 15 minutes and RTO 4 hours, DR tests	WAL archiving; warm standby; annual DR test	Restore test and DR test reports
Maintainability (NFR-20–21)	Modular design; maintained documentation	Modular monolith; document pack updated per release	Document acceptance
Usability and accessibility (NFR-22–24)	Consistent UI, browser support, WCAG 2.1 AA target	One design system; browser test matrix; accessibility check	UAT feedback; accessibility check report
Compatibility (NFR-25)	Standard APIs	REST/JSON, OpenAPI 3	API specification
Monitoring and logging (NFR-26–28)	System monitoring, alerting, central logs	Prometheus, Grafana, central log store	Alert test during SIT
Data (NFR-29–30)	Retention and data quality	Retention rules; validation and constraints	Retention configuration signed

5.7 Deliverables (RFP 7)

Documents are delivered in Word or Excel and PDF. IIFT reviews within five business days and we issue the final version within three business days of comments.

ID	Deliverable	Content and approach	Due	Acceptance evidence
DEL-01	Project charter	Objectives, scope, governance, stakeholders, assumptions, milestones	Wk 2	Steering Committee approval
DEL-02	Project plan	Activities, dependencies, resources, timeline; updated fortnightly	Wk 2	IIFT PM approval
DEL-03	Business requirement specification	Agreed processes from fit-gap workshops on the working system	Wk 4	Business owner sign-off
DEL-04	Functional requirement specification	Rules, screens, reports and configuration per product	Wk 5	Business owner sign-off
DEL-05	Requirements traceability matrix	Every RFP ID traced to design, test and acceptance	Wk 6, kept current	Final RTM at closure
DEL-06	Solution architecture	Application, infrastructure, security, integration, deployment	Wk 6	Design Authority sign-off
DEL-07	Technical design	Components, database, APIs, interfaces, configuration	Wk 6	Design Authority sign-off
DEL-08	UI/UX design	Screen designs and journeys based on the working screens	Wk 5	Business owner sign-off
DEL-09	Interface specification	APIs, mappings, methods, security, error handling per interface	Wk 6	System owner sign-off
DEL-10	Configured solution	SalesVerse 2.0 configured and integrated for IIFT	Wk 16	Build-complete demo (M3)
DEL-11	Source code	Source code of the deployed solution, build scripts, migrations	Wk 24 and each release	Repository handover checklist

ID	Deliverable	Content and approach	Due	Acceptance evidence
DEL-12	Test strategy	Levels, roles, environments, entry and exit criteria	Wk 6	IIFT approval
DEL-13	SIT test cases	Scenarios traced to requirements	Wk 14	IITH IT review
DEL-14	SIT results	Execution evidence, defect log, closure	Wk 19	SIT exit sign-off (M4)
DEL-15	UAT test cases	Business scenarios per product and role	Wk 18	Business owner review
DEL-16	UAT results	Execution evidence and business sign-off	Wk 22	UAT certificate (M5)
DEL-17	Security assessment	Independent VAPT, remediation, re-test	Wk 19	No open Critical or High findings
DEL-18	Performance test report	Load, stress and soak results against targets	Wk 19	Targets met
DEL-19	Data migration plan	Extraction, transformation, validation, reconciliation	Wk 8	Data owner sign-off
DEL-20	Deployment plan	Sequence, roles, rollback and contingency	Wk 21	IITH IT approval
DEL-21	Go-live checklist	Technical, operational, security, business, support readiness	Wk 23	Go/no-go decision
DEL-22	User training	Agents, bank officers, back-office users	Wk 21–23	Attendance and feedback
DEL-23	Administrator training	Administrators and support staff	Wk 22–23	Attendance and hands-on check
DEL-24	User manual	Portal and back-office guides with screenshots	Wk 21	Document acceptance
DEL-25	Administrator manual	Configuration, user administration, operations support	Wk 21	Document acceptance
DEL-26	Technical manual	Architecture, database, API, integration, deployment	Wk 23	Document acceptance
DEL-27	Operations manual	Monitoring, backup, incidents, routine operations	Wk 23	Document acceptance
DEL-28	Production deployment	Deployment per approved plan	Wk 24	Go-live confirmation (M6)
DEL-29	Go-live support	Four weeks of hypercare	Wk 25–28	Hypercare exit criteria met
DEL-30	Project closure report	Status, open items, lessons, acceptance	Wk 28	Closure sign-off (M7)

Table 7: Deliverables DEL-01 to DEL-30

5.8 Maintenance and support (RFP 8)

ID	Requirement	What we deliver and how	Acceptance evidence
MNT-01	Helpdesk support	Service desk via issue module, e-mail and phone for five years	Ticket records
MNT-02	Incident management	Record, classify, investigate, resolve, close	Monthly incident statistics
MNT-03	Problem management	Root-cause analysis for P1, P2 and recurring incidents	RCA reports within 5 business days
MNT-04	Service requests	User support, configuration and operational help	Request log
MNT-05	Critical incident SLA	P1: 30-minute response, 4-hour restoration, 24x7	SLA report
MNT-06	High priority SLA	P2: 2-hour response, 1 business day	SLA report
MNT-07	Medium/Low/Cosmetic SLA	P3: 4 business hours / 3 business days; P4: 1 business day / next release	SLA report
MNT-08	Bug fixing	Defects fixed under warranty, then under maintenance	Release notes
MNT-09	Corrective maintenance	Changes needed to keep functions working	Release notes
MNT-10	Preventive maintenance	Monthly health check of logs, jobs, capacity, queries, certificates	Health check in monthly report
MNT-11	Security patches	Monthly patch cycle; emergency patches as needed	Patch log
MNT-12	Vulnerability remediation	Critical 7 days, High 14, Medium 30, Low 90	Vulnerability register
MNT-13	Certificate management	Inventory, expiry alerts, renewal support	Certificate register
MNT-14	OS/platform compatibility	Compatibility with supported OS, Node.js, PostgreSQL	Compatibility statement per release
MNT-15	Browser compatibility	Current and previous browser versions	Browser test record
MNT-16	Database maintenance	Health checks, vacuum and index tuning, upgrades	Health check results
MNT-17	Performance monitoring	Monthly review and tuning advice	Performance section of monthly report
MNT-18	System monitoring	Availability and component monitoring with alerts	Availability figures
MNT-19	Backup verification	Quarterly restore test	Restore test record
MNT-20	Disaster recovery	DR planning and annual DR test	DR test report

ID	Requirement	What we deliver and how	Acceptance evidence
MNT-21	Release management	Tested, approved releases in agreed windows	Release records and approvals
MNT-22	Version upgrade	Application and platform upgrades, including SalesVerse 2.0 updates	Upgrade release notes
MNT-23	Minor enhancement	60 hours a year included; more at rate card	Enhancement log
MNT-24	Change request	Impact assessment, quotation, approval, delivery	Signed CR forms
MNT-25	Monthly service report	Incidents, SLA, availability, changes, open issues	Report issued monthly
MNT-26	Quarterly review	Service performance and improvement plan	Minutes and action log
MNT-27	Documentation updates	Documents updated after material change	Document version history
MNT-28	Knowledge transfer	Sessions for IIFT/IITH support staff	Session records
MNT-29	Onsite support	Onsite for major incidents or upgrades (rate card)	Visit reports
MNT-30	Transition support	Year 5 knowledge transfer, documents and technical help	Transition acceptance

Table 8: Maintenance requirements MNT-01 to MNT-30

6. Solution Fitment

Each functional and non-functional requirement is classified against SalesVerse 2.0 as it runs today. The four categories are:

Category	Meaning
Available in SalesVerse 2.0 today	Works in the current platform; IIFT can see it in the demonstration.
Configuration	Supported by the platform; needs IIFT values, rates, templates or rules.
Integration with IIFT systems	Adapter built on the platform's integration layer; completed once IIFT provides endpoints and credentials.
Delivered during implementation	Work or services performed during the project or, for MNT items, during maintenance.

RFP area	Available today	Configuration	Integration	During implementation	Total
Agent/Banca Portal (RFP 4.1)	49	12	1	0	62
Back-office (RFP 4.2)	27	6	0	0	33
Integration (RFP 4.3)	6	0	9	0	15
Shared platform (RFP 4.4)	8	2	0	0	10
Non-functional (RFP 5)	15	1	0	14	30
Deliverables (RFP 7)	0	0	0	30	30
Maintenance and support (RFP 8)	0	0	0	30	30
Total	105	21	10	74	210

Table 9: Fitment by RFP area (number of requirement IDs)

The 24 commercial requirements (RFP section 6) are contractual commitments rather than functions; they are answered in Section 19 and listed in Annex A without a fitment category.

6.1 What runs today and what is completed in the project

Capability	Available in SalesVerse 2.0 today	Completed during implementation
Products and rating	Seven Appendix 3 products (FTP-HP, FTP-NP, PFT, PHA, PRO, KHR, OSA) with configurable rating	IIFT actuarial rate tables replace the indicative rates; questionnaires and templates loaded
Underwriting referral	Financing above B\$150,000 referred to IIFT Sales	Further referral rules agreed in design
Payment control	Seven-day grace-period block on all agents of an agency	Bank-specific settlement rules, if any
Approvals	Maker-checker with configurable levels and amount thresholds, e.g. second approval from B\$300,000 sum covered	IIFT approval matrix and approver roles
AML/KYC	Watch-list screening with fuzzy name matching; Compliance review queue; a confirmed match rejects the subject's pending requests	Adapter to IIFT's AML provider or lists
e-Signature	On-screen signature and single-use e-mail link	Evidence format agreed with IIFT Legal
Documents	e-Policy and e-Receipt PDFs; encrypted document storage; ClamAV scan of uploads, mandatory in production	IIFT templates, wording and branding
End of day	EOD report, FIN interface file, receipt reconciliation; re-runs sent to FIN as numbered revisions	FIN file format sign-off; BRR reconciliation
Integration	Transactional outbox with retry and dead-letter	Adapters to IIFT core, FIN, AML provider, SMS gateway, SMTP and Active Directory
Reports	12 standard reports; Excel, CSV and PDF export; scheduling	Remaining reports from the agreed catalogue

Capability	Available in SalesVerse 2.0 today	Completed during implementation
Audit and data protection	Append-only audit enforced by a database trigger; AES-256-GCM encryption of IC and passport numbers	Retention and archival job; key rotation tooling; SIEM export
Access	Login, users, roles and permissions	IITH password and session values; MFA for back-office

Table 10: Capability status

6.2 Work that depends on IIFT

- Adapters to IIFT's core system, FIN, AML provider, SMS gateway, SMTP relay and Active Directory. Each needs IIFT's interface specification, test endpoint and credentials before it can be completed.
- IIFT's actuarial rate tables, questionnaires, product wording and document templates, replacing the indicative values in the working system.
- MFA for back-office users, production and DR infrastructure set-up, performance testing and the independent VAPT.
- Data migration, UAT, training and the IIFT-specific document set.

7. Personas

The personas below describe who uses the solution and what each needs. Roles and permissions are configurable; these are the defaults we propose for design.

Persona	Goals	Key tasks	Screens used	Role / permissions
Bank officer (branch banker)	Sell financing and annual plans quickly; keep customers informed	Quotations, participant capture, document upload, payment submission, claim notification	Dashboard, New quotation, Policies, Participants, Billing, Claims	Bank officer; own branch data
Bank supervisor	Oversee branch production and overdue payments	Review team pipeline, reports, bank-level requests	Dashboard, Team and hierarchy, Reports, My requests	Bank supervisor; whole bank
Agency principal / main agent	Grow the agency; avoid issuance blocks	Monitor sub-agents, payments, renewals, commission	Dashboard, Team and hierarchy, Billing, Commission, Reports	Main agent; whole agency
Sub-agent	Issue policies and get paid commission	Quotations, policies, payments, renewals, issues	New quotation, Policies, Renewals, Billing, Support issues	Agent; own records
IIFT Operations officer (maker)	Process Banca work accurately	Agent registrations, participant and policy changes, document checks	Agents, Participants, Policies, Document checks	Operations maker
Operations supervisor (checker)	Approve correct work promptly	Review and approve or reject requests with remarks	Approvals inbox, Approval detail, Dashboard	Operations checker
Underwriter / quality check	Accept good risk; refer the rest	Quality check, referrals above limits, endorsements, cancellations	Approvals inbox, Policies, Products	Underwriting checker
Finance officer	Match every receipt and close each day	Verify payment proofs, run EOD, review reconciliation	Payments, End of day, Integration monitor, Reports	Finance maker or checker
Compliance officer	Keep AML/KYC risk under control	Review possible matches, maintain watch-lists, compliance reports	AML/KYC cases and watch-lists, Audit trail, Reports	Compliance
Support desk	Resolve user problems within SLA	Triage, assign and resolve issues; release locked accounts on request	Issues, Users, Integration monitor	Support
Management	See production, risk and service at a glance	Read dashboards and scheduled reports	Dashboard (management KPIs), Reports and schedules	Management, read only
System administrator	Keep the platform configured and secure	Users, roles, workflows, products, parameters	Users, Roles and permissions, Workflows, Products, Parameters	Administrator
Participant (remote e-signature)	Sign without visiting a branch	Open link, confirm details, sign	Participant e-signature page	No account; single-use link
IITH core and finance systems (system actors)	Receive and send master and financial data	Agent master sync, FIN file intake, commission statements	Integration monitor (seen by IIFT IT)	System credentials; mTLS, API key or SFTP key

Table 11: Personas

8. User Journeys

Each journey shows who acts, on which screen, and what the system does. The diagram gives the flow; the table gives the detail. Journeys are rehearsed in the fit-gap workshops and become UAT scenarios.

8.1 Agent onboarding

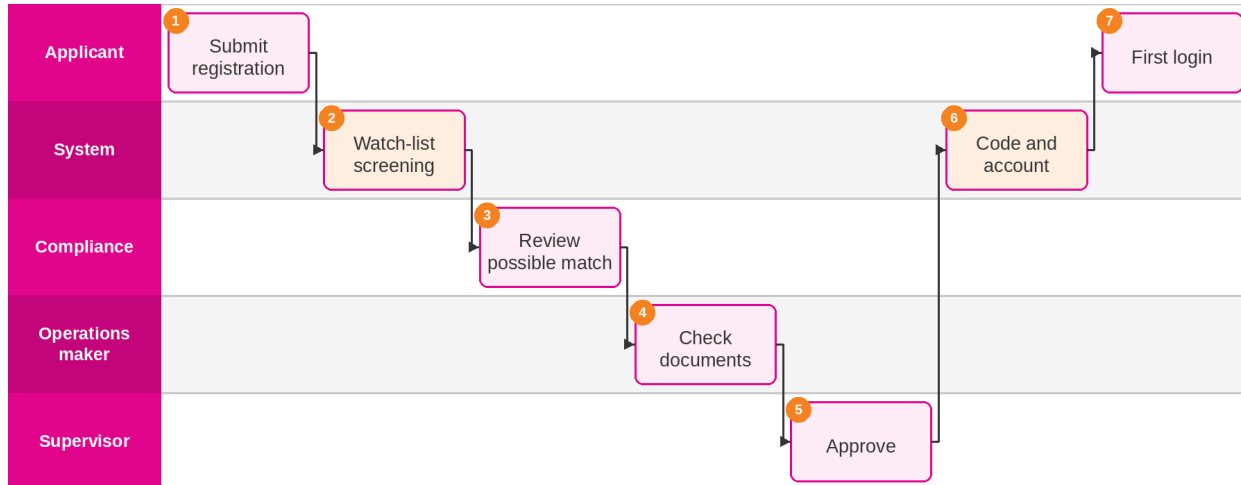


Figure 1: Agent onboarding

Step	Actor	Screen	What happens
1	Applicant	Agent registration	Enters personal, agency or bank and KYC data; uploads documents.
2	System	None	Screens the name with fuzzy matching; clear results pass straight on.
3	Compliance officer	AML/KYC cases	Clears or rejects a possible match with remarks.
4	Operations officer	Document checks	Validates documents and expiry dates.
5	Operations supervisor	Approvals inbox	Approves; the maker cannot approve their own request.
6	System	None	Issues the agent code, activates the account, e-mails the agent.
7	Agent	Sign-in, Change password	Signs in, sets a new password and lands on the dashboard.

8.2 Quotation to e-Policy for an annual product (FFR02 to FFR05)

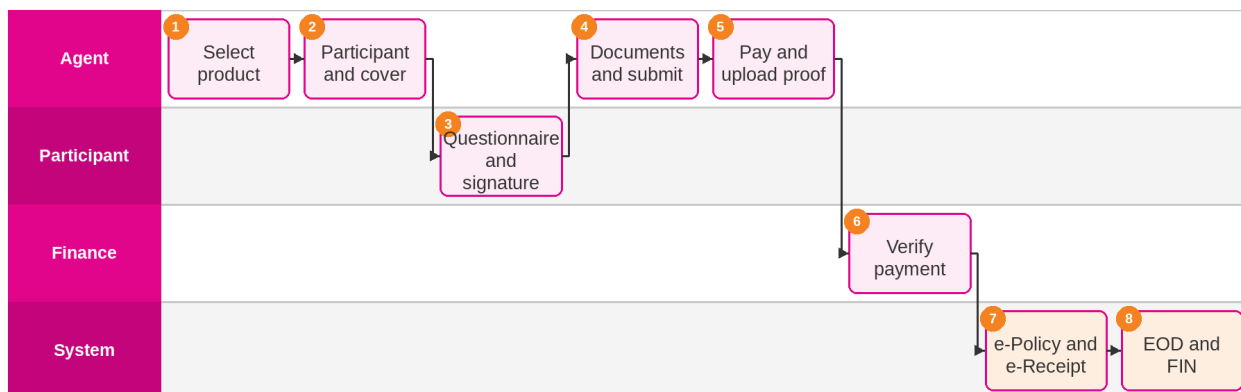


Figure 2: Quotation to e-Policy for an annual product (FFR02 to FFR05)

Step	Actor	Screen	What happens
1	Agent	New quotation	Chooses PHA, PRO, KHR or OSA from the authorised list.
2	Agent	New quotation	Captures participant and plan; the live quote shows the contribution.
3	Participant	Participant e-signature page	Completes the questionnaire and signs on screen or by e-mail link.
4	Agent	Policy detail	Uploads required documents; the checklist must be complete to submit.

Step	Actor	Screen	What happens
5	Agent	Billing	Pays and uploads proof of payment.
6	Finance officer	Payments	Verifies the proof against the bank statement.
7	System	None	Policy becomes Active; PDFs are produced and e-mailed.
8	System	End of day	The day's issuance goes into the EOD report and FIN file.

8.3 Banca hire-purchase financing with referral and grace period (FFR01)

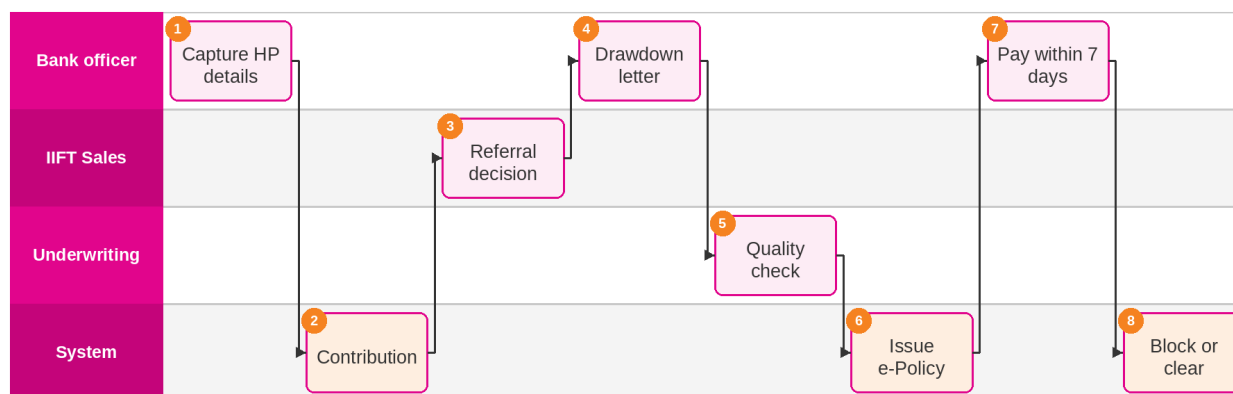


Figure 3: Banca hire-purchase financing with referral and grace period (FFR01)

Step	Actor	Screen	What happens
1	Bank officer	New quotation	Captures participant, financing amount, tenure and profit rate.
2	System	New quotation	Calculates the contribution; above B\$150,000 the case is referred.
3	IIFT Sales	Approvals inbox	Approves or declines cases above the high-risk limit.
4	Bank officer	Policy detail	Uploads the drawdown letter and remaining documents.
5	Underwriter	Approvals inbox	Approves the contract after quality check.
6	System	None	Issues e-Policy (issue-then-pay); payment due within seven days.
7	Bank officer	Billing	Submits payment with proof.
8	System	Dashboard	If unpaid on day 8, blocks issuance for the agency until settled.

8.4 Payment submission, verification and e-Receipt

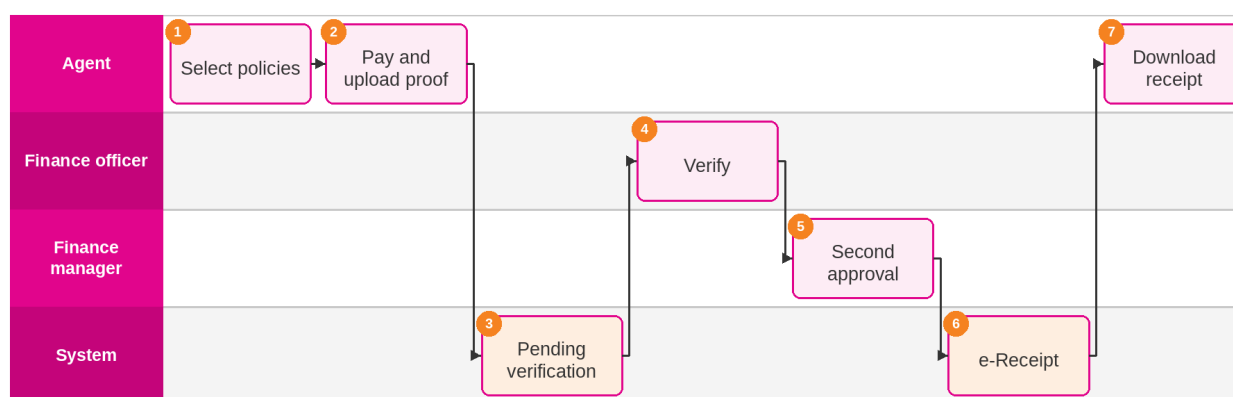


Figure 4: Payment submission, verification and e-Receipt

Step	Actor	Screen	What happens
1	Agent	Billing (outstanding)	Selects one or many outstanding policies.
2	Agent	Billing (bulk payment)	Allocates amounts per policy and uploads one proof.
3	System	Payment detail	Marks the payment Pending Verification and queues it.
4	Finance officer	Payments	Checks the proof; approves or rejects with remarks.
5	Finance manager	Approvals inbox	Approves when the amount is above the threshold.

Step	Actor	Screen	What happens
6	System	None	Produces the e-Receipt, marks policies Paid, lifts any agency block.
7	Agent	Payment detail	Downloads or e-mails the e-Receipt.

8.5 Endorsement or cancellation

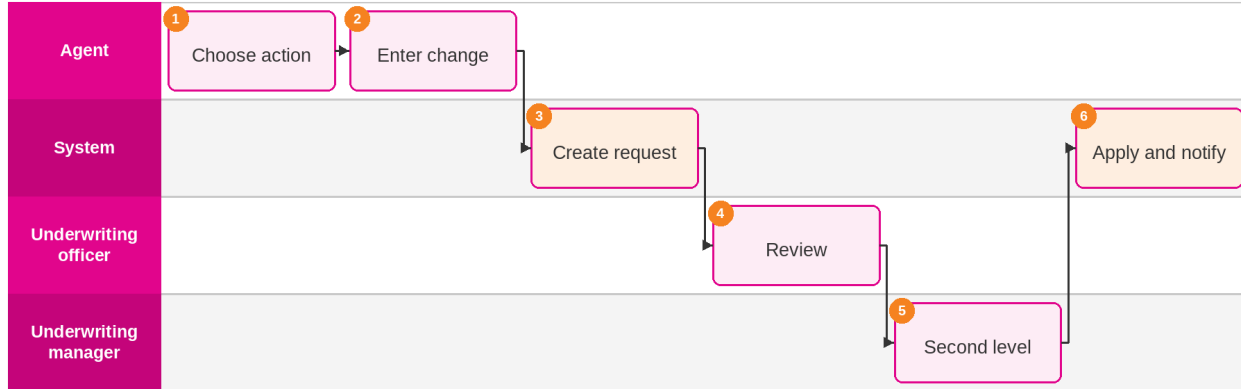


Figure 5: Endorsement or cancellation

Step	Actor	Screen	What happens
1	Agent	Policy detail (servicing)	Selects Endorse or Cancel on an active policy.
2	Agent	Policy detail (servicing)	Enters changes or reason and attaches documents.
3	System	My requests	Records before and after values; routes by workflow.
4	Underwriting officer	Approvals inbox	Reviews and approves or rejects with remarks.
5	Underwriting manager	Approval detail	Approves cancellations and endorsements above threshold.
6	System	Policy detail	Applies the change, records audit, notifies the agent.

8.6 Claim notification

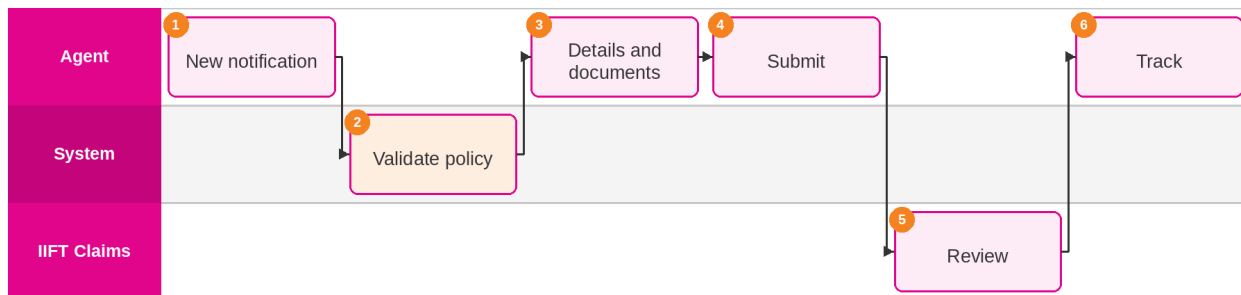


Figure 6: Claim notification

Step	Actor	Screen	What happens
1	Agent	Claims (notification)	Opens a new claim notification.
2	System	Claims (notification)	Checks the policy was in force on the event date.
3	Agent	Claims (notification)	Enters claim details and uploads documents.
4	Agent	Claims (notification)	Submits; a reference number is issued.
5	Claims team	Claims	Reviews the notification and updates its status.
6	Agent	Claims (detail)	Follows status and responds to requests.

8.7 Issue reporting with SLA

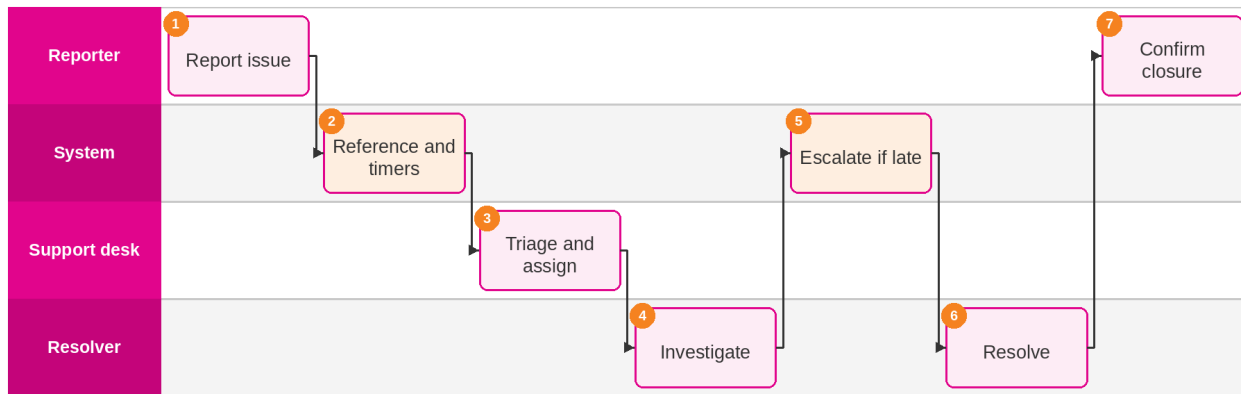


Figure 7: Issue reporting with SLA

Step	Actor	Screen	What happens
1	Agent or IIFT user	Support issues	Describes the problem and attaches screenshots.
2	System	None	Issues a reference number and starts SLA timers.
3	Support desk	Issues	Sets priority and assigns by rule.
4	Resolver	Issues	Updates the issue and the reporter.
5	System	None	Alerts the supervisor when a target is about to be missed.
6	Resolver	Issues	Records the fix.
7	Reporter	Support issues	Confirms or reopens.

8.8 End of day and FIN reconciliation

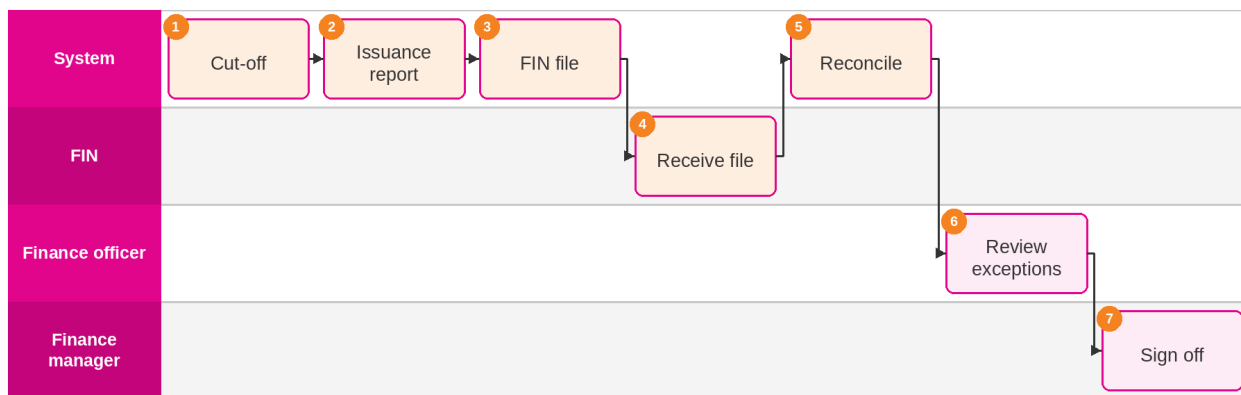


Figure 8: End of day and FIN reconciliation

Step	Actor	Screen	What happens
1	System	End of day	Closes the business day at the configured time.
2	System	End of day	Builds the issuance report by product, bank and agent.
3	System	Integration monitor	Writes the FIN interface file and queues delivery.
4	FIN system	None	Receives and posts the entries.
5	System	End of day	Reconciles receipts; lists exceptions.
6	Finance officer	End of day	Investigates and comments on exceptions.
7	Finance manager	End of day	Signs off the run; history is kept.

8.9 Maker-checker approval

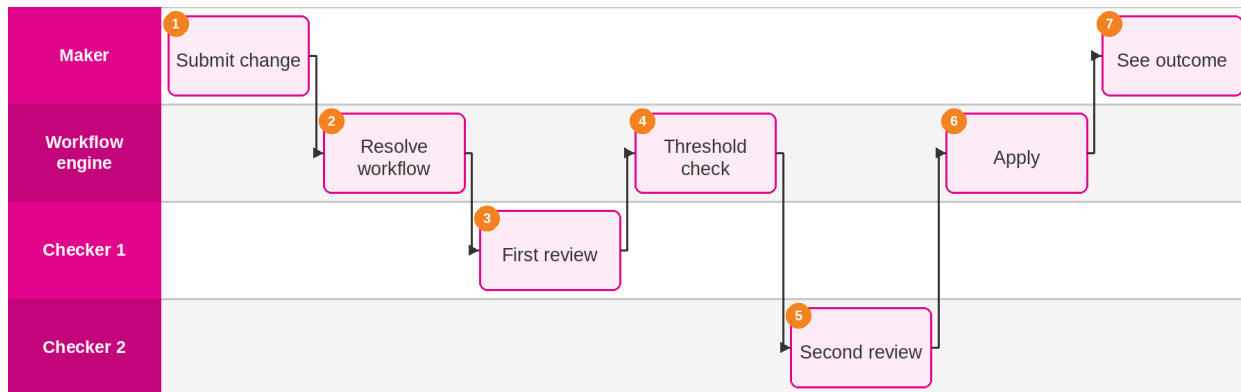


Figure 9: Maker-checker approval

Step	Actor	Screen	What happens
1	Maker	Any maintenance screen	Submits a change; before and after values are stored.
2	Workflow engine	None	Finds the definition, levels and thresholds.
3	Checker 1	Approval detail	Approves or rejects with mandatory remarks.
4	Workflow engine	None	Adds a level, e.g. sum covered from B\$300,000.
5	Checker 2	Approval detail	Approves or rejects with remarks.
6	Workflow engine	None	Applies the change with audit record and notification.
7	Maker	My requests	Sees the result and any remarks.

9. Functional Solution

SalesVerse 2.0 is iorta TechNXT's distribution platform for Takaful and insurance. For IIFT it is configured as the IIFT Agent/Banca Portal & Back-office: one portal for agents and bank officers and one back-office for IIFT and IITH staff, sharing a single database, workflow engine, document store and audit trail. A transaction submitted in the portal appears in the back-office at once, and every action on it is recorded.

Functions are listed by RFP area in Section 5, users in Section 7, journeys in Section 8 and screens in Section 11. This chapter covers the rules that tie them together.

9.1 Policy lifecycle

Only the server changes a policy's status, after validation and any required approval. Validation covers product eligibility, mandatory documents, the AML result, the agency payment block and the authority limit. Each change is written to the policy history and the audit trail.

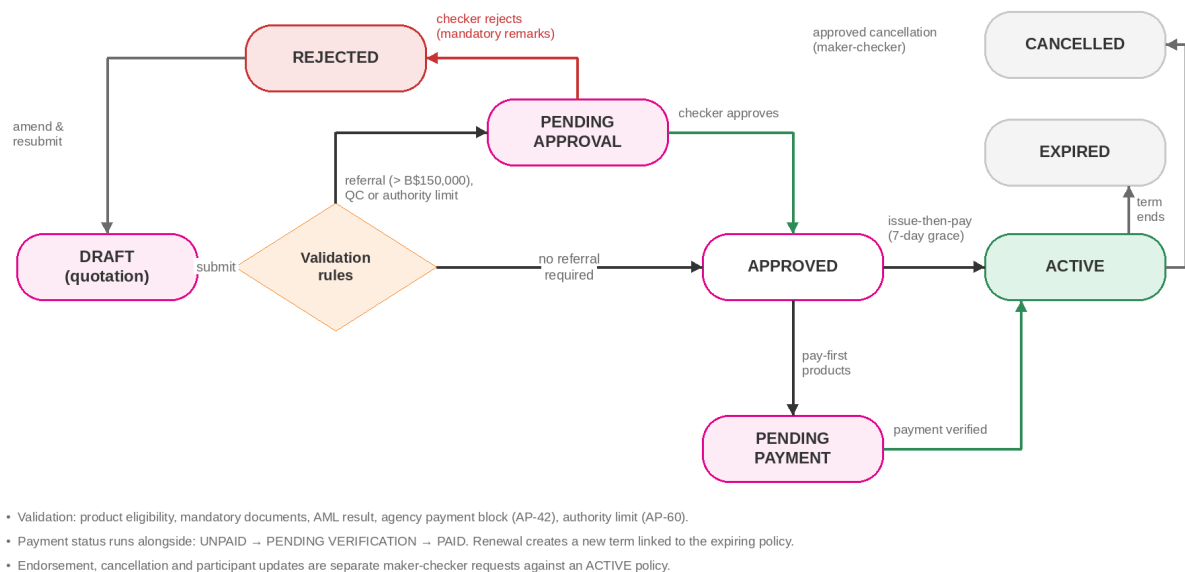


Figure 10: Policy lifecycle and status transitions

Status	Meaning	Next actions
DRAFT	Quotation saved; editable by the maker.	Submit
PENDING APPROVAL	Referral above the authority limit, quality check or other approval.	Approve or reject
PENDING PAYMENT	Approved; waiting for verified payment (pay-first products).	Payment verified
ACTIVE	In force; e-Policy and e-Receipt issued.	Endorse, cancel, renew, expire
REJECTED	Rejected with mandatory remarks.	Amend and resubmit
EXPIRED	Cover period ended.	Renew where enabled
CANCELLED	Cancellation approved.	None

9.2 Workflow engine and maker-checker

Approvals are data, not code. A workflow definition sets the transaction type, the number of levels, the permission needed at each level and amount thresholds that add a level, for example a second approval from B\$300,000 sum covered. On final approval a handler applies the change in the same database transaction as the audit record and the notification event. The API rejects any attempt by the maker to approve their own request. When Compliance confirms an AML match, the pending requests for that agent

or participant are rejected automatically with the Compliance remarks and the maker is notified. Journey 9 in Section 8 shows the flow.

Transaction	Level 1 checker	Level 2 checker	Rule
Agent / Banca registration	Banca/Sales officer	Banca/Sales manager	Always two levels
Agent profile update	Banca/Sales officer	None	None
Participant update	Underwriting officer	None	None
Quotation above B\$150,000 financing	IIFT Sales	Underwriting manager	Referral rule
New policy, sum covered from B\$300,000	Underwriting officer	Underwriting manager	Amount threshold
Quality check (Mortgage Takaful)	Underwriting / QC	None	Product rule
Endorsement	Underwriting officer	Underwriting manager	Contribution change threshold
Cancellation	Underwriting officer	Underwriting manager	Always two levels
Payment verification	Finance officer	Finance manager	Amount threshold
AML possible match	Compliance officer	Compliance manager	On escalation

Table 12: Default workflow definitions, confirmed in design

9.3 Seven-day grace period

A nightly job finds issued policies unpaid for more than seven days and blocks issuance for every agent of that agency. Agents see the reason on their dashboard. The block lifts automatically once Finance verifies the payment.

9.4 Notifications

Event	Recipients	Channels
Request submitted, approved or rejected (with remarks)	Maker; checker queue	Portal, e-mail
Policy issued with e-Policy and e-Receipt	Agent; participant	E-mail with PDF, portal
Payment overdue: reminder on day 5, block on day 8	Agent; agency principal; Finance	Portal, e-mail, SMS
Agency block lifted	All agents of the agency	Portal, e-mail
Renewal due (60, 30 and 7 days) and expiry	Agent	Portal, e-mail
Agent document expiring	Agent; Banca/Sales officer	Portal, e-mail
AML possible match	Compliance	Back-office, e-mail
Issue assigned, updated or SLA breached	Assignee; reporter; supervisor	Portal, e-mail
Dead-letter or EOD failure	Support team; IIFT IT	E-mail, monitoring alert

Table 13: Notification events (templates and channels configurable)

9.5 Reports

The platform has 12 standard reports with filters, Excel, CSV and PDF export and scheduled e-mail delivery. The go-live catalogue below is confirmed in design; reports not yet built are delivered during implementation within the fixed price.

Report	Primary users	Content
Daily issuance (EOD) report	Finance, Banca	Issued policies by product, bank and agent
FIN interface report	Finance	Entries sent to FIN with control totals
Receipt and BRR reconciliation	Finance	Matched and unmatched receipts and issuance
Policy register by status	Banca, Underwriting	Policies with status, dates and contribution
Production report	Management, Banca	Production by agent, agency, bank, product, period

Report	Primary users	Content
Outstanding payments and ageing	Finance, Banca	Unpaid policies by ageing bucket
Grace-period breaches and agency blocks	Finance, Banca	Policies unpaid over 7 days; blocked agencies
Renewals due and expiries	Banca, agents	Policies due for renewal or expired
Claims notification register	Claims, Banca	Notifications and their status
AML screening and review	Compliance	Screenings, possible matches, outcomes
Agent register and document expiry	Banca/Sales	Agents, hierarchy, status, licence and document validity
Approval turnaround	Management	Requests by type, ageing and turnaround
Issue log and SLA	IT, Management	Issues by priority, status and SLA result
User access review	IT security, Audit	Users, roles, last login, dormant accounts
Commission / referral fee statement	Agents, Finance	Statement retrieved from FIN (INT-05)

Table 14: Report catalogue for go-live

10. Products, Rating and Process Flows

All seven products in RFP section 1.1 and Appendix 3 are configured in the working application. Each product definition holds plans, sums covered, eligibility, required documents, questionnaires, renewal availability, payment mode (pay first, or issue then pay within the grace period) and its rating engine.

10.1 Product configuration

Code	Product	Line	Rating inputs	Key rules
FTP-HP	Financing Takaful Plan – Hire Purchaser	Mortgage	Financing amount, tenure, profit rate, age next birthday	Above B\$150,000 referred to IIFT Sales; QC before contract; issue-then-pay with 7-day grace
FTP-NP	Financing Takaful Plan – Non-Participating	Mortgage	As FTP-HP	Same engine and referral rule
PFT	Property Financing Takaful Plan	Mortgage	As FTP-HP (decreasing term)	Same engine; property financing documents
PHA	Personal Home Assistant Takaful Plan	Annual	Cover period 1 or 2 years	Employer IC, employee IC and details, passport, labour licence
PRO	Professional Takaful Plan	Annual	Plan A/B/C B\$15k/30k/50k; additional cover B\$90/120/140	Occupational Class I only; nominee, beneficiary or executor IC
KHR	Khairat Takaful Plan	Annual	Plan A/B/C B\$5k/10k/15k; Individual or Wider (Child)	Renewal switched off
OSA	Overseas Student Assist Takaful Plan	Annual	Basic B\$20k or Tertiary B\$50k	Brunei citizen, registered student, age 65 or under; student ID

Table 15: Product configuration

10.2 Contribution calculation

The Mortgage Takaful engine (FTP-HP, FTP-NP, PFT) calculates contribution from financing amount, financing period, profit rate and age next birthday using decreasing-term rate tables. Annual products use plan-based tables with riders and cover periods.

Rate tables, loadings, minimum contributions, rounding and authority limits are master data, changed in the back-office with every change audited. Maker-checker approval and effective-dated versions of rate tables, with the version recorded on each quotation, are added during implementation; until then a second person checks each rate change against IIFT's signed rate sheet. The working application holds indicative rates; IIFT's actuarial rates and wakalah fee and tabarru' split are loaded and verified during design and SIT.

10.3 Process flows (Appendix 3)

FFR01: Financing Takaful Plan, Hire Purchaser (FTPHP)

Step	RFP step	Portal / participant	Back-office / system
FTPHP01	Capture participant details	Full name, IC number, address, phone; duplicate check and AML screening.	Possible matches go to Compliance.
FTPHP02	Attach documents	Proposal form, product disclosure sheet, IC copy, HP agreement or financier approval; HP statement for existing HP.	Checklist enforces mandatory documents per case.
FTPHP03	Contribution	Calculated from financing amount, period, profit rate and age next birthday.	Above B\$150,000: referred to IIFT Sales.
FTPHP04	Drawdown letter	Agent uploads the drawdown letter and remaining documents and enters drawdown details.	Back-office validates documents.
FTPHP05	Issue receipt	e-Receipt issued for the contribution collected.	Receipt number assigned; FIN entry queued.
FTPHP06	Quality check	Status shows Pending Approval (QC).	Underwriting approves or rejects with remarks.

Step	RFP step	Portal / participant	Back-office / system
FTPHP07	e-Policy and e-Receipt	PDFs available to download and e-mailed to the participant.	Stored in the encrypted document store.
FTPHP08	EOD and final issuance	None	EOD report, master report, FIN file, reconciliation.

FFR02: Personal Home Assistant Takaful Plan (APHA)

Step	RFP step	Portal / participant	Back-office / system
APHA01	Capture participant details	Full name, IC number, address, phone.	Duplicate check, AML screening.
APHA02	Select cover	New or renewal; 1 or 2 years.	Plan rules from configuration.
APHA03	Quotation and questionnaire	Summary shown; Appendix II questionnaire completed.	Questionnaire version stored.
APHA04	Requirements	Employer IC, employee IC, employee details, passport, labour licence.	Checklist and expiry dates.
APHA05	Payment	Single or bulk payment with proof.	Finance verifies.
APHA06	e-Policy and e-Receipt	Issued on payment confirmation.	Policy becomes ACTIVE.
APHA07	FIN reporting	None	Daily report, FIN file, BRR reconciliation.

FFR03: Professional Takaful Plan (APPT)

Step	RFP step	Portal / participant	Back-office / system
APPT01	Capture participant details	Full name, IC number, address, phone.	Duplicate check, AML screening.
APPT02	Quotation, new or renewal	Plan A B\$15,000, Plan B B\$30,000, Plan C B\$50,000.	Occupational Class I only.
APPT03	Additional cover	Optional: Plan A B\$90, Plan B B\$120, Plan C B\$140.	Configured as rider.
APPT04	Summary and questionnaire	Summary shown; questionnaires completed.	Answers can trigger referral rules.
APPT05	Documents	Questionnaire, IC copy, IC of nominee, beneficiary or executor.	Checklist validation.
APPT06	Payment	Payment with proof.	Finance verifies.
APPT07	Policy and receipt	e-Policy and e-Receipt on confirmation.	Policy becomes ACTIVE.
APPT08	EOD reporting	None	EOD report, FIN entries, BRR reconciliation.

FFR04: Khairat Takaful Plan (APKT)

Step	RFP step	Portal / participant	Back-office / system
APKT01	Capture participant details	Full name, IC number, address, phone.	Duplicate check, AML screening.
APKT02	Quotation	Plan A B\$5,000, B B\$10,000, C B\$15,000; Individual or Wider (Child).	Renewal switched off.
APKT03	Summary and questionnaire	Summary shown; questionnaires completed.	Stored with quotation.
APKT04	Documents	Questionnaires, IC copy, nominee's IC copy.	Checklist validation.
APKT05	Payment	Payment with proof.	Finance verifies.
APKT06	Policy issuance	e-Policy and e-Receipt on confirmation.	Policy becomes ACTIVE.
APKT07	EOD reporting	None	EOD report to FIN; BRR reconciliation.

FFR05: Overseas Student Assist Takaful Plan (APAOS)

Step	RFP step	Portal / participant	Back-office / system
APAOS01	Capture participant details	Full name, IC number, address, phone.	Duplicate check, AML screening.
APAOS02	Select cover	Basic B\$20,000 or Tertiary B\$50,000.	Plan rules from configuration.
APAOS03	Eligibility	Brunei citizen, registered student, age 65 or under.	Ineligible cases cannot proceed.
APAOS04	Summary and questionnaire	Summary shown; Appendix IV questionnaire.	Stored with quotation.
APAOS05	Documents	Questionnaires, IC copy, student ID, IC of nominee, beneficiary or executor.	Checklist validation.
APAOS06	Payment	Payment with proof.	Finance verifies.
APAOS07	e-Policy and e-Receipt	Issued on confirmation.	Policy becomes ACTIVE.
APAOS08	EOD reporting	None	EOD report, FIN file, BRR reconciliation.

10.4 End-of-day processing

EOD runs on a schedule and can be re-run from the back-office. One locked job runs across application nodes. Each posting to FIN carries a revision number and its own idempotency key, and a re-run is marked as replacing the previous revision, so FIN can reject duplicates and keep only the latest figures for the day.

#	Step	Description
1	Cut-off	Configurable daily cut-off; later transactions roll to the next business day.
2	Issuance report	Policies issued, endorsed and cancelled and receipts approved, by product, bank and agent.
3	Final issuance	Master report for Mortgage Takaful final issuance (FFR01 step 8).
4	FIN interface file	Contribution and receipt entries, with wakalah fee and tabarru' split where applicable.
5	Reconciliation	Receipts reconciled today; BRR reconciliation added once IIFT confirms the BRR format.
6	Grace-period check	Policies unpaid for more than seven days found; agency blocks applied or lifted.
7	Sign-off	Finance reviews exceptions and signs off the run; run history kept.

11. Screen Catalogue

The tables list every screen in SalesVerse 2.0 as configured for IIFT, with its purpose and the RFP requirements it serves. Sample screens follow; all data shown is test data.

11.1 Agent/Banca Portal screens

Screen	Purpose	RFP IDs
Sign-in	Secure login	AP-01, AP-03, AP-04
Change password	Password change under policy	AP-02
Dashboard	Profile summary, pending actions, production, payments	AP-37, AP-52, AP-53
New quotation	Product, participant, coverage and live quote	AP-17 to AP-20
Quotations and policies	List and search by status	AP-21 to AP-23
Policy detail	Checklist, questionnaire, nominees, documents, signatures, submit; servicing actions; history tabs	AP-24 to AP-36, AP-44 to AP-47, AP-62
Renewals	Policies due for renewal	AP-26
Participants	List, registration and detail	AP-11 to AP-16
Billing	Outstanding, bulk payment, history	AP-37 to AP-42
Payment detail	Allocation, proof, status, e-Receipt	AP-39 to AP-41
Claims	List, notification and detail	AP-43
My requests	Status of submitted requests and rejection remarks	AP-49 to AP-51
Team and hierarchy	Main agent, sub-agents and bank officers	AP-09, AP-10
Agent registration	Online registration and onboarding	AP-07, AP-08, AP-46, AP-48
Profile	Profile and change requests	AP-05, AP-06
Commission	Commission and referral fee statement	INT-05
Reports	Role-scoped reports and exports	AP-58
Support issues	Report and track issues	AP-55 to AP-57
Notifications	In-portal notifications	AP-34, AP-54
Participant e-signature page	Remote signature by single-use link	AP-62

11.2 Back-office screens

Screen	Purpose	RFP IDs
Dashboard	Pending actions and management KPIs	BO-20, BO-21
Approvals inbox and detail	Maker-checker queue with before and after values	BO-16 to BO-19
Agents	List, registration and detail	BO-05 to BO-08, BO-10
Agencies and banks	Agency, bank and branch records	BO-09
Policies	Policy search and detail	AP-21 to AP-31 (back-office view)
Participants	Participant search and detail	AP-11 to AP-15
Payments	Verification and receipts	AP-40, AP-41
Claims	Claim notifications	AP-43
AML/KYC cases and watch-lists	Screening results, review, lists	BO-13 to BO-15
Document checks	Validation, approval and expiry	BO-11, BO-12
Audit trail	Audit search and export	BO-26 to BO-28
Issues	Assignment, priority, SLA	BO-29 to BO-31
Reports and schedules	Report runs, exports, schedules	BO-22 to BO-25
End of day	EOD runs, FIN file, reconciliation, sign-off	FFR EOD, INT-04
Integration monitor and reconciliation	Outbox messages, retries, dead-letters, reconciliation	INT-11 to INT-15
Users	User administration	BO-03
Roles and permissions	Role builder and data scopes	BO-02, BO-04
Workflows	Workflow definitions and thresholds	BO-16, COM-04

Screen	Purpose	RFP IDs
Products	Products, plans, rates, questionnaires	BO-32
Parameters and master data	System parameters and reference data	BO-32, BO-33, COM-09

11.3 Sample screens

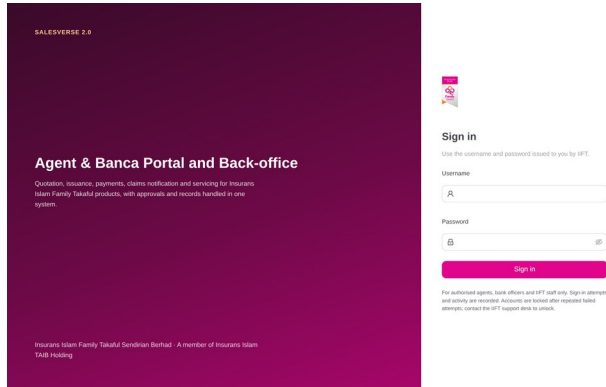


Figure 11: Secure login

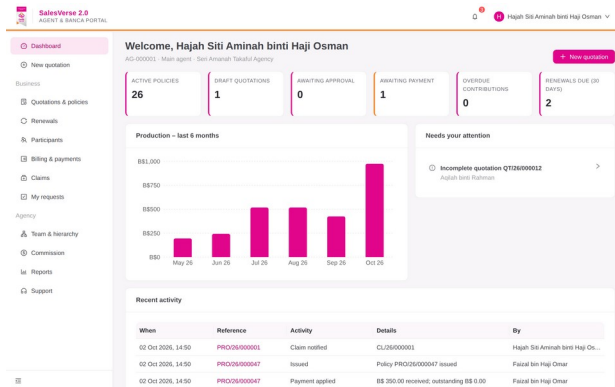


Figure 12: Agent/Banca dashboard

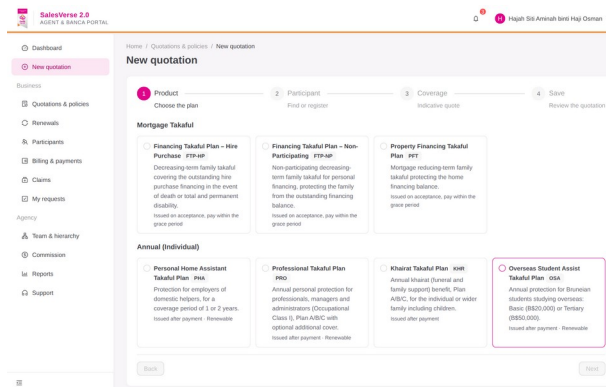


Figure 13: Product selection

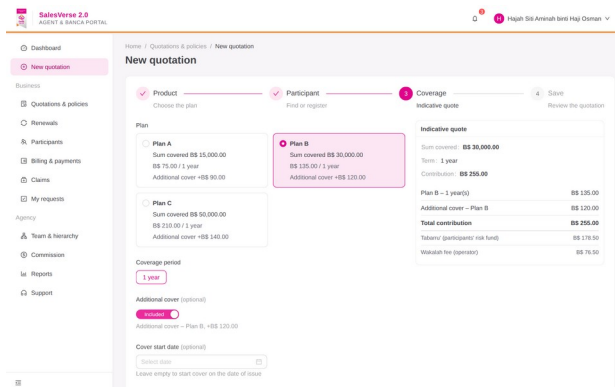


Figure 14: Quotation – participant & coverage

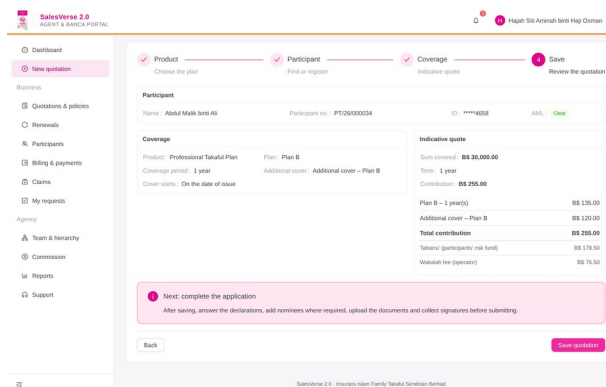


Figure 15: Quotation summary & contribution

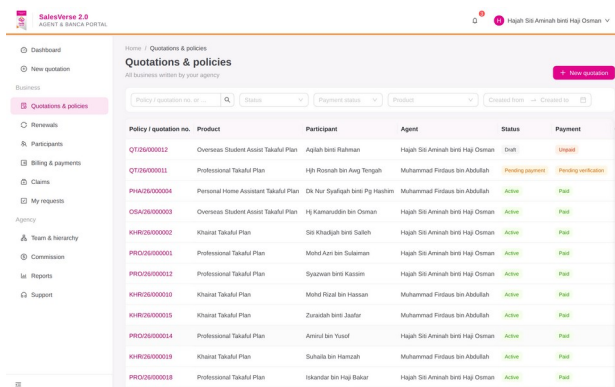


Figure 16: Policy listing & search

Figure 17: Policy details, documents & history

Figure 18: Participant management

Figure 19: Payment submission (single/bulk)

Figure 20: Claim notification

Figure 21: Issue reporting & tracking

Figure 22: Back-office management dashboard

Figure 23: Maker-checker approvals inbox

Figure 24: Agent/Banca administration

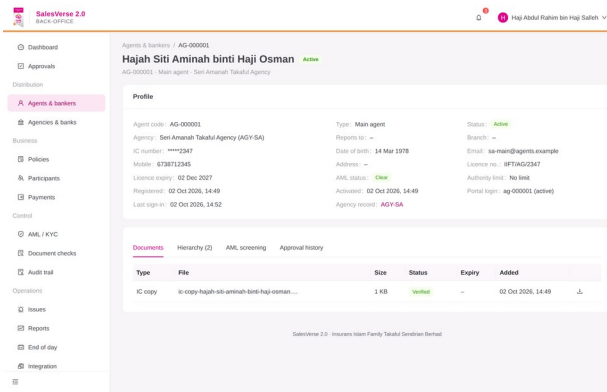


Figure 25: Agent profile, hierarchy & documents

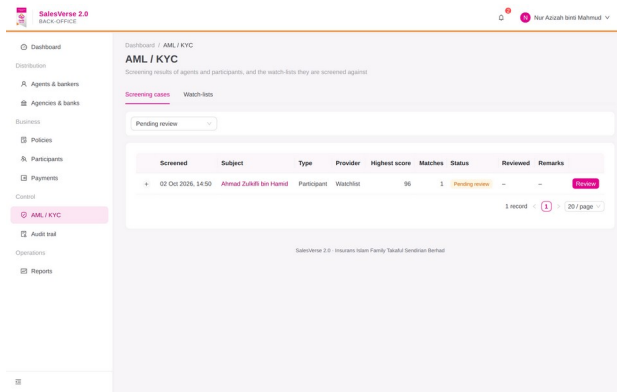


Figure 26: AML/KYC compliance review

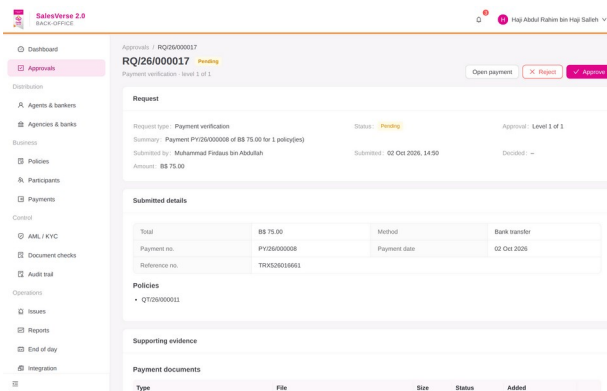


Figure 27: Payment verification & receipts

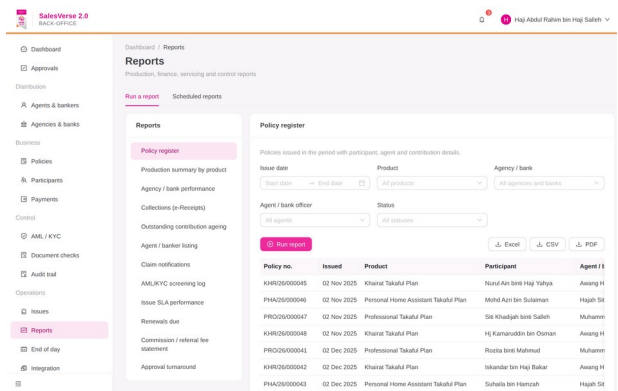


Figure 28: Reports & export

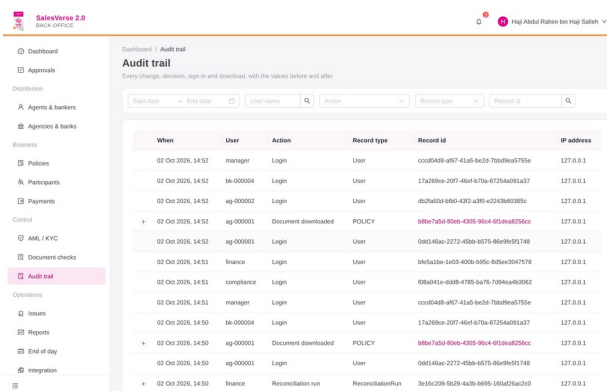


Figure 29: Audit trail search

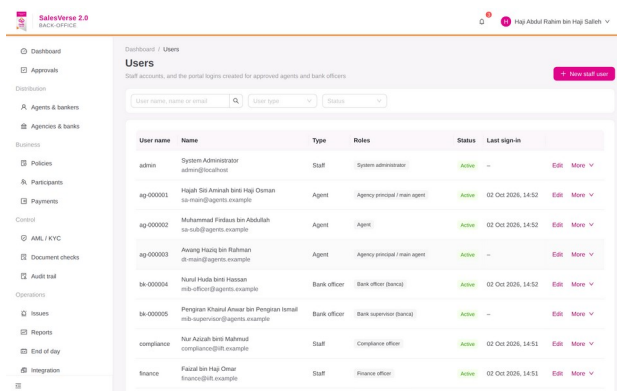


Figure 30: Users, roles & permissions

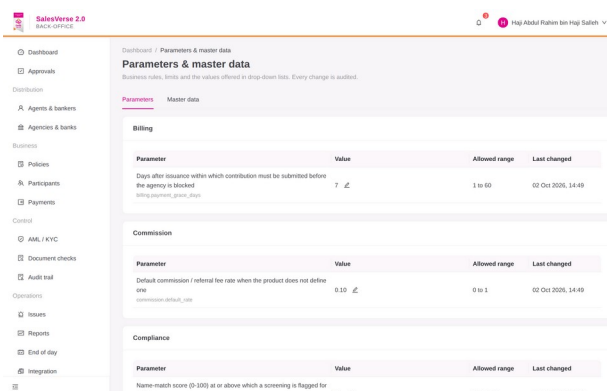


Figure 31: System parameters & master data

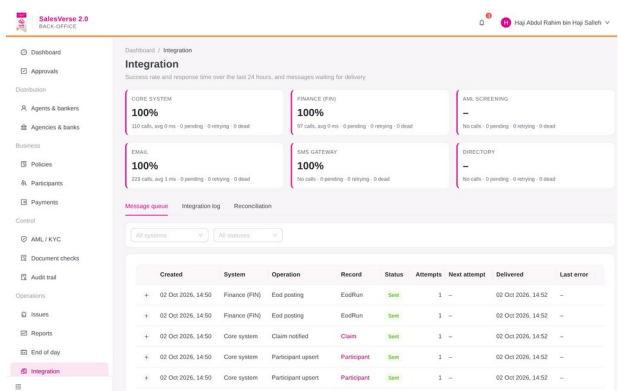


Figure 32: Integration monitor & reconciliation

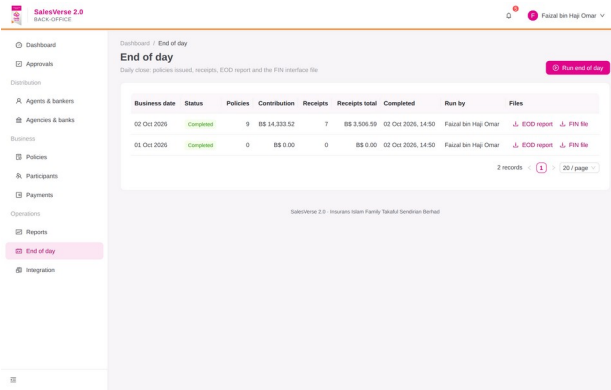


Figure 33: End-of-day processing

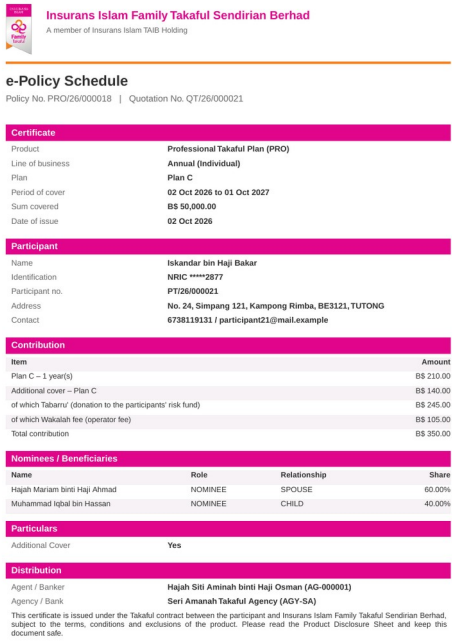


Figure 34: e-Policy schedule (PDF)

12. Solution Architecture

12.1 SalesVerse 2.0 logical architecture

SalesVerse 2.0 is a modular monolith: one deployable application split into modules that follow the RFP functional areas. For 26 named users and about 600 policies a year this is simpler to run than microservices, and a module can be split out later without a rewrite (NFR-20).

Users reach two React applications through a reverse proxy and web application firewall. The back-office path is published only on the internal network. Both applications call one versioned REST API. The integration layer isolates IITH systems through a transactional outbox, and PostgreSQL is the system of record.

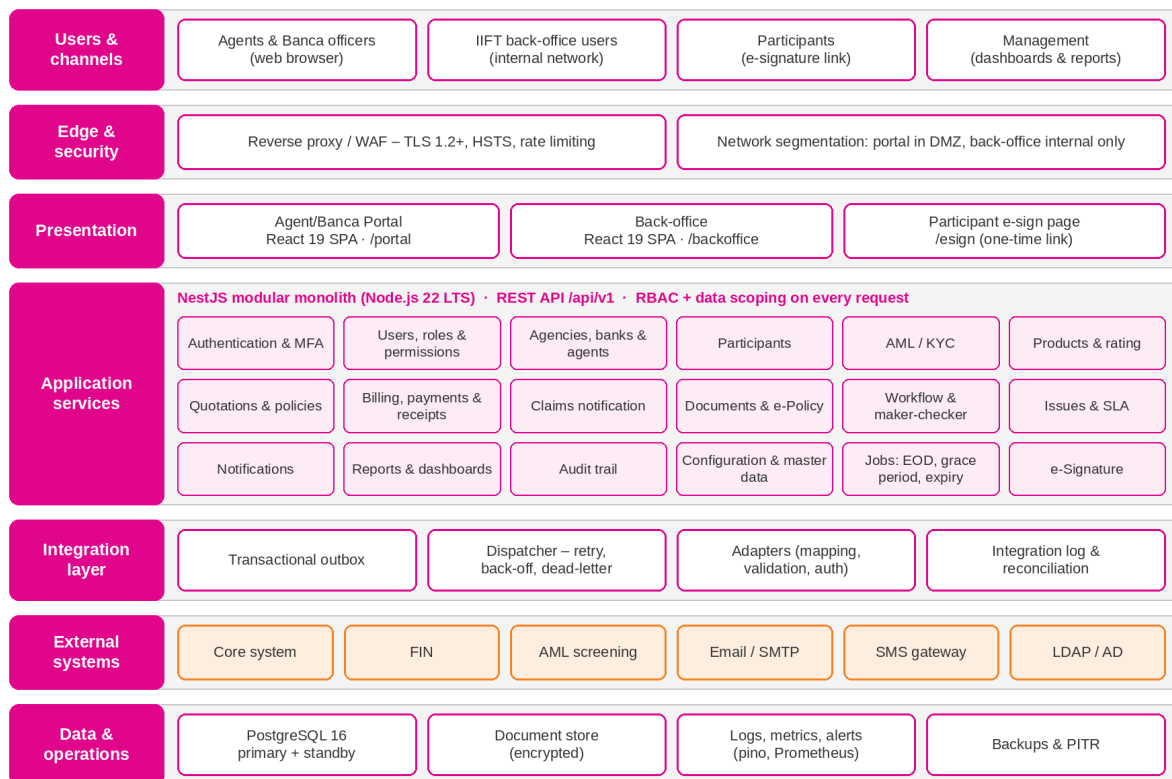


Figure 35: SalesVerse 2.0 logical architecture

12.2 SalesVerse 2.0 components

Module	Responsibilities	RFP requirements
auth, users-roles	Login, sessions, password policy, roles, permissions, data scopes	AP-01–04, AP-59, BO-01–04
agencies, agents	Agencies, banks, branches, agents, hierarchy, status, onboarding	AP-05–12, BO-05–12
participants, aml	Participant record, duplicate checks, watch-list screening, Compliance review	AP-11–16, AP-48, BO-13–15
products	Catalogue, plans, rate tables, rating engines, questionnaires	AP-17–19, BO-32
policies	Quotations, issuance, status, renewal, endorsement, cancellation, history	AP-20–36, AP-60
billing	Payments, allocation, proofs, verification, e-Receipts, grace-period rule	AP-37–42
claims, issues	Claim notifications; issues with priority and SLA timers	AP-43, AP-55–57, BO-29–31
documents, esign	Encrypted store, upload checks, PDF output, e-mail, on-screen and e-mail-link signatures	AP-44–47, AP-62, COM-06

Module	Responsibilities	RFP requirements
workflow	Workflow definitions, approval requests and actions, handlers	AP-33, AP-49–51, BO-16–19, COM-04
notifications	Templates; e-mail, SMS and in-portal delivery; delivery log	AP-34, AP-54, COM-05
reports, dashboard	Report definitions, exports, schedules, KPIs	AP-52–53, AP-58, BO-20–25, COM-08
audit, config	Append-only audit; parameters, master data, numbering	BO-26–28, BO-32–33, COM-03, COM-09
integration, jobs	Outbox, dispatcher, adapters, integration log; EOD, grace period, expiry, reminders	INT-01–15, AP-42

Table 16: Application modules

12.3 Data architecture

Entity group	Purpose
Agency, Bank, Branch, Agent	Distribution structure, agent code, type, status, parent agent, licence
User, Role, Permission, Session	Access control; sessions held on the server
Participant, Nominee	Shared participant record; IC and passport numbers encrypted
Product, Plan, RateTable, Questionnaire	Versioned product configuration
Quotation, Policy, PolicyTransaction	Policy lifecycle and history
Payment, PaymentAllocation, Receipt	Billing with exact NUMERIC amounts
Claim, Issue, Document	Claim notifications, issues with SLA timestamps, document metadata and checksums
WorkflowDefinition, ApprovalRequest, ApprovalAction	Maker-checker
AmlScreening, AuditEvent	Screening results; append-only audit with before and after values
OutboxEvent, IntegrationLog, EodRun, AgencyBlock	Integration delivery, EOD runs and grace-period blocks

Table 17: Key data entities

We chose PostgreSQL over MongoDB because IIFT's data is relational and financial. The comparison below summarises the reasons.

Criterion	PostgreSQL 16 (selected)	MongoDB
Data shape	Relational: agency, main agent, sub-agent; policy, participant, nominees, payments, receipts, claims	Suits self-contained documents with few relationships
Transaction integrity (NFR-16)	Issuance, receipt, audit and outbox commit in one ACID transaction; row locks for numbering	Multi-document transactions exist but fit this model less naturally
Financial accuracy	Exact NUMERIC plus foreign keys, CHECK and UNIQUE constraints	Decimal128; integrity rules mostly in application code
Reporting (BO-22–25)	Joins and aggregates written directly in SQL	Aggregation pipelines; joins via \$lookup
Flexible data	JSONB for questionnaires, product settings and audit values	Native
Operations	Streaming replication and point-in-time recovery; managed on every cloud and on-premise	Replica sets; a second skill set for IITH
Licence	PostgreSQL licence, no fee	SSPL; enterprise features are paid
Fit to volume	Under 5 GB of structured data in five years	Strength is very large schemaless data, not needed here

Table 18: Database selection

12.4 Integration architecture

Each outbound message is written to an outbox table in the same transaction as the business change. A policy therefore cannot be issued without its FIN entry being queued, and a failed call loses nothing. A

dispatcher delivers messages through adapters, retries with back-off, logs every attempt and moves exhausted messages to a dead-letter queue shown in the integration monitor. The outbox, retry and dead-letter handling run in the working application; the IIFT adapters are built during implementation against the interface specification (DEL-09).

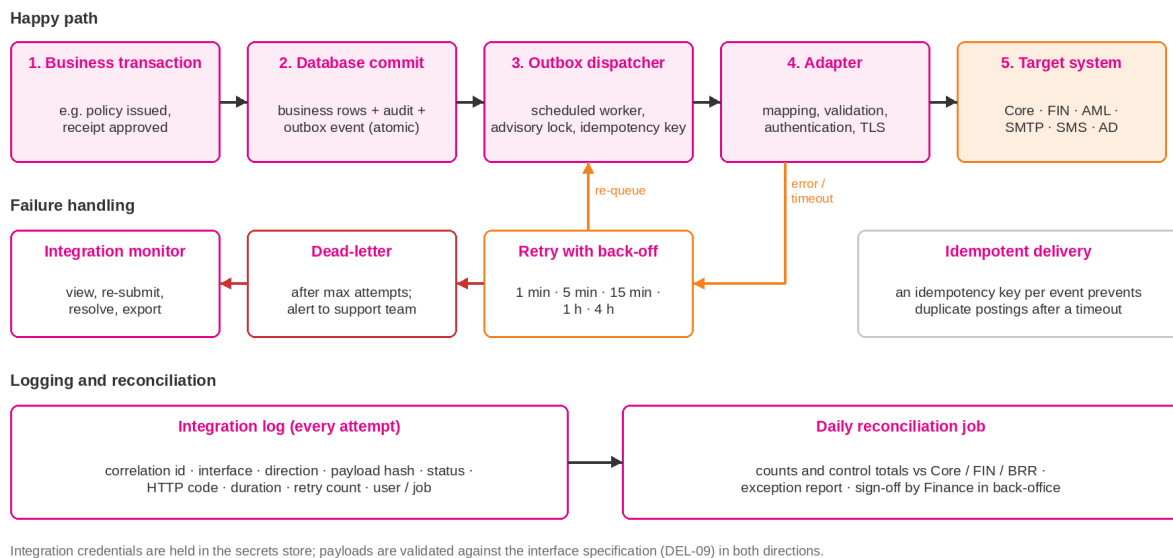


Figure 36: Integration flow with outbox, retry and reconciliation

Interface	Direction	Method	Frequency	Data	Security
Core system	Both ways	REST API or scheduled file	Event-driven plus daily sync	Agent, agency, bank master; status; policy summary	mTLS or API key; IP allow-list
FIN	Out (commission in)	Interface file over SFTP, or API	EOD; statements on demand	Contribution, receipts, allocations; commission and referral fees	SFTP keys or mTLS
AML screening	Request / response	REST API	On onboarding and change; scheduled re-screen	Name, IC, date of birth, nationality; match result	API key or OAuth2; TLS
Active Directory	Inbound	LDAPS	At login	Back-office authentication and groups	LDAPS service account
SMS gateway	Out	HTTPS API	Real time	OTP and alerts	API key; TLS
E-mail	Out	SMTP with STARTTLS	Real time	Notifications, documents, scheduled reports	Authenticated relay

Table 19: Interface catalogue

12.5 Technology stack

Layer	Technology and version	Reason
Front end	React 19, TypeScript, Vite, Ant Design 6, TanStack Query, React Router	Widely used; accessible components themed to Insurans Islam TAIB colours
API	Node.js 22 LTS, NestJS 12 (TypeScript), modular monolith	Module boundaries follow the RFP functional areas
Data access	Prisma 7 with the PostgreSQL driver adapter; versioned SQL migrations	Typed, parameterised queries only
Database	PostgreSQL 16	See data architecture
Documents and reports	pdfkit for PDFs, exceljs for Excel; encrypted local volume, MinIO or S3	No native dependencies
Jobs	NestJS scheduler with PostgreSQL advisory locks	One runner across replicas for EOD, grace period and outbox

Layer	Technology and version	Reason
Observability	pino JSON logs with correlation id; health endpoints; Prometheus metrics; Grafana	Central monitoring and alerting
Testing	Vitest (unit and API tests against PostgreSQL; web component tests with Testing Library), Playwright, k6	Run in the CI pipeline
Delivery	Non-root Docker images; docker compose on-premise; Kubernetes or ECS optional; GitHub Actions or GitLab CI	Same images in every environment

Table 20: SalesVerse 2.0 technology stack

All components are open source under permissive licences. The CI pipeline produces a CycloneDX software bill of materials of the production dependencies on every build; it is delivered with the technical manual (COM-17).

12.6 Non-functional requirements

Security requirements (NFR-08 to NFR-14) are covered in Section 13. The remaining targets are:

Area	Target and approach
Performance (NFR-01)	Page load under 3 s; standard API calls p95 under 1 s; reports on 12 months of data under 10 s. Proven by k6 tests (DEL-18).
Capacity (NFR-02, 03, 06, 07)	100 concurrent and 500 named users; ten times Appendix 2 volume. Stateless API nodes scale out.
Availability (NFR-04, 05)	99.5% overall and 99.9% in IIFT business hours; two application nodes and a PostgreSQL standby.
Backup and DR (NFR-17 to 19)	Daily full backup and continuous WAL archiving; 35 days online, monthly copies for 12 months; RPO 15 minutes, RTO 4 hours; annual DR test.
Integrity and errors (NFR-15, 16)	ACID transactions and idempotency keys; users see a generic message with a correlation id.
Monitoring and logs (NFR-26 to 28)	Health, metrics, job and integration monitors; alerts on downtime, error rate, latency, failed jobs and dead-letters; logs to a central store.
Retention and quality (NFR-29, 30)	Policy and financial records kept 7 years after expiry by default; validation on every input plus database constraints.
Usability (NFR-22 to 24)	One design system; WCAG 2.1 AA target; current and previous versions of Chrome, Edge, Firefox and Safari.
API standards (NFR-25)	REST and JSON over HTTPS, OpenAPI 3, ISO 8601 dates, versioned URLs (/api/v1).

13. Security Architecture & Controls

This chapter is the single description of security controls; other chapters refer to it. Controls are aligned with OWASP ASVS Level 2, the OWASP Top 10, Brunei's Personal Data Protection Order 2025 (PDPO) and AMBD's technology risk expectations for Takaful operators. We claim alignment, not certification. IIFT's policies are mapped in detail in the Solution Architecture (DEL-06).

13.1 Controls

Control	Implementation (defaults configurable)	RFP reference
Authentication	Argon2id password hashing; rate-limited login; MFA by e-mail OTP or authenticator app for back-office (optional for portal); Active Directory single sign-on	AP-01, BO-01, NFR-09, INT-06
Password policy	12 characters minimum with complexity; last 5 passwords blocked; 90-day expiry; change at first login. All configurable	AP-02
Sessions	Sessions held on the server; HttpOnly, Secure, SameSite=Strict cookie; CSRF token; new session id at login; 15-minute idle and 8-hour absolute timeout; optional single session; revoked at logout	AP-03
Lockout	Locked for 30 minutes after 5 failed attempts (configurable); alert on repeated lockouts	AP-04
Authorisation	Permissions per role plus data scoping by agency, bank, branch and hierarchy, checked by the API on every call; deny by default	AP-59, BO-02, COM-02, NFR-10
Encryption in transit	TLS 1.2 or later with HSTS; TLS also for database, SMTP, LDAPS and APIs	COM-10, NFR-08
Encryption at rest	IC and passport numbers encrypted per field with AES-256-GCM, keys held outside the database; HMAC-SHA256 blind index for exact search; masked on screen; encrypted document storage and backups	COM-01, COM-10, NFR-08
Input and output	Whitelist validation of every request body; parameterised queries only; output encoding; CSP and security headers	NFR-14, NFR-30
File uploads	Type allow-list with magic-byte check, size limit, SHA-256 checksum, ClamAV malware scan (mandatory in production); files served only through authorised API calls	AP-46, COM-06
Audit	Append-only table; a database trigger rejects UPDATE and DELETE; the application role can only insert and read; before and after values, user, time, IP, correlation id	BO-26–28, COM-03, NFR-11
Errors and secrets	Generic error messages with correlation id; secrets in environment or vault, rotated on staff change and yearly	NFR-15, INT-10
System APIs	mTLS or OAuth2 client credentials for integrations; rate limiting	INT-09, INT-10

Table 21: Security controls

The audit trigger, field-level encryption, encrypted document storage and the ClamAV upload scan already run in the working application; the API will not start in production without the scanner configured. MFA, Active Directory sign-on and the remaining parameters are set up during implementation. Personal data is limited to what products and AML/KYC rules need; a data inventory with purpose and retention is delivered with the Solution Architecture.

13.2 Secure development lifecycle

Stage	Practice	Mechanism
Design	Threat model for login, payment, approval and e-signature flows	Workshop, ASVS checklist
Build	Secure coding standard aligned with OWASP ASVS Level 2 and Top 10; peer review of every change	Pull-request review
Pipeline	Static analysis, dependency, secret and container image scans; a failed scan blocks release	CI security gates
Testing	Authorisation tests per role, negative tests, DAST baseline in SIT	API tests, DAST

Stage	Practice	Mechanism
Pre-production	Independent VAPT, remediation and re-test	Independent tester
Operation	Monthly patching, vulnerability monitoring, log review	Maintenance service

13.3 Penetration testing and remediation

An independent tester assesses the portal, back-office, APIs and hosting set-up before production (NFR-13, DEL-17). The cost is in the fixed price. Findings are fixed and re-tested, and no Critical or High finding may be open at go-live. After go-live, vulnerabilities are fixed within these limits (NFR-12, MNT-12):

Severity	Description	Timeline
Critical	Remotely exploitable; data exposure or system compromise	Contained within 24 hours; fixed within 7 days
High	Significant impact, exploitation likely	Within 14 days
Medium	Limited impact or hard to exploit	Within 30 days
Low	Minimal impact; hardening advice	Within 90 days or next release

Table 22: Vulnerability remediation timelines

13.4 Regulatory alignment

For PDPO we provide consent and purpose notices at participant capture, access and correction support, retention rules and breach-notification support. For AMBD expectations we cover access control, change management, outsourcing and cloud considerations (infrastructure options), DR testing and audit trails. AML/CFT record keeping is met by the screening history and review decisions. Screens and documents use Shariah-appropriate terms: participant, contribution, Takaful operator, wakalah and tabarru'.

13.5 Security incident response

Phase	Action
Detect	Alert, log review or user report; logged as a P1 security incident.
Contain	Within 30 minutes: assess scope, isolate the component, revoke sessions or credentials.
Notify	IIFT IT security told immediately; iorta supports regulatory and PDPO notifications.
Recover	Patch, restore from a clean backup if needed, verify integrity, monitor.
Review	Root-cause report within 5 business days; actions tracked to closure.

14. Deployment & Release Management

14.1 Environments

Environment	Purpose	Hosted by	Data	Users
DEV	Development and sprint demos	iorta	Synthetic	iorta team
SIT	Integration and performance testing	IIFT (iorta-hosted until the IIFT VM is ready)	Synthetic and masked samples	iorta, IIFT IT
UAT	Acceptance testing and training	IIFT data centre	Masked or migrated test data	IIFT users
PROD	Live operation	IIFT/IITH data centre	Production	IIFT users; iorta under change control
DR	Disaster recovery	IIFT/IITH DR site	Replicated production	Activated per DR plan

Table 23: Environments

14.2 CI/CD pipeline

Every environment runs the same versioned container images, so what passes UAT is what runs in production. The pipeline can run on IITH's GitLab or on GitHub Actions; the repository is held for IIFT.

Stage	Activities
Commit and review	Branch, pull request, mandatory peer review
Build	Type check, lint, compile front end and API, build container images
Test	Unit and API tests against PostgreSQL; web component tests; coverage report
Scan	CodeQL static analysis, npm audit, gitleaks secret scan and Trivy image scan
Package	Versioned, signed images with release notes and SBOM
Deploy to SIT/UAT	Automated deployment and migration; Playwright smoke tests
Approve	Test sign-off and IIFT change approval
Deploy to PROD	Scripted deployment in the approved window; health checks; post-deployment checks

14.3 Release management and rollback

Every production change follows the agreed procedure (MNT-21): change request, impact assessment, SIT and UAT, release notes, IIFT approval and deployment in an approved window.

Release type	Content	Cadence
Emergency fix	P1/P2 defect or security patch	As needed, expedited approval
Maintenance release	Fixes, minor enhancements, patches	Monthly
Feature release	Change requests	Quarterly or as agreed
Platform upgrade	Node.js, PostgreSQL, framework and SalesVerse 2.0 updates	Typically yearly

The previous container images are kept, so the application rolls back in minutes. Database migrations follow the expand-and-contract pattern, which lets the previous version run on the new schema. A verified backup and a recovery marker are taken before each release. Rollback criteria and decision owners are set in the deployment plan (DEL-20).

14.4 Go-live checklist

Area	Readiness criteria
Business	UAT signed off; users trained; procedures and contacts issued; agents and banks informed
Technical	Production built and hardened; performance test passed; backups and monitoring running
Security	VAPT findings fixed and re-tested; certificates installed; privileged access reviewed
Data	Two migration rehearsals; reconciliation signed; master data and rates verified
Integration	All interfaces tested end to end; credentials in the vault
Operations	Operations manual, runbooks and DR plan; support rota and escalation agreed
Decision	Steering Committee go/no-go recorded with open items and rollback criteria

14.5 Cut-over plan

When	Activities
T-10 days	Scope frozen; deployment plan approved; agents and banks informed
T-5 days	Deployment rehearsal; final migration dry-run and reconciliation
T-2 days	Legacy data frozen; manual process continues where needed
T-1 day	Final extract, migration load, reconciliation, business check
T-0	Deployment, smoke tests, integration checks, go/no-go, users enabled
T+1 to T+20 business days	Hypercare: daily health checks and stand-ups, priority fixes

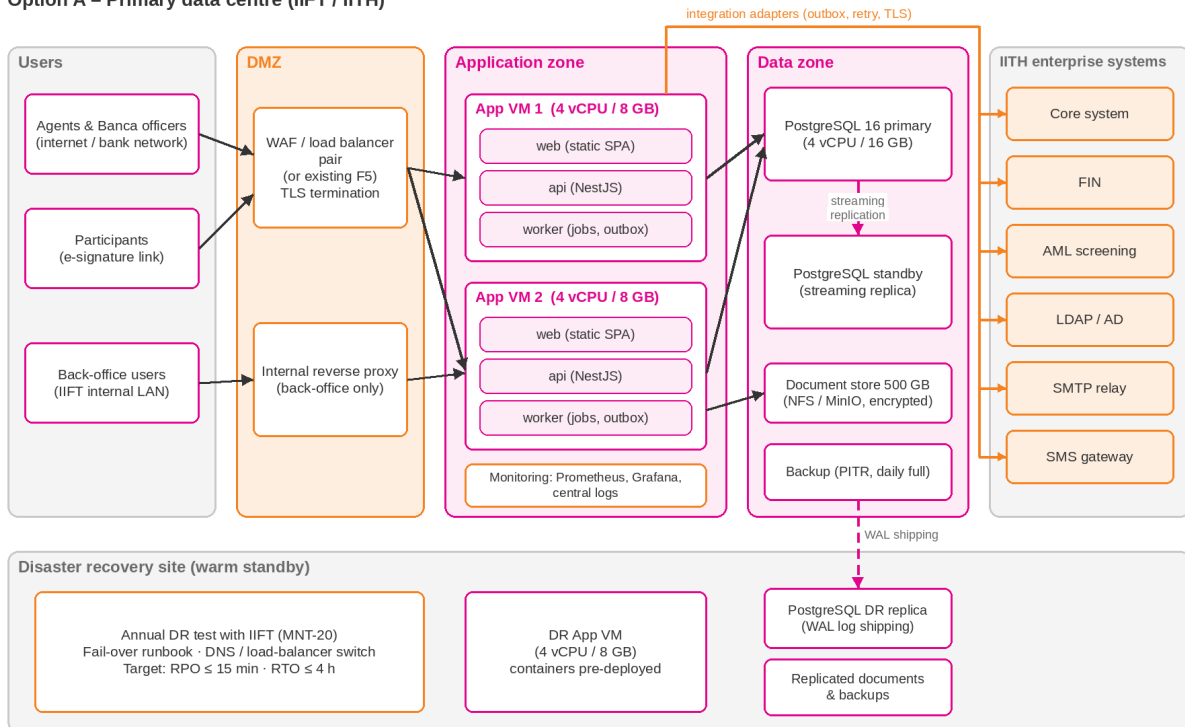
15. Infrastructure Options

SalesVerse 2.0 ships as standard containers and runs unchanged on-premise or in the cloud. We present three options and recommend Option A. Infrastructure is not part of iorta's fees under Option A; under Option B, cloud charges are passed through at cost.

15.1 Option A: on-premise in the IIFT/IITH data centre (recommended)

Two application VMs run behind the load balancer. PostgreSQL runs as a primary with a streaming standby, and a warm standby at the DR site receives WAL log shipping. The solution uses IITH's network security, backup and monitoring services where capacity allows. Production needs two 4 vCPU / 8 GB application servers, two 4 vCPU / 16 GB database servers and 500 GB of document storage.

Option A – Primary data centre (IIFT / IITH)



Sizing per environment is in the infrastructure chapter. Hardware, OS and network are IIFT/IITH-provided; iorta installs, configures and operates the application stack.

Figure 37: Option A: on-premise deployment

Component	Quantity	Specification	Storage	Provided by
Load balancer / WAF	Pair or existing F5	2 vCPU / 4 GB each	–	IIFT/IITH
Application server (PROD)	2	4 vCPU / 8 GB	100 GB	IIFT/IITH
Database server (PROD)	2 (primary + standby)	4 vCPU / 16 GB	200 GB SSD	IIFT/IITH
Document storage	NFS share or MinIO	–	500 GB	IIFT/IITH
Backup storage	Per IIFT backup policy	–	About 1 TB	IIFT/IITH
Monitoring	1 (or existing tooling)	2 vCPU / 4 GB	100 GB	IIFT/IITH
SIT	1 (app + DB)	4 vCPU / 16 GB	200 GB	IIFT/IITH
UAT	1 (app + DB)	4 vCPU / 16 GB	200 GB	IIFT/IITH
DR site	1 app + 1 DB	4 vCPU / 8 GB; 4 vCPU / 16 GB	200 GB + 500 GB	IIFT/IITH

Table 24: Option A sizing per environment

Item	IIFT / IITH	iorta TechNXT
Servers, storage, network, firewall, load balancer	Provide	Specify, review
Linux OS (RHEL 9 or Ubuntu 24.04 LTS) and licences	Provide, patch	Specify hardening
Container runtime, application, proxy configuration	Approve	Install, configure, operate
PostgreSQL install, replication, backup set-up	Approve; run backup infrastructure	Install, configure, tune
Monitoring and alerting	Infrastructure monitoring; receive alerts	Application monitoring
Certificates, DNS, SMTP relay, SMS account, AML subscription	Provide	Configure; support renewal
DR site and DR tests	Provide site; lead test	Replication; take part in tests

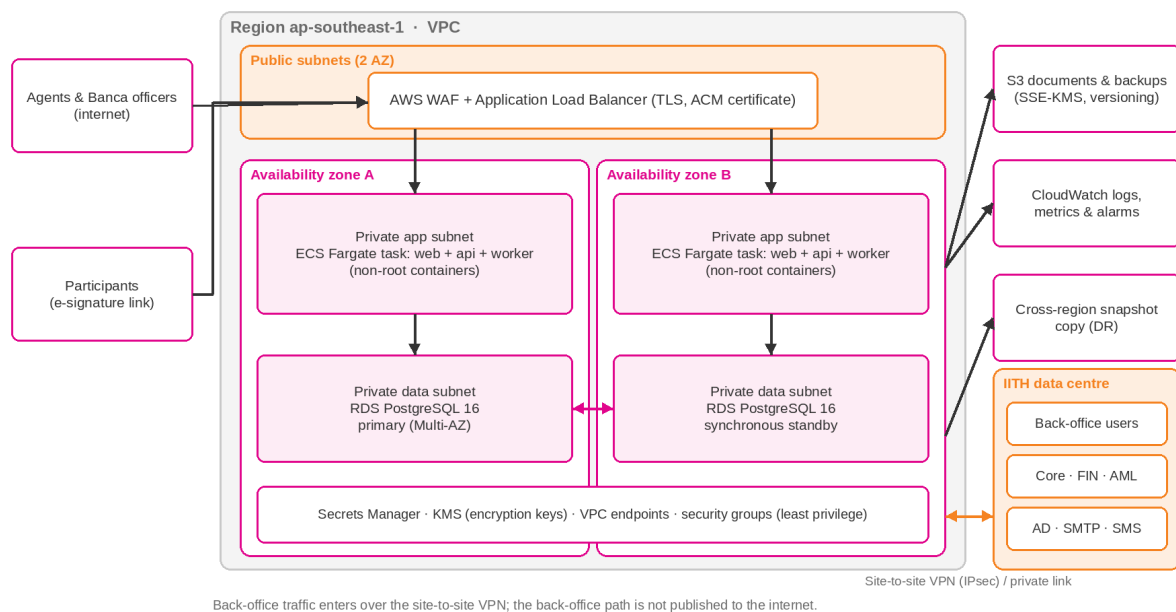
Table 25: Option A responsibilities

15.2 Option B: public cloud

The solution can run in AWS Asia Pacific (Singapore) or Azure Southeast Asia on managed services. Cloud use depends on IIFT's data-residency decision and on AMBD's outsourcing and cloud notification requirements; we will prepare the supporting documents.

Service	AWS reference (Azure equivalent)
Compute	ECS Fargate across two availability zones (AKS on Azure), or two EC2 instances
Database	RDS for PostgreSQL 16 Multi-AZ (Azure Database for PostgreSQL, zone-redundant)
Storage	S3 with SSE-KMS, versioning and lifecycle rules (Azure Blob)
Edge	Application Load Balancer with AWS WAF (Azure Application Gateway with WAF)
Monitoring	CloudWatch logs, metrics and alarms (Azure Monitor)
Connectivity	Site-to-site VPN to the IITH data centre for Core, FIN, AD, SMTP and back-office users
DR	Automated backups with point-in-time recovery and cross-region snapshot copy

Option B – Cloud (AWS Asia Pacific – Singapore; Azure Southeast Asia equivalent)



Indicative run cost B\$ 950 per month (pass-through). Subject to AMBD outsourcing / cloud notification and IIFT data-residency approval.

RPO ≤ 15 min (PITR) · RTO ≤ 4 h (restore in secondary AZ or region)

Figure 38: Option B: cloud deployment (AWS reference)

Service	Purpose	B\$ / month
Container compute (2 tasks across 2 availability zones)	Portal, back-office and API	B\$ 180
Managed PostgreSQL, Multi-AZ, with storage and automated backups	Database and PITR	B\$ 330
Application load balancer and web application firewall	TLS termination, OWASP rules	B\$ 120
NAT gateway and data transfer	Outbound integration and updates	B\$ 120
Object storage (SSE-KMS) and cross-region snapshot copy	Documents, backups, DR copies	B\$ 60
Logging, metrics and alarms	Monitoring and alerting	B\$ 60
Secrets and key management	Credentials and encryption keys	B\$ 20
Scaled-down UAT environment	Non-production testing	B\$ 60
Total per month		B\$ 950
Total per year (pass-through, indicative)		B\$ 11,400

Table 26: Option B indicative cloud cost

Managed cloud operations by iorta (patching, monitoring, backup checks, cost control) cost B\$ 4,800 a year. Cloud prices move with usage and exchange rates and are confirmed before provisioning.

15.3 Option C: hybrid

The back-office, database and documents stay in the IITH data centre. Only the portal's web tier (static application and reverse proxy) sits in the DMZ or at a cloud edge and forwards API calls over an encrypted private link. Personal data never leaves the data centre. This suits IIFT if internet-facing capacity in the data centre is limited.

15.4 Comparison and recommendation

Criterion	A: On-premise	B: Cloud	C: Hybrid
Data residency	Data stays in Brunei	Singapore region; needs IIFT and AMBD acceptance	Personal data on-premise; portal web tier in DMZ or cloud
Regulatory steps	Internal approval only	AMBD outsourcing and cloud notification	Depends on what sits in the cloud
Group assets	Uses IITH data-centre services and operating model	New cloud account and skills	Partly
Provisioning time	Depends on IITH VM procurement	Days	Mixed
Resilience	Two nodes, standby database, DR site	Multi-AZ; cross-region copies	Per component
Hosting charge from iorta	B\$0	About B\$ 11,400 a year pass-through plus B\$ 4,800 managed operations	Between A and B
Who runs it	IITH infrastructure team; iorta application support	iorta managed cloud operations	Shared

Table 27: Infrastructure options compared

Recommendation: Option A, on-premise in the IITH/IIFT data centre

Participant and agent data stays in Brunei and no extra regulatory step is needed.

IITH's existing data-centre services and operating model are reused, as for other group systems, which keeps cost and support effort down.

iorta charges nothing for hosting. Option B remains available if IIFT prefers an operating-expense model.

16. Technical Document Pack

The technical documents below accompany this proposal in the folder docs/technical. They describe SalesVerse 2.0 as it stands today and are updated for IIFT during the project. This chapter summarises them; it does not repeat their content.

Document	Content	RFP references
SalesVerse-2.0-Solution-Architecture.docx	Logical, functional and deployment architecture; integration and security architecture; on-premise and cloud infrastructure sizing	DEL-06, DEL-07, DEL-09 (baseline)
SalesVerse-2.0-Data-Dictionary.xlsx and .docx	Every table and field with type, constraints, encryption and retention	DEL-07, DEL-26
SalesVerse-2.0-Code-Standards-and-Quality-Report.docx	Coding standards, review process, automated test coverage and static analysis results	NFR-14, NFR-20
SalesVerse-2.0-Security-Assessment-Report.docx	Internal vulnerability assessment and security test results; the independent third-party penetration test follows before go-live	NFR-12 to NFR-14; DEL-17 (precursor)
SalesVerse-2.0-Production-Support-Handover.docx	Support process, runbooks, knowledge transfer and release management	DEL-27, MNT-01 to MNT-04, MNT-21, MNT-28

Table 28: Technical document pack

The security assessment report covers iorta's internal testing. It does not replace the independent penetration test, which is performed by a third party on the IIFT environment before go-live (DEL-17).

17. Implementation Methodology

17.1 Delivery approach: configure and integrate

SalesVerse 2.0 already runs IIFT's products and rules, so the project configures and integrates an existing platform instead of building one. Discovery starts with fit-gap workshops on the working system: IIFT users walk through each journey with their own products, confirm what fits, and record the gaps. The gaps become the configuration backlog.

Governance follows formal gates, so IIFT signs off scope, design, test exit and go-live at fixed points and pays against them. Configuration and integration run in two-week sprints, each ending with a demonstration to IIFT key users. SIT, an independent VAPT, UAT, two migration rehearsals and a rehearsed cut-over follow, then four weeks of hypercare.

The Steering Committee reviews each gate: design sign-off (M2), build complete (M3), SIT exit and security clearance (M4), UAT sign-off (M5), go-live (M6) and hypercare exit (M7).

17.2 Phases and activities

#	Phase	Weeks	Key activities	Outputs
0	Mobilisation	1–2	Kick-off, governance set-up, environment and data requests, detailed plan	Project charter, project plan (DEL-01, 02)
1	Discovery and fit-gap confirmation	2–6	Workshops per product and process on the working system; collect IIFT rates, wording, questionnaires, templates; confirm gaps	BRS, FRS, RTM, SAD, TDD, UI/UX, interface specification, test strategy (DEL-03 to 09, 12)
2	Configuration sprints	6–16	Five two-week sprints configuring products, rates, workflows, roles, reports and templates; demo every sprint	Configured solution, source code (DEL-10, 11)
3	Integration with IIFT systems	8–17	Adapters to core, FIN, AML provider, SMS, SMTP and Active Directory; joint tests with system owners	Working interfaces, test evidence
4	SIT, performance and VAPT	16–19	SIT, regression, performance test, independent VAPT, remediation, re-test	SIT results, performance report, security assessment (DEL-13, 14, 17, 18)
5	UAT, migration rehearsals, training	19–23	UAT support, two migration rehearsals, train-the-trainer and end-user training, manuals	UAT results, migration plan, training, manuals (DEL-15, 16, 19, 22 to 27)
6	Cut-over and go-live	24	Production deployment, final migration, go/no-go	Deployment plan, go-live checklist, production deployment (DEL-20, 21, 28)
7	Hypercare and closure	25–28	Hypercare, stabilisation, handover to support, closure	Go-live support, closure report (DEL-29, 30)

Table 29: Implementation phases

Configuration sprints

Sprint	Scope
Sprint 1 (wk 6–7)	Access set-up to IITH policy, MFA, roles and data scopes; agent, agency and bank master data loaded
Sprint 2 (wk 8–9)	Onboarding and approval matrix; participant rules; AML connection to IIFT's screening service
Sprint 3 (wk 10–11)	IIFT actuarial rates, questionnaires and document templates for all seven products
Sprint 4 (wk 12–13)	Referral and QC rules, policy servicing, e-Policy and e-Receipt wording and branding
Sprint 5 (wk 14–15)	Payment rules, EOD and FIN file sign-off, BRR reconciliation, remaining reports, notifications
Hardening (wk 16)	Performance tuning, security hardening, defect fixing, documentation; build complete (M3)

All 30 deliverables in RFP section 7 are provided; Section 5 gives the due week and acceptance evidence of each.

17.3 Project governance

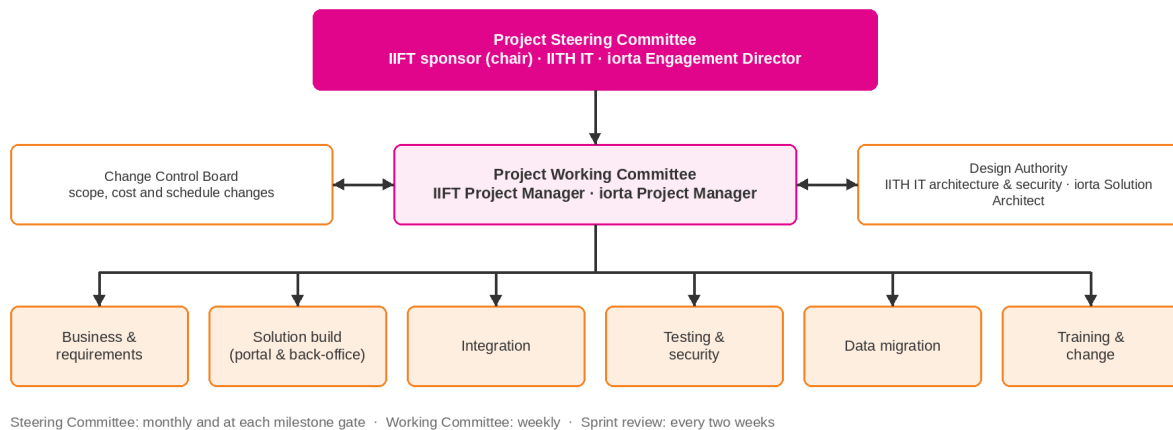


Figure 39: Project governance structure

Forum	Members	Frequency	Purpose
Steering Committee	IIFT sponsor (chair), IITH IT head, IIFT business heads, iorta Engagement Director, both PMs	Monthly and at milestone gates	Direction, milestone approval, escalated risks and changes, go/no-go
Project Working Committee	IIFT PM, iorta PM, business and IT leads	Weekly	Progress, RAID log, dependencies, decisions
Sprint review & planning	Product owner, key users, iorta team	Every two weeks	Demo of completed features, backlog priorities
Design Authority	IITH IT architecture & security, iorta Solution Architect	As needed in phases 1–4	Architecture, security and integration decisions
Change Control Board	IIFT PM, sponsor delegate, iorta PM	As needed	Assess and approve change requests

Progress is reported weekly (status, milestones and the RAID log of risks, assumptions, issues and dependencies) and monthly to the Steering Committee. Issues that cannot be resolved by the Working Committee within five business days are escalated to the Steering Committee.

RACI matrix

R = Responsible, A = Accountable, C = Consulted, I = Informed.

Activity	IIFT Sponsor	IIFT PM	IIFT business	IITH IT	iorta PM	iorta team
Project governance & reporting	A	R	I	I	R	C
Requirements & process sign-off	A	R	R	C	C	R
Solution design & architecture	I	C	C	A	R	R
Build & configuration	I	I	C	C	A	R
Infrastructure provisioning (Option A)	I	C	I	A/R	C	C
Interface availability (Core, FIN, AML, AD)	I	C	I	A/R	C	R
SIT & performance testing	I	I	I	C	A	R
VAPT (independent)	I	C	I	C	A	R
UAT execution & sign-off	A	R	R	C	C	C
Data extraction from legacy sources	I	C	R	A	C	C
Data transformation, load & reconciliation	I	C	R	C	A	R
Training delivery	I	C	R	C	A	R

Activity	IIFT Sponsor	IIFT PM	IIFT business	IITH IT	iorta PM	iorta team
Go-live decision	A	R	C	C	C	I
Hypercare & support	I	C	C	C	A	R

17.4 Quality assurance

A feature is done when its code is reviewed, its automated tests and security scans pass, its documentation is updated and the product owner has seen it working. The traceability matrix (DEL-05) links every RFP requirement ID to design, build, test case and acceptance evidence. Each major document is reviewed internally by someone outside the delivery team before it reaches IIFT.

17.5 Testing strategy

Level	Scope	Responsibility	Environment	Tools / exit criteria
Unit	Business rules, rating engines, validators, web components	iorta developers	DEV / CI	Vitest and Testing Library; coverage ≥ 70% on core modules
API / integration	Endpoints, permissions per role, database constraints	iorta	CI with PostgreSQL	Vitest e2e; every endpoint has positive and negative tests
SIT	End-to-end flows across portal, back-office and interfaces	iorta QA, IITH system owners	SIT	No open Critical/High defects; ≥ 95% pass
Regression	Automated suite re-run every sprint and release	iorta QA	SIT	Playwright; 100% of critical paths automated
Performance	Load (100 concurrent), stress and soak	iorta	SIT (production-like)	k6; non-functional targets met
Security	SAST/SCA/DAST in pipeline; independent VAPT and re-test	iorta, independent tester	SIT / pre-prod	No open Critical/High findings
UAT	Business scenarios per product and role	IIFT users with iorta support	UAT	Signed UAT certificate; no open Critical/High defects
Migration	Dry-runs, reconciliation, sample verification	iorta, IIFT data owners	UAT	100% record count and control-total match

Table 30: Test levels

Defect severity	Definition	Resolution in test phases
Critical	System unusable or data integrity at risk; no workaround	Fix before exit of current test phase
High	Key function fails; workaround difficult	Fix before exit of current test phase
Medium	Function impaired; workaround exists	Fix before go-live or agreed deferral
Low	Cosmetic or minor	Planned release

17.6 Data migration approach

Migration covers existing agent, agency, bank and branch records, participant records needed for servicing, and reference data. Sources and volumes are confirmed in the Data Migration Plan (DEL-19).

Step	Description
Discovery & profiling	Identify sources (spreadsheets, existing agent records, IIGT/core extracts), profile quality, agree scope
Mapping & rules	Field mapping to target model, transformation and cleansing rules, ownership of corrections
Tooling	Repeatable scripts (TypeScript/SQL) with validation reports; no manual re-keying
Dry-run 1 & 2	Full loads in UAT; exceptions corrected at source; reconciliation reports

Step	Description
Reconciliation	Record counts, control totals, mandatory-field completeness, sample verification by business owners
Production load	Executed during cut-over; signed reconciliation before users are enabled

17.7 Training plan

Audience	Format	Duration	Participants	Materials
Agents & bank officers (Portal)	Train-the-trainer plus two hands-on sessions	Half day per session	Up to 21 users (12 bank, 4 agent, 5 IIFT portal)	Quick reference guide, video walkthroughs, user manual
Back-office users	Hands-on workshop per role	1 day	IIFT admin, Finance, Compliance, Underwriting users	Role-based user manual, exercises
System administrators	Workshop	1 day	IITH/IIFT administrators	Administrator manual, configuration guide
Technical team	Technical handover sessions	2 days	IITH IT support, infrastructure, security	Technical and operations manuals, runbooks
Management	Briefing and dashboard demo	1 hour	IIFT management	Dashboard guide

Table 31: Training plan

17.8 Change management

- Stakeholder map and communication plan covering IIFT teams, bank partners and agencies.
- Early involvement of key users in design workshops and sprint demos to build ownership.
- Bank and agent onboarding pack: invitation e-mail, account activation guide, FAQ and support contacts.
- Super-user network within IIFT to provide first-line help after go-live.
- Adoption metrics on the management dashboard (active users, digital submissions, turnaround times).

17.9 Hypercare

- Four weeks immediately after go-live (weeks 25–28) with the core delivery team on standby.
- Daily health checks of application, jobs, EOD and integrations; daily stand-up with IIFT during the first two weeks.
- Priority fixing of go-live defects; daily defect and usage report.
- Onsite presence during go-live week (one of the four planned visits).
- Formal handover to the maintenance team and hypercare exit criteria: no open P1/P2 issues, EOD stable for ten consecutive business days, all interfaces reconciled.

18. Delivery & Implementation Timeline

Production go-live is in week 24 counted from kick-off, followed by four weeks of hypercare. The six-month warranty and Year 1 maintenance both start at go-live; defects found under warranty are fixed free of charge.

Delivery plan – 24 weeks to go-live, 4 weeks hypercare

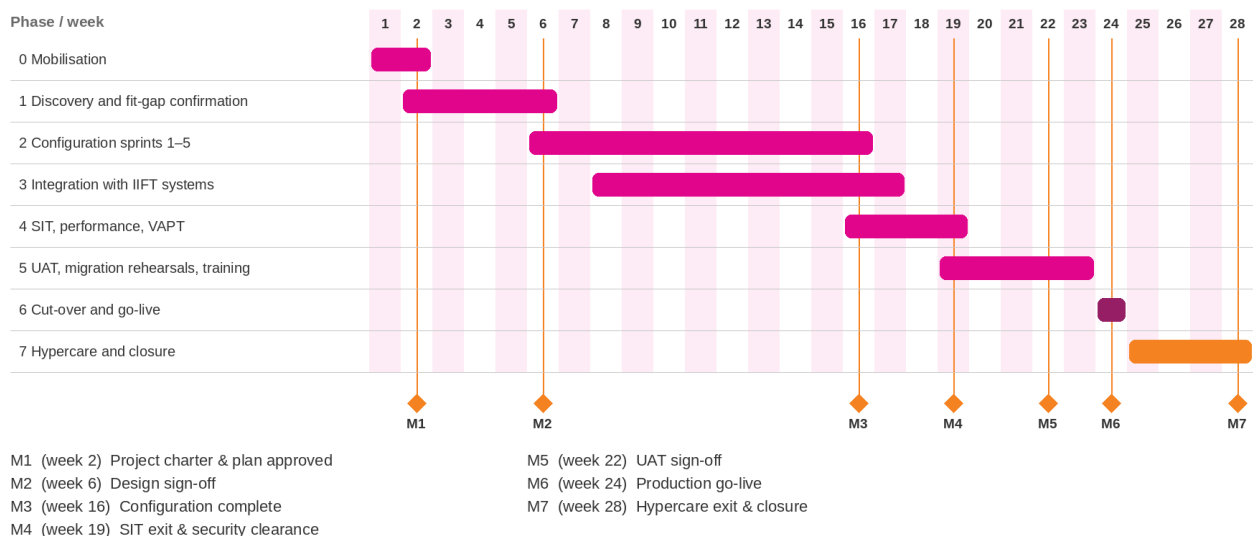


Figure 40: Delivery plan (weeks from kick-off)

18.1 Key milestones

Milestone	Description	Target	Payment
M1	Project charter & plan approved	Week 2	15%
M2	Design sign-off	Week 6	15%
M3	Configuration complete	Week 16	–
M4	SIT exit & security clearance	Week 19	25%
M5	UAT sign-off	Week 22	25%
M6	Production go-live	Week 24	15%
M7	Hypercare exit & closure	Week 28	5%

Table 32: Milestones and linked payments

18.2 Dependencies on the critical path

- Contract signature and kick-off date (week 1 starts at kick-off).
- Availability of IIFT business owners for workshops in weeks 2–5 and sign-off of design by week 6.
- SIT and UAT environments (VMs, network, database access) available by week 8 and week 16 respectively.
- Interface specifications, test endpoints and credentials for Core, FIN, AML and AD available by week 8.
- IIFT actuarial rate tables and product documents (proposal forms, PDS, questionnaires) by week 6.
- UAT participants available in weeks 19–22; legacy data extracts available by week 12.

19. Commercial Proposal

Fees are shown first in the format IIFT requested, then in the RFP section 10 breakdown. The pricing workbook (IIFT-Commercial-Pricing-iorta-TechNXT.xlsx) holds the same figures with live formulas.

19.1 Fee summary in IIFT's requested format

Description	Fee	WHT	OPE	Total
Agent/Banca Portal	B\$ 91,500 / USD 70,385	Inclusive	Included	B\$ 91,500 / USD 70,385
Back-office solution	B\$ 74,500 / USD 57,308	Inclusive	Included	B\$ 74,500 / USD 57,308
Total one-time fee	B\$ 166,000 / USD 127,692	Inclusive	Included	B\$ 166,000 / USD 127,692

Table 33: One-time implementation fee (fixed price)

Description	Fee	WHT	OPE	Total
Agent/Banca Portal	B\$ 11,900 / USD 9,154	Inclusive	Included	B\$ 11,900 / USD 9,154
Back-office solution	B\$ 9,700 / USD 7,462	Inclusive	Included	B\$ 9,700 / USD 7,462
Total per year (Years 1–5, flat)	B\$ 21,600 / USD 16,615	Inclusive	Included	B\$ 21,600 / USD 16,615

Table 34: Annual maintenance and support fee, Years 1 to 5 (0% escalation)

Notes

Both lines are modules of SalesVerse 2.0: Agent/Banca Portal (SalesVerse 2.0) and Back-office solution (SalesVerse 2.0). The SalesVerse 2.0 licence is bundled at B\$0.

WHT, inclusive: fees are quoted gross; withholding tax deducted under Brunei law is borne by iorta.

OPE, included: travel and subsistence for the 4 planned onsite visits; no separate OPE is invoiced.

Third-party charges, excluded: SMS messages, AML data subscription, SSL certificates, e-mail relay and infrastructure, at cost (section 19.8).

USD amounts are indicative (USD 1 = B\$ 1.30), each rounded to the nearest dollar.

19.2 Pricing basis

The price is sized to IIFT's scope as stated in the RFP: 26 named users, about 600 policies a year and seven products. It does not include per-user, per-policy or subscription charges, so it does not rise as IIFT adds agents, bank branches or group companies.

Most of the functionality already exists in SalesVerse 2.0. IIFT pays for configuring, integrating, testing and supporting it for IIFT, not for building a platform. That is why the price is lower than a bespoke build of the same scope, and why delivery takes 24 weeks.

Day rates and the maintenance fee reflect Brunei market levels for application support, and the five-year cost is spread so that the annual fee stays affordable for a business of IIFT's volume. Maintenance is flat for five years. **[Bid owner to confirm the market comparison statement.]**

19.3 Pricing principles

Item	Basis
Currency	Brunei Dollars. USD figures are indicative at USD 1 = B\$1.30, rounded to the nearest dollar; the B\$ amounts are binding.
Fixed price	The one-time fee covers the full scope of this proposal. It is not time and materials.

Item	Basis
Software licence	SalesVerse 2.0 licence bundled at B\$0: perpetual, royalty-free and enterprise-wide for IIFT and IITH group companies, unlimited named and concurrent users, no renewal fee. The SalesVerse 2.0 core platform remains iorta TechNXT's intellectual property.
Source code	Source code of the deployed solution is delivered for maintenance and enhancement; escrow is available as an option.
Subscriptions	None (COM-03). Maintenance and support is the only recurring iorta charge.
WHT	Inclusive. Fees are quoted gross; any Brunei withholding tax deducted by IIFT is borne by iorta, and IIFT pays the invoice net of WHT without grossing up.
OPE	Included for the 4 planned onsite visits (kick-off, design workshop, UAT and training, go-live). Unplanned visits requested by IIFT follow the rate card, travel at cost.
Third-party charges	Excluded. Listed in section 19.8; paid by IIFT directly or passed through at cost without mark-up.
Infrastructure	Under the recommended on-premise option IIFT/IITH provides hardware and OS and iorta charges no hosting fee. PostgreSQL carries no licence fee.

19.4 Commercial pricing breakdown (RFP section 10)

No	Component	Basis	Portal (B\$)	Back-office (B\$)	Total (B\$) / rate	Remarks
1	SalesVerse 2.0 software licence (Agent/Banca Portal and Back-office)	One-off	0	0	0	Bundled at B\$0. Perpetual, royalty-free, enterprise-wide licence for unlimited users; source code of the deployed solution delivered. No per-user or renewal fees.
2	Implementation (project management, business analysis, design, configuration, development, deployment)	One-off	56,000	44,000	100,000	Fixed price for the full functional scope in RFP section 4 and Appendix 3.
3	Integration (per interface; see integration breakdown)	Per interface	17,500	13,000	30,500	Core system, FIN, AML screening, LDAP/AD, SMS gateway and Email/SMTP.
4	Data migration	One-off	3,000	4,500	7,500	Extraction support, transformation, cleansing, two dry runs, load and reconciliation of agreed agent, agency, bank and participant data.
5	Infrastructure / cloud	Annual	–	–	On-premise (recommended): B\$ 0, IIFT/IITH-provided	Optional cloud hosting: approx. B\$ 11,400 per year pass-through plus B\$ 4,800 per year managed cloud operations. Not included in totals.
6	Database	Annual	–	–	B\$ 0: PostgreSQL community edition, no licence fee	Optional commercial PostgreSQL support available at cost if IIFT requires it.
7	Security (independent VAPT and re-test)	One-off	4,500	4,000	8,500	Independent vulnerability assessment and penetration test before go-live, remediation and re-test.
8	Testing (SIT, performance, regression, UAT support)	One-off	5,500	4,500	10,000	Test strategy, SIT cases and execution, performance test, automated regression suite, UAT support and defect fixing.
9	Training	One-off	2,500	2,000	4,500	End-user, administrator and technical training including materials.
10	Documentation	One-off	2,500	2,500	5,000	User, administrator, technical and operations manuals; maintained throughout the contract.
11	Warranty (6 months from go-live)	Included	Included	Included	Included	Defect correction at no charge for 6 months after go-live.

No	Component	Basis	Portal (B\$)	Back-office (B\$)	Total (B\$) / rate	Remarks
	Subtotal one-time (items 1–11)	One-off	91,500	74,500	166,000	Fixed price; WHT inclusive; planned OPE included
12	Year 1 maintenance: Stabilisation & Hypercare	Annual	11,900	9,700	21,600	Intensive post-go-live support, defect resolution, performance tuning, monitoring and stabilisation of integrations.
13	Year 2 maintenance: Operational Support	Annual	11,900	9,700	21,600	BAU incident management, preventive maintenance, security patches, performance monitoring and minor enhancements.
14	Year 3 maintenance: Optimisation	Annual	11,900	9,700	21,600	System optimisation, reporting improvements, integration improvements and technology review.
15	Year 4 maintenance: Technology Refresh	Annual	11,900	9,700	21,600	Platform compatibility review, security hardening, version upgrades and capacity assessment.
16	Year 5 maintenance: Sustainability & Transition	Annual	11,900	9,700	21,600	Comprehensive health assessment, documentation refresh, knowledge transfer, technology roadmap and exit/renewal preparation.
	Subtotal maintenance Years 1–5 (items 12–16)	Annual	59,500	48,500	108,000	Flat, 0% escalation
17	Enhancements and change requests	Per man-day / per hour	–	–	B\$ 680 per man-day; B\$ 90 per hour	Blended rate for all roles; quoted per change request after impact assessment.
18	Onsite support (unplanned visits)	Per day	–	–	B\$ 850 per day + travel at cost	4 planned visits are already included in the fixed price.
19	After-hours support	Per hour / Included	–	–	P1 (Critical): included; other after-hours work: B\$ 120 per hour	After-hours work other than P1 is performed only on IIFT request.
20	Third-party charges	Annual / usage	–	–	Excluded; payable at cost	SMS messages, AML data subscription, SSL certificates, email relay; see third-party schedule.
21	Exit / transition	Included	–	–	Included in Year 5 maintenance	Knowledge transfer, data extraction and technical handover at contract end.

Table 35: RFP section 10 commercial pricing breakdown

19.5 Integration cost per interface

Interface	Scope	Portal (B\$)	Back-office (B\$)	Total (B\$)
Core system	Agent, agency, bank and policy master synchronisation; agent information exchange	4,250	4,250	8,500
Financial system (FIN)	End-of-day postings, receipts, commission / referral fee information, BRR reconciliation	4,250	4,250	8,500
AML screening service	Participant and agent screening, result capture, re-screening	2,500	2,500	5,000
LDAP / Active Directory	Single sign-on for back-office users; optional directory look-up for portal administration	2,000	2,000	4,000
SMS gateway	OTP and alert delivery through IIFT-approved SMS provider	2,500	0	2,500
Email / SMTP relay	Notification and document e-mail delivery through IIFT mail relay	2,000	0	2,000

Interface	Scope	Portal (B\$)	Back-office (B\$)	Total (B\$)
Total integration (item 3)		17,500	13,000	30,500

Table 36: Integration cost per interface (COM-05)

19.6 Five-year maintenance and support

Year	Focus (RFP section 9)	Portal (B\$)	Back-office (B\$)	Total (B\$)
Year 1	Stabilisation & Hypercare	11,900	9,700	21,600
Year 2	Operational Support	11,900	9,700	21,600
Year 3	Optimisation	11,900	9,700	21,600
Year 4	Technology Refresh	11,900	9,700	21,600
Year 5	Sustainability & Transition	11,900	9,700	21,600
Total	Five years	59,500	48,500	108,000

Table 37: Maintenance fees by year

Maintenance is billed quarterly in advance, from go-live (B\$ 5,400 per quarter). Section 20 describes the services.

19.7 Infrastructure, hosting and database

Item	Basis	Option A: on-premise	Option B: cloud (optional)
Hosting / cloud infrastructure (item 5)	Annual	B\$ 0 (IIFT/IITH-provided)	≈ B\$ 11,400 pass-through (B\$ 950/month)
Managed cloud operations	Annual	Not applicable	B\$ 4,800
Database licence (item 6)	Annual	B\$ 0 (PostgreSQL)	Included in managed service above
Total per year		B\$ 0	≈ B\$ 16,200

Table 38: Infrastructure and database costs

19.8 Third-party charges (excluded, at cost)

Third-party charges are excluded from iorta's fees and payable at cost. Amounts are indicative and are confirmed by each provider.

Item	Basis	Provider / note	Indicative amount
SMS messages	Per message	IIFT-approved SMS gateway or telco tariff	About B\$0.05–0.10 per SMS; roughly B\$300–600 a year at expected volumes
AML screening data subscription	Annual / usage	IIFT's existing or chosen screening list provider	B\$0 if IIFT's existing service is reused; otherwise from about B\$3,000 a year
SSL/TLS certificates	Annual	IIFT-approved certificate authority	About B\$150–500 a year per certificate
Email relay service	Annual / usage	IIFT/IITH mail infrastructure	B\$0 with the existing IITH relay
Servers, storage, network and OS licences (on-premise)	One-off / annual	IIFT/IITH data centre	Per IITH standard; Linux subscriptions about B\$500–1,200 per server a year if RHEL is used, B\$0 for Ubuntu
Cloud hosting (if Option B is chosen)	Monthly	Cloud provider, passed through at cost	About B\$11,400 a year
Optional commercial PostgreSQL support	Annual	Only if IIFT elects vendor-backed database support	About B\$5,000–10,000 a year
Optional PKI digital signature certificates	Per certificate / annual	Only if certificate-based signing is required in future	Quoted by the certificate authority

Table 39: Third-party dependencies and charges (COM-17, item 20)

Third-party components inside SalesVerse 2.0 are open source (MIT, Apache 2.0, BSD or PostgreSQL licence) and carry no fees. A software bill of materials, produced by the CI pipeline on every build, comes with the technical manual.

19.9 Rate card

Item	Rate	Notes
Enhancements / change requests (item 17)	B\$ 680 per man-day; B\$ 90 per hour	60 hours per year included in maintenance
Onsite support, unplanned (item 18)	B\$ 850 per day + travel at cost	4 planned visits included in the fixed price
After-hours support (item 19)	P1: included. Other: B\$ 120 per hour	Other after-hours work only at IIFT's request
Exit / transition (item 21)	Included in Year 5 maintenance	See Section 20

Table 40: Rate card (COM-16)

19.10 Payment milestones

Milestone	Payment trigger	Share	Amount (B\$)
M1	Contract signing and project kick-off	15%	24,900
M2	Design sign-off (BRS, FRS, SAD, TDD, UI/UX, interface specification)	15%	24,900
M4	SIT exit and security clearance (VAPT remediated)	25%	41,500
M5	UAT sign-off	25%	41,500
M6	Production go-live	15%	24,900
M7	Hypercare exit and project closure	5%	8,300
Total		100%	166,000

Table 41: One-time fee payment schedule

Invoices are payable within 30 days. Maintenance is invoiced quarterly in advance from go-live.

19.11 Five-year total cost of ownership

Cost element	Option A: on-premise (B\$)	Option B: cloud (B\$)
One-time implementation	166,000	166,000
Software licence (5 years)	0	0
Maintenance & support (5 years)	108,000	108,000
Cloud infrastructure pass-through (5 years, indicative)	0	57,000
Managed cloud operations (5 years)	0	24,000
Five-year TCO (B\$)	274,000	355,000
Five-year TCO (USD, indicative)	210,769	273,077

Table 42: Five-year TCO (excluding IIFT-provided infrastructure and third-party charges)

20. Five-Year Maintenance & Support

Maintenance and support start at go-live and run for five years at B\$ 21,600 a year with no escalation. In the first six months the warranty covers defect fixes; the maintenance fee pays for the service desk, SLA, monitoring, patches, upgrades, 60 enhancement hours a year and the services below.

20.1 Support model

Element	Description
Channels	Issue module in the portal/back-office (preferred), dedicated support e-mail and telephone hotline; P1 incidents must be reported by telephone in addition to the ticket.
Support hours	IIFT business hours as defined in the 'Insurans Islam TAIB Business Hours (Family Takaful)' schedule (COM-14); P1 incidents 24 x 7, including weekends and public holidays.
Incident management	Record, classify, prioritise, investigate, resolve and close per ITIL-aligned procedure; IIFT confirms closure (MNT-02).
Problem management	Root-cause analysis for every P1 and P2 and for recurring incidents, with report within five business days and tracked corrective actions (MNT-03).
Service requests	User support, configuration changes, data corrections and operational assistance (MNT-04).
Preventive maintenance	Monthly health check: logs, error trends, job runs, capacity, slow queries, certificate expiry, backup status (MNT-10, 16, 17).
Security maintenance	Monthly patch cycle for application dependencies and container images; emergency patches per remediation timelines; certificate management support (MNT-11 to 13).
Technology currency	Compatibility with supported OS, browser, Node.js and PostgreSQL versions; upgrades included (MNT-14, 15, 22).
Minor enhancements	60 hours per year included; further changes via change request at the rate card (MNT-23, 24).
Onsite support	Remote by default; onsite for major incidents, upgrades or critical activities at the rate card (MNT-29).

20.2 Service levels

Priority	Definition	Response	Resolution / restoration	Coverage
P1 Critical	Production down or critical function unavailable for all users; data integrity or security breach	30 minutes	4 hours (restore / workaround)	24 x 7
P2 High	Major function impaired for many users; no reasonable workaround	2 hours	1 business day	IIFT business hours
P3 Medium	Function impaired with workaround; limited users affected	4 business hours	3 business days	IIFT business hours
P4 Low / Cosmetic	Cosmetic issue, question or minor defect	1 business day	Next scheduled release	IIFT business hours

Table 43: Incident service levels (MNT-05 to MNT-07, COM-13)

Availability target: 99.5% per month overall and 99.9% during IIFT business hours, excluding approved maintenance windows and infrastructure outside iorta's responsibility. Response is measured from ticket logging; resolution may be a workaround with a permanent fix scheduled. Service credits for repeated breaches are set out in Section 26.

20.3 Escalation matrix

Level	Role	Contact	Escalation trigger
Level 1	Service Desk / Support Engineer	[Name, phone, e-mail]	Immediately on logging

Level	Role	Contact	Escalation trigger
Level 2	Support Lead / Senior Engineer	[Name, phone, e-mail]	P1 not restored in 1 h; P2 not responded in 2 h
Level 3	Account / Project Manager	[Name, phone, e-mail]	P1 not restored in 2 h; P2 not resolved in 1 business day
Level 4	Engagement Director	[Name, phone, e-mail]	P1 not restored in 4 h; repeated SLA breach

Table 44: Critical incident escalation (COM-15)

20.4 Year-by-year focus

Year	Focus	Activities
Year 1	Stabilisation & Hypercare	Intensive post-go-live support, defect resolution, performance tuning, monitoring and stabilisation of integrations.
Year 2	Operational Support	BAU incident management, preventive maintenance, security patches, performance monitoring and minor enhancements.
Year 3	Optimisation	System optimisation, reporting improvements, integration improvements and technology review.
Year 4	Technology Refresh	Platform compatibility review, security hardening, version upgrades and capacity assessment.
Year 5	Sustainability & Transition	Comprehensive health assessment, documentation refresh, knowledge transfer, technology roadmap and exit/renewal preparation.

Table 45: Five-year maintenance structure (RFP section 9)

20.5 Service reporting and reviews

- Monthly service report (MNT-25): incidents and service requests by priority, SLA performance, availability, changes and releases, security patches and vulnerabilities, capacity trends, outstanding issues.
- Quarterly service review (MNT-26): service performance, improvement plan, roadmap, upcoming platform changes.
- Annual DR test participation (MNT-20) and quarterly backup restore verification (MNT-19) with written results.
- Documentation updated after every material change (MNT-27) and ongoing knowledge-transfer sessions for IIFT/IITH support staff (MNT-28).

20.6 Exit and transition plan

Activity	Description
Transition planning	Exit plan agreed in Year 5 (or within 30 days of a termination notice): scope, timeline, receiving party.
Knowledge transfer	Structured sessions on architecture, code, configuration, operations and known issues.
Documentation	Final refresh of technical, operations, administrator and user manuals; runbooks.
Source code & tooling	Final source code, pipelines, infrastructure scripts and credentials handed over.
Data extraction	Full export of business data, documents and audit records in open formats (SQL dump, CSV, original files).
Parallel support	Shadow support to the new provider or IIFT team for up to 30 days.
Confirmation	Signed transition acceptance; secure deletion of IIFT data held by iorta, with certificate.

20.7 iorta business continuity

Source code, documentation and infrastructure scripts are kept in version-controlled repositories with off-site backup. At least two engineers know each module, support can run from more than one location, and every support role has a named backup (COM-24). [Confirm or extend with iorta's corporate BCP details.]

21. Project Team & Organisation

The team pairs Takaful business analysts with engineers who know the platform. CVs are in Annex B. Key personnel are not replaced without IIFT's consent, and any replacement has equivalent experience.

Role	Name	Responsibilities
Engagement Director	[Name]	Executive sponsor; Steering Committee member; commercial and escalation owner
Project Manager	[Name]	Plan, governance, RAID, reporting, change control; single point of contact
Solution Architect	[Name]	Architecture, security and integration design; Design Authority member
Business Analyst (Takaful)	[Name]	Workshops, BRS/FRS, product rules, RTM, UAT preparation, training
UI/UX Designer	[Name]	Screen designs, user journeys, accessibility
Technical Lead	[Name]	Code quality, technical design, reviews, deployment
Software Engineers	[Names]	Portal, back-office and API development
Integration Engineer	[Name]	Core, FIN, AML, AD, SMS and e-mail adapters; interface testing
QA Lead / Test Engineer	[Name]	Test strategy, SIT, regression automation, performance testing
DevOps & Security Engineer	[Name]	Environments, CI/CD, monitoring, hardening, VAPT coordination
Support Manager	[Name]	Hypercare, maintenance service, SLA reporting

Table 46: Proposed team

21.1 Involvement by phase

● = lead / high involvement, ○ = supporting / part-time, – = not involved.

Role	0 Mob.	1 Design	2 Build	3 Integr.	4 SIT	5–6 UAT & go-live	7 Hypercare
Engagement Director	●	○	○	○	○	●	○
Project Manager	●	●	●	●	●	●	●
Solution Architect	●	●	○	●	○	○	–
Business Analyst	●	●	○	○	●	○	○
UI/UX Designer	–	●	○	–	–	–	–
Technical Lead	○	●	●	●	●	●	●
Software Engineers	–	○	●	●	●	○	●
Integration Engineer	–	●	●	●	○	○	○
QA Lead / Test Engineer	–	○	●	●	●	○	○
DevOps & Security Engineer	○	○	●	●	○	●	○
Support Manager	–	–	–	○	○	●	●

21.2 Expected IIFT roles

IIFT role	Responsibility
Project Sponsor	Chairs Steering Committee; final decisions
Project Manager	Coordinates IIFT resources, approvals and logistics
Business owners / SMEs	Banca, Underwriting, Finance and Compliance: requirements, rules, UAT
IITH IT	Infrastructure, network, security, interfaces, environments

IIFT role	Responsibility
UAT testers / super-users	Execute UAT; support colleagues after go-live

22. Assumptions, Dependencies & Exclusions

22.1 Assumptions

- Scope is as described in the RFP (sections 3–8 and Appendices 1–4) and this proposal; changes follow the change request process.
- IIFT provides timely access to business owners and decisions; document reviews are completed within five business days.
- Each external system (Core, FIN, AML, AD, SMS gateway, SMTP) offers a usable interface (API, file or protocol) with test environments; iorta does not modify those systems.
- Rate tables, product rules, document templates and questionnaires are provided by IIFT and signed off by week 6.
- Data migration covers agent, agency, bank, branch, participant and reference data, estimated below 10,000 records from no more than three source extracts; legacy data cleansing at source is IIFT's responsibility.
- Under Option A, hardware, OS licences, network, firewall, load balancer, backup infrastructure and DR site are provided by IIFT/IITH to the sizing in Section 15.
- The user interface and documents are in English; Malay translation of screens can be added as a change request.
- Delivery is predominantly remote with four planned onsite visits (kick-off, design workshop, UAT/training, go-live), whose travel and subsistence are included in the price.
- IIFT business hours are as published in the 'Insurans Islam TAIB Business Hours (Family Takaful)' schedule.

22.2 Dependencies

- Contract signature and kick-off date; project timeline counts from kick-off.
- SIT and UAT environments by weeks 8 and 16; production and DR environments by week 20.
- Interface specifications, endpoints and credentials by week 8.
- Legacy data extracts by week 12; UAT testers available weeks 19–22.
- Independent VAPT provider access to the SIT/pre-production environment in weeks 17–19.

22.3 Exclusions

- Hardware, operating system licences, network equipment and data-centre services (Option A).
- Cloud hosting charges (Option B), passed through at cost if chosen.
- Third-party charges: SMS messages, AML data subscription, SSL certificates, e-mail relay services, optional commercial PostgreSQL support, optional PKI digital certificates.
- Changes to the Core system, FIN, AML service or other third-party systems.
- Native mobile applications (the portal is responsive and works on tablets and mobile browsers).
- Payment gateway / online card payments (can be added later as a change request).
- Full claims administration and policy administration in the core system beyond claim notification and the portal/back-office functions described.
- Unplanned onsite visits (charged at the rate card) and work outside IIFT business hours other than P1 incidents.

23. Risk Register

The initial risk register is below. It is maintained by the project managers, reviewed weekly by the Working Committee and monthly by the Steering Committee.

ID	Risk	Impact	Likelihood	Mitigation
R1	Delayed availability or incomplete specification of Core/FIN interfaces	High	Medium	Early interface workshops (week 3); mock services so build continues; interface specification sign-off by week 6
R2	Late provision of SIT/UAT/production environments	High	Medium	Environment request at kick-off; iorta-hosted SIT as interim; containerised deployment reduces set-up time
R3	Rate tables or product rules not finalised in time	High	Medium	Configurable rating with placeholder rates; rate validation as an SIT exit criterion
R4	Scope growth during design	Medium	Medium	Workshops on working screens; scope baseline at M2; formal change control
R5	Legacy data quality lower than expected	Medium	Medium	Early profiling; two dry-runs; cleansing rules owned by IIFT data owners
R6	Limited availability of IIFT users for UAT	Medium	Low	UAT plan agreed at M2; scenario scripts prepared by iorta; super-user model
R7	VAPT reveals significant findings close to go-live	High	Low	Secure SDLC with SAST/SCA/DAST from sprint 1; VAPT in weeks 17–19 leaves time for remediation
R8	AML provider integration constraints (rate limits, formats)	Medium	Low	Adapter with queueing and retry; manual screening fallback with audit
R9	Low adoption by bank officers and agents	Medium	Low	Change management, train-the-trainer, simple UX, onboarding pack, adoption KPIs
R10	Regulatory feedback on cloud hosting (Option B)	Medium	Low	Recommend Option A; early AMBD engagement if Option B is chosen
R11	Key personnel unavailability	Medium	Low	Named backups; shared code ownership; documentation as we go
R12	Performance issues with document-heavy transactions	Low	Low	Streaming uploads, object storage, performance tests on production-like data

Table 47: Risk register

24. Quotation Validity

This proposal and the prices quoted are valid for **90 days from the date of submission** ([submission date] to [expiry date]).

On award, the one-time fee is fixed for the implementation and the maintenance fee is fixed for five years. Indicative cloud charges (Option B) and third-party charges follow the providers' prices at the time of order.

25. Supporting Information

25.1 Value-added items beyond the RFP

Item	Benefit to IIFT
Requirements compliance matrix	All 234 requirement IDs answered line by line, with fitment (Annex A).
Working application before contract	IIFT can test SalesVerse 2.0 with its own products before signing.
Multi-factor authentication	E-mail OTP or authenticator app for back-office users, delivered during implementation.
Remote participant e-signature	Single-use e-mail link, so participants can sign without visiting a branch.
Integration monitor	One screen for outbox messages, retries, dead-letters and reconciliation results.
End-of-day run screen	Run, check and sign off EOD, the FIN file and reconciliation, with run history.
Retention and archival policy	Retention period per data class, with archival of audit records and documents.
PDPO 2025 and AMBD alignment	Controls mapped to Brunei data protection law and AMBD expectations (Section 13).
Shariah-appropriate terminology	Contribution, participant, Takaful operator, wakalah and tabarru' throughout.
Accessibility	WCAG 2.1 AA target.
Source code escrow option	Independent escrow in addition to source code delivery, if IIFT wants it.
Exit plan	Transition, data extraction and knowledge transfer defined now (Section 20).
Hardware sizing	Sizing for every environment for the on-premise option (Section 15).
Unlimited users	More banks, agents and group companies at no licence cost.

Table 48: Value-added items included at no extra cost

Further supporting material: the technical document pack (Section 16), the screen catalogue with sample screens (Section 11) and the pricing workbook.

26. Terms & Conditions

We propose the key terms below and are willing to work from IIFT's standard contract in negotiation.

26.1 Basis of contract. This proposal, the RFP and agreed clarifications form the basis for a definitive agreement to be negotiated in good faith. In case of conflict, the signed agreement prevails.

26.2 Prices and taxes. Prices are in Brunei Dollars, fixed as set out in Section 19. Fees are inclusive of withholding tax, which is borne by iorta TechNXT. Planned out-of-pocket expenses are included. Third-party charges are excluded and payable at cost. Should new taxes be introduced by law after submission, the parties will agree their treatment.

26.3 Payment terms. One-time fees are invoiced on achievement of the payment milestones in Section 19.10 and are payable within 30 days of invoice. Maintenance fees are invoiced quarterly in advance from go-live.

26.4 Acceptance. Deliverables and milestones are accepted by written sign-off against agreed acceptance criteria. IIFT will review within ten business days; a deliverable is deemed accepted if no material non-conformity is notified within that period or if it is used in production.

26.5 Intellectual property. The SalesVerse 2.0 core platform remains the intellectual property of iorta TechNXT. IIFT receives a perpetual, royalty-free, enterprise-wide licence to use SalesVerse 2.0 for unlimited users, covering IIFT and IITH group companies; the licence fee is B\$0 and is bundled in this proposal. Configurations, reports, documentation and other work products produced specifically for IIFT belong to IIFT. Open-source components remain under their own licences.

26.6 Source code and escrow. The full source code of the deployed solution, with build scripts, database migrations and deployment configuration, is delivered at go-live and with every later release, so that IIFT or a provider it appoints can maintain and enhance the solution under the licence above. Deposit with an independent escrow agent is available as an option; escrow fees are payable by IIFT.

26.7 Data ownership and protection. All business data, including participant, agent and transaction data and documents, always remains the property of IITH/IIGT/IIFT. iorta TechNXT processes personal data only on IIFT's instructions and for the purpose of the contract, applies the security controls in Section 13, supports IIFT's obligations under the Personal Data Protection Order 2025 and returns or securely deletes data at contract end.

26.8 Confidentiality. Each party keeps the other's confidential information confidential, uses it only for the contract and discloses it only to personnel and approved subcontractors bound by equivalent obligations. These obligations survive termination for [five] years, and indefinitely for personal data.

26.9 Warranty. iorta TechNXT warrants that the solution will perform materially in accordance with the approved specifications for 6 months from production go-live and will correct reported defects at no charge within the SLA. The warranty does not cover issues caused by changes made by others, misuse, infrastructure or third-party products outside iorta's scope.

26.10 Service levels and service credits. Service levels are as defined in Section 20. Where a P1 restoration target is missed, service credits apply: [e.g. 5% of the quarterly maintenance fee per breach, capped at 20% of the quarterly fee]. Service credits are IIFT's financial remedy for SLA failure without prejudice to termination rights for persistent failure.

26.11 Limitation of liability. Each party's total aggregate liability under the contract is limited to [the total contract value (fees paid and payable for implementation and the current maintenance year)]. Neither party is liable for indirect or consequential loss. The limitation does not apply to liability for fraud, wilful misconduct, breach of confidentiality or infringement of intellectual property.

26.12 Change requests. Changes to scope follow the change request process: written request, impact assessment (scope, cost, schedule, risk) within five business days, quotation using the rate card, written approval before work starts.

26.13 Price protection. Maintenance fees are fixed for five years with 0% escalation. [Confirm: rate card held without escalation for the five-year term.]

26.14 Personnel and subcontracting. Key personnel are not replaced without IIFT's consent. The independent VAPT is performed by a qualified third-party tester engaged by iorta with IIFT's approval; iorta remains responsible for its subcontractors.

26.15 Term and termination. Either party may terminate for material breach not remedied within 30 days of notice. IIFT may terminate for convenience on [90] days' notice, paying for work performed and accepted up to the termination date. On any termination, iorta provides the exit assistance in Section 20 and hands over work in progress and the source code of the deployed solution.

26.16 Force majeure. Neither party is liable for delay or failure caused by events beyond its reasonable control (including natural disasters, pandemics, war, government action or widespread utility or network failure), provided it notifies the other party promptly and uses reasonable efforts to mitigate. If force majeure continues for more than 60 days, either party may terminate the affected services.

26.17 Governing law and disputes. The contract is governed by the laws of Brunei Darussalam. Disputes are first escalated through the governance structure; unresolved disputes are referred to [mediation / arbitration in Brunei Darussalam] and, failing that, to the courts of Brunei Darussalam.

26.18 Validity. This proposal is valid for 90 days from the submission date.

Annex A – Requirements Compliance and Fitment Matrix

Every requirement ID in the RFP is listed. **Compliance:** Fully Compliant is a standard capability of the delivered solution; Compliant – configuration is met by setting rules, parameters or master data; Compliant – custom is met by IIFT-specific work inside the fixed price. **Fitment** uses the four categories in Section 6. Priority follows RFP Appendix 4. The RFP uses the prefix COM for both shared platform (section 4.4) and commercial (section 6) requirements; each set is listed under its section, and commercial items carry no fitment.

ID	Requirement	Priority	Compliance	Fitment	How addressed
RFP 4.1 – Agent/Banca Portal functional requirements					
AP-01	Secure login	Must	Fully Compliant	Available today	Username/password with Argon2id hashing, rate-limited login, optional MFA, server-side sessions.
AP-02	Password management	Must	Compliant – configuration	Configuration	Self-service change and reset; length, complexity, history (last 5) and 90-day expiry are parameters.
AP-03	Session management	Must	Compliant – configuration	Configuration	15-min idle and 8-h absolute timeout, logout revokes server session, ID regenerated at login, single-session option.
AP-04	Account lockout	Must	Compliant – configuration	Configuration	Lock after 5 failed attempts for 30 min (configurable); manual release by an administrator is audited.
AP-05	View agent/banca profile	Must	Fully Compliant	Available today	My Profile page: personal, contact, agency/bank, licence, hierarchy and document status.
AP-06	Maintain profile (approval)	Must	Fully Compliant	Available today	Profile change request with before/after values routed through maker-checker workflow.
AP-07	Agent/banca registration	Must	Fully Compliant	Available today	Online registration wizard with AML/KYC data, document upload (IC copy, or passport copy for passport holders) and back-office approval.
AP-08	Agent ID	Must	Compliant – configuration	Configuration	Unique agent/banca code produced on approval from a configurable numbering pattern.
AP-09	Main/sub-agent relationships	Must	Fully Compliant	Available today	Hierarchy view of main agent, sub-agents and bank officers scoped to the user's position.
AP-10	Agency/banca information	Must	Fully Compliant	Available today	Agency/bank details, branches, contacts and status shown on dashboard and profile.
AP-11	Cross-agency participant sharing	Must	Compliant – custom	Available today	Single participant master with de-duplication by IC/registration no.; sharing governed by RBAC and policy rules.
AP-12	Agency-wide participant visibility	Must	Compliant – custom	Available today	Consolidated participant 360 view (policies, payments, claims, interactions) for authorised roles.
AP-13	Participant registration	Must	Fully Compliant	Available today	Individual and corporate participant forms with validation and duplicate check.
AP-14	Participant search	Must	Compliant – configuration	Available today	Search by name, IC/passport (blind index), phone, policy no.; criteria configurable.
AP-15	Participant maintenance (approval)	Must	Fully Compliant	Available today	Update requests go through maker-checker; approved values applied with audit history.
AP-16	AML screening	Must	Compliant – custom	Available today	Screening on registration and update via AML adapter; hits routed to compliance review queue.
AP-17	Product selection	Must	Compliant – configuration	Configuration	Product catalogue from configuration showing products the agent/bank is authorised to sell.
AP-18	New quotation	Must	Compliant – custom	Available today	Product-specific quotation wizard for all seven Appendix 3 products.
AP-19	Automatic contribution calculation	Must	Compliant – custom	Configuration	Rating engines per product using configurable rate tables supplied by IIFT actuarial.

ID	Requirement	Priority	Compliance	Fitment	How addressed
AP-20	Draft quotation	Must	Fully Compliant	Available today	Quotations saved as DRAFT at any step and resumed later.
AP-21	Policy status	Must	Fully Compliant	Available today	Draft, Pending Approval, Pending Payment, Active, Rejected, Expired and Cancelled statuses.
AP-22	Policy listing	Must	Fully Compliant	Available today	Paginated policy list with status chips, scoped to agent, agency or bank.
AP-23	Policy search	Must	Compliant – configuration	Available today	Filter by policy no., participant, product, status, date range; criteria configurable.
AP-24	Policy details	Must	Fully Compliant	Available today	Coverage, participant, nominees, contribution, payment status, documents and history.
AP-25	Policy issuance	Must	Fully Compliant	Available today	Submit eligible quotation for issuance after validation; e-Policy produced on approval/payment.
AP-26	Policy renewal	Must	Compliant – configuration	Configuration	Renewal-due list and one-click renewal quotation inside the renewal window (notice period before expiry to 30 days after); renewal enable/disable per product (e.g. Khairat).
AP-27	Policy endorsement	Must	Fully Compliant	Available today	Endorsement request with change details and documents, subject to approval rules.
AP-28	Policy cancellation	Must	Fully Compliant	Available today	Cancellation request with reason and supporting documents, subject to approval.
AP-29	Policy documents	Must	Fully Compliant	Available today	View/download e-Policy, schedule, receipts, certificates and correspondence.
AP-30	Payment status	Must	Fully Compliant	Available today	Payment status and outstanding amount shown on policy and billing screens.
AP-31	Policy history	Should	Fully Compliant	Available today	Timeline of every transaction and status change with user and timestamp.
AP-32	Policy validation	Must	Compliant – configuration	Available today	Server-side rules for product, eligibility, documents, payment and AML before submission.
AP-33	Approval workflow	Must	Compliant – configuration	Available today	Transactions needing approval routed by workflow definition to authorised checkers.
AP-34	Notifications	Should	Compliant – configuration	Configuration	In-portal, email and SMS notifications for issuance, approval, rejection, expiry and renewal.
AP-35	Audit trail	Must	Fully Compliant	Available today	Append-only audit of policy creation, updates and approvals with before/after values.
AP-36	Policy restrictions	Must	Compliant – configuration	Available today	Issuance blocked when payment, AML or document requirements are not met.
AP-37	Payment dashboard	Must	Fully Compliant	Available today	Billing dashboard with collected, outstanding and overdue amounts and ageing.
AP-38	Bulk payment (per policy)	Must	Fully Compliant	Available today	Single or bulk payment submission with amounts allocated per policy.
AP-39	Upload payment proof	Must	Fully Compliant	Available today	Upload bank slip / transfer evidence against one or many policies.
AP-40	Payment verification	Must	Fully Compliant	Available today	Payment proof routed to Finance for verification (approve/reject with remarks).
AP-41	Receipt generation	Must	Fully Compliant	Available today	PDF e-Receipt with unique number produced on approval and e-mailed.
AP-42	Payment rules (7-day grace, agency block)	Must	Compliant – configuration	Available today	Daily job blocks all agents of an agency when any policy is unpaid > 7 days; auto-lifted on settlement.
AP-43	Claims notification	Must	Fully Compliant	Available today	Claim notification: validate policy, capture details, upload documents, submit to IIFT.
AP-44	Policy schedule / receipt	Must	Fully Compliant	Available today	Branded PDF policy schedule and receipt produced after issuance.
AP-45	Document download / e-mail	Must	Fully Compliant	Integration with IIFT systems	Download and e-mail documents to participant from the portal.
AP-46	Document upload	Must	Fully Compliant	Available today	Upload during registration and maintenance with type allow-list, size limit, checksum and ClamAV malware scan (mandatory in production).
AP-47	Document status	Should	Fully Compliant	Available today	Document status, validity and expiry dates displayed with reminders.
AP-48	AML information	Must	Compliant – configuration	Configuration	AML/KYC fields captured in onboarding (configurable questionnaire).

ID	Requirement	Priority	Compliance	Fitment	How addressed
AP-49	Application submission	Must	Fully Compliant	Available today	Save, submit and track applications and requests.
AP-50	Application status	Must	Fully Compliant	Available today	Draft, Submitted, Pending Approval, Approved, Rejected and Completed statuses.
AP-51	Rejection remarks	Must	Fully Compliant	Available today	Rejection reasons displayed; maker can amend and resubmit.
AP-52	Agent dashboard	Must	Fully Compliant	Available today	Dashboard with profile, applications, pending actions, production and payments.
AP-53	Pending actions	Must	Fully Compliant	Available today	Action list of incomplete drafts, rejected items, overdue payments and expiring documents.
AP-54	Notifications (applications)	Should	Compliant – configuration	Configuration	Notification templates per workflow event; channels configurable.
AP-55	Report issue	Must	Fully Compliant	Available today	Issue form with category, priority, description and attachments.
AP-56	Issue tracking	Must	Fully Compliant	Available today	Unique issue reference and status tracking with comments.
AP-57	Issue attachment	Should	Fully Compliant	Available today	Screenshots, documents and evidence attached to issues.
AP-58	Agency/agent/banca reports	Should	Compliant – configuration	Configuration	Role-scoped reports (production, payments, renewals) with export.
AP-59	Role-based access	Must	Compliant – configuration	Configuration	Permissions by agent type, hierarchy and role; data scoping enforced in the API.
AP-60	Authority limit	Must	Compliant – configuration	Available today	Financing above B\$150,000 is referred to IIFT Sales; further thresholds per product and role are configurable.
AP-61	Approval history	Must	Fully Compliant	Available today	Approval actions with remarks retained and displayed; immutable audit.
AP-62	Electronic signature	Must	Compliant – custom	Available today	On-screen signature by agent or participant, signature by single-use e-mail link, or upload of signed documents.
RFP 4.2 – Back-office (Admin) functional requirements					
BO-01	Secure login	Must	Fully Compliant	Available today	Internal login with optional AD/LDAP SSO and MFA (email OTP / TOTP).
BO-02	Role-based access control	Must	Compliant – configuration	Available today	Roles, permissions and data scopes configured by administrators.
BO-03	User administration	Must	Fully Compliant	Available today	Create, update, deactivate and reset back-office users and release lockouts; changes audited.
BO-04	Role & permission management	Must	Compliant – configuration	Available today	Role builder with fine-grained permissions grouped by module; maker-checker optional.
BO-05	Agent administration	Must	Fully Compliant	Available today	Create, view, update, activate, suspend and terminate agent records.
BO-06	Agent search	Must	Compliant – configuration	Available today	Search by agent/banca ID, name, agency/bank, status and configurable criteria.
BO-07	Agent status	Must	Fully Compliant	Available today	Pending, Active, Inactive, Suspended and Terminated with controlled transitions.
BO-08	Main/sub-agent management	Must	Fully Compliant	Available today	Maintain hierarchy (agency → main agent → sub-agent / bank → branch → officer).
BO-09	Agency administration	Must	Fully Compliant	Available today	Maintain agency/bank details, branches, contacts and associated agents.
BO-10	Agent approval (maker-checker)	Must	Compliant – configuration	Configuration	Registration and profile changes approved through configurable workflow levels.
BO-11	Agent documents	Must	Fully Compliant	Available today	View, validate, approve, reject and manage agent documents.
BO-12	Document validation	Should	Compliant – configuration	Configuration	Mandatory document checklist per entity type; expiry tracking and reminders.
BO-13	AML screening	Must	Compliant – custom	Available today	Watch-list screening with fuzzy name matching for agents and participants; manual re-screen.
BO-14	AML status	Must	Fully Compliant	Available today	Screening result, score, status, reviewer and dates recorded and displayed.

ID	Requirement	Priority	Compliance	Fitment	How addressed
BO-15	Compliance review	Must	Fully Compliant	Available today	Compliance queue to clear, escalate or reject flagged cases with remarks; a confirmed match rejects the subject's pending approval requests.
BO-16	Approval workflow	Must	Compliant – configuration	Configuration	Configurable levels and amount thresholds per transaction type (e.g. second approval from B\$300,000 sum covered).
BO-17	Maker-checker	Must	Fully Compliant	Available today	Server-side rule: maker cannot approve own transaction; enforced for every workflow.
BO-18	Workflow status	Must	Fully Compliant	Available today	Tracking of every request through Submitted, Pending, Approved, Rejected, Completed.
BO-19	Rejection & remarks	Must	Fully Compliant	Available today	Mandatory remarks on rejection; visible to maker and in audit.
BO-20	Management dashboard	Should	Fully Compliant	Available today	KPIs, trends, production by product/channel, outstanding actions and SLA.
BO-21	Pending action dashboard	Must	Fully Compliant	Available today	Approvals inbox, incomplete applications and outstanding actions per user/role.
BO-22	Standard reports	Must	Compliant – custom	Available today	12 standard reports in the working application; remaining IIFT reports added from the catalogue agreed in design.
BO-23	Report filtering	Must	Fully Compliant	Available today	Filters by date, agent, agency/bank, product, status and other criteria.
BO-24	Report export	Must	Fully Compliant	Available today	Export to Excel, CSV and PDF.
BO-25	Scheduled reports	Should	Compliant – configuration	Available today	Report schedules (daily/weekly/monthly) e-mailed to authorised recipients.
BO-26	Audit trail	Must	Fully Compliant	Available today	Records user activities, transactions, approvals, rejections and system events.
BO-27	Change history	Must	Fully Compliant	Available today	Previous and new values, user, date/time and IP retained for critical changes.
BO-28	Audit search	Must	Fully Compliant	Available today	Audit search by user, entity, action and date with export for authorised roles.
BO-29	Issue assignment	Must	Compliant – configuration	Available today	Assignment to support personnel or teams with assignment rules.
BO-30	Issue priority	Must	Compliant – configuration	Configuration	Critical, High, Medium, Low priorities (configurable).
BO-31	Issue SLA monitoring	Should	Compliant – configuration	Configuration	Response/resolution timers per priority with breach alerts and SLA report.
BO-32	Master data management	Must	Fully Compliant	Available today	Maintain products, plans, rates, branches, banks, occupations, document types, reasons, etc.
BO-33	System configuration	Must	Compliant – configuration	Configuration	Business rules, statuses, approval levels and parameters maintainable without redevelopment.
RFP 4.3 – Integration functional requirements					
INT-01	Core system integration	Must	Compliant – custom	Integration with IIFT systems	Core system adapter to retrieve/update agent and policy information per interface specification.
INT-02	Agent/banca master data synchronisation	Must	Compliant – custom	Integration with IIFT systems	Scheduled and event-driven synchronisation of agent, agency, bank and status data.
INT-03	Agent information exchange	Must	Compliant – custom	Integration with IIFT systems	Agent-related events exchanged between portal, back-office and core via outbox.
INT-04	Financial system integration	Must	Compliant – custom	Integration with IIFT systems	EOD postings, receipts and commission data exchanged with FIN.
INT-05	Commission / referral fee information	Must	Compliant – custom	Integration with IIFT systems	Commission/referral statements retrieved from FIN and shown to authorised agents.
INT-06	Enterprise authentication	Should	Compliant – custom	Integration with IIFT systems	LDAP/Active Directory SSO for back-office; SAML/OIDC possible if IITH IdP exists.

ID	Requirement	Priority	Compliance	Fitment	How addressed
INT-07	Email integration	Should	Compliant – configuration	Integration with IIFT systems	SMTP relay integration with TLS; templates configurable.
INT-08	SMS integration	Should	Compliant – custom	Integration with IIFT systems	SMS gateway adapter for OTP and alerts via IIFT-approved provider.
INT-09	API gateway / API services	Must	Fully Compliant	Available today	Versioned REST APIs (OpenAPI 3) exposed and consumed through the integration layer.
INT-10	API security	Must	Fully Compliant	Available today	Mutual TLS or API keys/OAuth2 client credentials, RBAC, input validation, rate limiting.
INT-11	API monitoring	Should	Fully Compliant	Available today	Integration monitor with availability, volumes, failures and response times; metrics endpoint.
INT-12	Data validation	Must	Fully Compliant	Available today	Schema validation of inbound and outbound payloads; rejects logged with reason.
INT-13	Error handling & retry	Must	Fully Compliant	Available today	Transactional outbox, exponential back-off retry, dead-letter and alerting.
INT-14	Integration logging	Must	Fully Compliant	Available today	Every attempt logged with correlation id, status, duration and payload hash.
INT-15	Data reconciliation	Should	Fully Compliant	Integration with IIFT systems	Receipt reconciliation in the working application; BRR and Core reconciliation added during implementation.
RFP 4.4 – Shared / common platform requirements					
COM-01	Sensitive data protection	Must	Fully Compliant	Available today	TLS 1.2+, AES-256-GCM field encryption for IC/passport, masking, encrypted storage.
COM-02	Access control	Must	Fully Compliant	Available today	Least-privilege RBAC with data scoping by agency/bank/hierarchy.
COM-03	Comprehensive audit	Must	Fully Compliant	Available today	Common append-only audit service used by all modules.
COM-04	Configurable workflow	Must	Compliant – configuration	Configuration	Workflow definitions by transaction type, role, level and business rule.
COM-05	Notification framework	Should	Fully Compliant	Available today	Common notification service for email, SMS and in-portal channels with templates.
COM-06	Document repository	Must	Fully Compliant	Available today	Encrypted document store with metadata, versioning, checksum and access control.
COM-07	Global search	Should	Fully Compliant	Available today	Global search across participants, policies, agents and requests for authorised users.
COM-08	Reporting framework	Must	Fully Compliant	Available today	Common report engine with filters, scheduling and Excel/CSV/PDF export.
COM-09	Parameter management	Must	Compliant – configuration	Configuration	System parameters and master data maintained in the back-office without code change.
COM-10	Data encryption	Must	Fully Compliant	Available today	Encryption in transit (TLS) and at rest (volume/DB/object storage + field level).
RFP 5 – Non-functional requirements					
NFR-01	Response time	Must	Fully Compliant	Delivered during implementation	Targets: page < 3 s, standard API p95 < 1 s, reports < 10 s for 12 months of data.
NFR-02	Concurrent users	Must	Fully Compliant	Delivered during implementation	Sized for 100 concurrent / 500 named users against 26 named users in Appendix 1.
NFR-03	Transaction volume	Must	Fully Compliant	Available today	Sized for > 10x the Appendix 2 five-year projection (~628 policies per year).
NFR-04	System availability	Must	Fully Compliant	Delivered during implementation	99.5% overall, 99.9% during IIFT business hours, excluding approved maintenance.
NFR-05	High availability	Should	Fully Compliant	Delivered during implementation	Two application nodes, PostgreSQL primary + standby, redundant load balancer.
NFR-06	User scalability	Must	Fully Compliant	Available today	Stateless API nodes scale horizontally; no per-user licences.
NFR-07	Transaction scalability	Must	Fully Compliant	Available today	Indexed relational model, partition-ready audit tables, archival policy.
NFR-08	Encryption	Must	Fully Compliant	Available today	TLS 1.2+ in transit; encrypted volumes, backups and field-level encryption at rest.

ID	Requirement	Priority	Compliance	Fitment	How addressed
NFR-09	Authentication	Must	Compliant – configuration	Delivered during implementation	Password policy, lockout, MFA and AD integration aligned to IITH security policy.
NFR-10	Authorisation	Must	Fully Compliant	Available today	Least privilege RBAC with deny-by-default API guards.
NFR-11	Audit logging	Must	Compliant – configuration	Available today	Security and business events logged; retention period configurable (default 7 years).
NFR-12	Vulnerability management	Must	Fully Compliant	Available today	CodeQL static analysis, dependency audit, secret and container scanning in CI; remediation within severity-based timelines.
NFR-13	Penetration testing	Must	Fully Compliant	Delivered during implementation	Independent VAPT before go-live with remediation and re-test (included in price).
NFR-14	Secure development	Must	Fully Compliant	Available today	OWASP ASVS Level 2 aligned SDLC, OWASP Top 10 controls, peer review.
NFR-15	Error handling	Must	Fully Compliant	Available today	Generic user messages with correlation id; no stack traces or internals exposed.
NFR-16	Transaction integrity	Must	Fully Compliant	Available today	ACID transactions in PostgreSQL; issuance, receipt, audit and outbox commit atomically.
NFR-17	Data backup	Must	Fully Compliant	Delivered during implementation	Daily full backup, continuous WAL archiving (PITR), configuration backup, restore tests.
NFR-18	DR capability	Must	Fully Compliant	Delivered during implementation	Warm standby DR with RPO ≤ 15 min and RTO ≤ 4 h.
NFR-19	DR testing	Must	Fully Compliant	Delivered during implementation	Annual DR test with IIFT, documented results and improvement actions.
NFR-20	Modular architecture	Should	Fully Compliant	Available today	Modular monolith with module boundaries per functional area; modules can be split later.
NFR-21	Documentation	Must	Fully Compliant	Delivered during implementation	Technical, configuration, operations and user documentation maintained for 5 years.
NFR-22	User interface	Must	Fully Compliant	Available today	Consistent Ant Design based UI themed to Insurans Islam TAIB branding.
NFR-23	Browser compatibility	Must	Fully Compliant	Available today	Current and previous major versions of Chrome, Edge, Firefox and Safari.
NFR-24	Accessibility	Should	Fully Compliant	Delivered during implementation	Designed to WCAG 2.1 AA (keyboard navigation, contrast, labels).
NFR-25	API standards	Must	Fully Compliant	Available today	REST/JSON over HTTPS, OpenAPI 3 contracts, ISO 8601 dates, UTF-8.
NFR-26	System monitoring	Must	Fully Compliant	Delivered during implementation	Health endpoints, Prometheus metrics, integration monitor and infrastructure checks.
NFR-27	Alerting	Must	Fully Compliant	Delivered during implementation	Alert rules for availability, errors, latency, job failures and dead-letters to support teams.
NFR-28	Centralised logs	Should	Fully Compliant	Delivered during implementation	Structured JSON logs with correlation id shipped to central log store.
NFR-29	Data retention	Must	Compliant – configuration	Configuration	Retention and archival rules per data class, configurable to IIFT policy.
NFR-30	Data quality	Must	Fully Compliant	Available today	Validation on every DTO plus database constraints (FK, CHECK, UNIQUE).
RFP 6 – Commercial requirements					
COM-01	Implementation cost	Must	Fully Compliant	Commercial	Fixed price B\$166,000 for the full scope (Commercial Proposal).
COM-02	Software licence	Must	Fully Compliant	Commercial	SalesVerse 2.0 licence bundled at B\$0: perpetual, royalty-free, enterprise-wide, unlimited users, no renewal fees.
COM-03	Subscription	Must	Fully Compliant	Commercial	No SaaS or platform subscription; optional cloud hosting shown separately.

ID	Requirement	Priority	Compliance	Fitment	How addressed
COM-04	Infrastructure	Must	Fully Compliant	Commercial	On-premise and cloud infrastructure, database and third-party costs identified.
COM-05	Integration cost	Must	Fully Compliant	Commercial	Cost per interface itemised (six interfaces, B\$30,500).
COM-06	Implementation services	Must	Fully Compliant	Commercial	PM, BA, design, configuration, development and deployment within item 2.
COM-07	Data migration	Must	Fully Compliant	Commercial	Scope, assumptions, tools and cost (B\$7,500) stated.
COM-08	Testing	Must	Fully Compliant	Commercial	SIT, UAT support, performance (B\$10,000) and security testing (B\$8,500) itemised.
COM-09	Training	Must	Fully Compliant	Commercial	User, administrator and technical training included (B\$4,500).
COM-10	Documentation	Must	Fully Compliant	Commercial	Preparation and maintenance of documentation included (B\$5,000).
COM-11	Warranty period	Must	Fully Compliant	Commercial	6 months from production go-live, included.
COM-12	5-year maintenance	Must	Fully Compliant	Commercial	Detailed five-year maintenance plan at B\$21,600 per year, flat.
COM-13	Service level agreement	Must	Fully Compliant	Commercial	Response, restoration and availability targets committed (Maintenance and Support chapter).
COM-14	Support hours	Must	Fully Compliant	Commercial	Aligned to IIFT business hours with 24x7 for P1 incidents.
COM-15	Critical incident escalation	Must	Fully Compliant	Commercial	Four-level escalation matrix with time-based triggers.
COM-16	Change request	Must	Fully Compliant	Commercial	Rate card and formal CR process (impact, quote, approval).
COM-17	Third-party dependencies	Must	Fully Compliant	Commercial	All third-party products and charges disclosed; open-source components listed.
COM-18	Annual increase	Should	Fully Compliant	Commercial	0% escalation for five years.
COM-19	Exit assistance	Must	Fully Compliant	Commercial	Transition, data extraction and knowledge transfer included in Year 5 or on termination.
COM-20	IP ownership	Must	Fully Compliant	Commercial	SalesVerse 2.0 core remains iorta IP under a perpetual licence; IIFT-specific configuration, reports, documentation and data belong to IIFT.
COM-21	Source code	Should	Fully Compliant	Commercial	Full source code of the deployed solution delivered for maintenance and enhancement; escrow optional.
COM-22	Data ownership	Must	Fully Compliant	Commercial	All business data remains the property of IITH/IIGT/IIFT.
COM-23	Vendor liability	Must	Fully Compliant	Commercial	Responsibilities, liability and service credits defined in Terms and Conditions.
COM-24	Vendor BCP	Must	Fully Compliant	Commercial	iorta business continuity arrangements: code repositories, backup staff, remote support.
RFP 7 – Project deliverables					
DEL-01	Project charter	Must	Fully Compliant	Delivered during implementation	Week 2: objectives, scope, governance, stakeholders, assumptions, milestones.
DEL-02	Project plan	Must	Fully Compliant	Delivered during implementation	Week 2: activities, dependencies, resources, timeline; updated fortnightly.
DEL-03	Business requirement specification	Must	Fully Compliant	Delivered during implementation	Week 4: agreed business requirements and processes (as-is / to-be).
DEL-04	Functional requirement specification	Must	Fully Compliant	Delivered during implementation	Week 5: detailed functional behaviour, rules, screens and reports.
DEL-05	Requirements traceability matrix	Must	Fully Compliant	Delivered during implementation	Week 6, maintained to closure: requirement → design → build → test → acceptance.
DEL-06	Solution architecture	Must	Fully Compliant	Delivered during implementation	Week 6: application, infrastructure, security, integration and deployment architecture.
DEL-07	Technical design	Must	Fully Compliant	Delivered during implementation	Week 6: components, database, APIs, interfaces and configuration.

ID	Requirement	Priority	Compliance	Fitment	How addressed
DEL-08	UI/UX design	Must	Fully Compliant	Delivered during implementation	Week 5: clickable screen designs, navigation and user journeys.
DEL-09	Interface specification	Must	Fully Compliant	Delivered during implementation	Week 6: APIs, data mappings, methods, security and error handling per interface.
DEL-10	Configured solution	Must	Fully Compliant	Delivered during implementation	Week 16: complete portal and back-office, demonstrated sprint by sprint.
DEL-11	Source code	Must	Fully Compliant	Delivered during implementation	Week 24 and every release: source code of the deployed solution, build scripts and migrations.
DEL-12	Test strategy	Must	Fully Compliant	Delivered during implementation	Week 6: levels, responsibilities, environments, entry/exit and acceptance criteria.
DEL-13	SIT test cases	Must	Fully Compliant	Delivered during implementation	Week 14: scenarios and expected results traced to requirements.
DEL-14	SIT results	Must	Fully Compliant	Delivered during implementation	Week 19: execution evidence, defect log and closure.
DEL-15	UAT test cases	Must	Fully Compliant	Delivered during implementation	Week 18: business scenarios prepared with IIFT users.
DEL-16	UAT results	Must	Fully Compliant	Delivered during implementation	Week 22: execution evidence and business sign-off.
DEL-17	Security assessment	Must	Fully Compliant	Delivered during implementation	Week 19: VAPT report, remediation evidence and re-test results.
DEL-18	Performance test report	Must	Fully Compliant	Delivered during implementation	Week 19: load, stress and soak results against NFR targets.
DEL-19	Data migration plan	Must	Fully Compliant	Delivered during implementation	Week 8: extraction, transformation, validation, migration and reconciliation.
DEL-20	Deployment plan	Must	Fully Compliant	Delivered during implementation	Week 21: deployment sequence, responsibilities, rollback and contingency.
DEL-21	Go-live checklist	Must	Fully Compliant	Delivered during implementation	Week 23: technical, operational, security, business and support readiness.
DEL-22	User training	Must	Fully Compliant	Delivered during implementation	Weeks 21–23: agents, banca officers, back-office users and stakeholders.
DEL-23	Administrator training	Must	Fully Compliant	Delivered during implementation	Weeks 22–23: system administrators and support personnel.
DEL-24	User manual	Must	Fully Compliant	Delivered during implementation	Week 21: portal and back-office user guides with screenshots.
DEL-25	Administrator manual	Must	Fully Compliant	Delivered during implementation	Week 21: configuration, user administration and operational support.
DEL-26	Technical manual	Must	Fully Compliant	Delivered during implementation	Week 23: architecture, database, API, integration and deployment.
DEL-27	Operations manual	Must	Fully Compliant	Delivered during implementation	Week 23: monitoring, backup, incident handling and routine operations.
DEL-28	Production deployment	Must	Fully Compliant	Delivered during implementation	Week 24: production deployment per approved deployment plan.

ID	Requirement	Priority	Compliance	Fitment	How addressed
DEL-29	Go-live support	Must	Fully Compliant	Delivered during implementation	Weeks 25–28: hypercare with daily checks and priority defect fixing.
DEL-30	Project closure report	Must	Fully Compliant	Delivered during implementation	Week 28: final status, outstanding items, lessons learned, formal acceptance.
RFP 8 – Five-year maintenance & support requirements					
MNT-01	Helpdesk support	Must	Fully Compliant	Delivered during implementation	Service desk via portal issue module, e-mail and phone for five years.
MNT-02	Incident management	Must	Fully Compliant	Delivered during implementation	Record, classify, investigate, resolve and close incidents per ITIL-aligned procedure.
MNT-03	Problem management	Must	Fully Compliant	Delivered during implementation	Root-cause analysis for P1/P2 and recurring incidents with corrective actions.
MNT-04	Service request	Must	Fully Compliant	Delivered during implementation	User support, configuration and operational assistance requests handled.
MNT-05	Critical incident SLA	Must	Fully Compliant	Delivered during implementation	P1: 30-min response, 4-h restoration, 24x7, immediate escalation.
MNT-06	High-priority SLA	Must	Fully Compliant	Delivered during implementation	P2: 2-h response, resolution within 1 business day.
MNT-07	Medium/Low/Cosmetic SLA	Must	Fully Compliant	Delivered during implementation	P3: 4 business hours / 3 business days; P4: 1 business day / next release.
MNT-08	Bug fixing	Must	Fully Compliant	Delivered during implementation	Defects fixed under warranty (6 months) and maintenance thereafter.
MNT-09	Corrective maintenance	Must	Fully Compliant	Delivered during implementation	Corrective changes to maintain functionality included.
MNT-10	Preventive maintenance	Should	Fully Compliant	Delivered during implementation	Monthly health checks, log and capacity review, proactive fixes.
MNT-11	Security patches	Must	Fully Compliant	Delivered during implementation	Application, framework and dependency patches; monthly cycle, emergency as needed.
MNT-12	Vulnerability remediation	Must	Fully Compliant	Delivered during implementation	Critical ≤ 7 days, High ≤ 14 days, Medium ≤ 30 days, Low ≤ 90 days.
MNT-13	Certificate management	Must	Fully Compliant	Delivered during implementation	Certificate inventory, expiry alerts and renewal support.
MNT-14	OS/platform compatibility	Must	Fully Compliant	Delivered during implementation	Compatibility with supported OS, container runtime, Node.js and PostgreSQL versions.
MNT-15	Browser compatibility	Must	Fully Compliant	Delivered during implementation	Compatibility maintained with supported browser versions.
MNT-16	Database maintenance	Must	Fully Compliant	Delivered during implementation	Health checks, vacuum/index optimisation, query tuning, upgrade support.
MNT-17	Performance monitoring	Should	Fully Compliant	Delivered during implementation	Monthly performance review and optimisation recommendations.
MNT-18	System monitoring	Must	Fully Compliant	Delivered during implementation	Availability and critical component monitoring with alerting.

ID	Requirement	Priority	Compliance	Fitment	How addressed
MNT-19	Backup verification	Must	Fully Compliant	Delivered during implementation	Quarterly restore test and backup job verification.
MNT-20	Disaster recovery	Must	Fully Compliant	Delivered during implementation	Participation in DR planning and annual DR test.
MNT-21	Release management	Must	Fully Compliant	Delivered during implementation	Releases follow agreed testing, CAB approval and deployment procedures.
MNT-22	Version upgrade	Must	Fully Compliant	Delivered during implementation	Application and platform upgrades during the maintenance period included.
MNT-23	Minor enhancement	Should	Fully Compliant	Delivered during implementation	60 hours per year of minor enhancements included; beyond that per rate card.
MNT-24	Change request	Must	Fully Compliant	Delivered during implementation	Formal CR: impact assessment, quotation, approval, delivery.
MNT-25	Monthly service report	Must	Fully Compliant	Delivered during implementation	Incidents, SLA performance, availability, changes and outstanding issues.
MNT-26	Quarterly review	Should	Fully Compliant	Delivered during implementation	Quarterly service review with improvement plan.
MNT-27	Documentation updates	Must	Fully Compliant	Delivered during implementation	System, technical and operational documents updated after material changes.
MNT-28	Knowledge transfer	Must	Fully Compliant	Delivered during implementation	Ongoing knowledge transfer sessions to IIFT/IITH support staff.
MNT-29	Onsite support	Should	Fully Compliant	Delivered during implementation	Onsite support for major incidents/upgrades per rate card (planned visits included).
MNT-30	Transition support	Must	Fully Compliant	Delivered during implementation	Year 5 knowledge transfer, documentation and technical assistance for transition.

Table 49: Requirements compliance and fitment matrix

Annex B – Curricula Vitae of Key Personnel

Key personnel from Section 21 are summarised below; full CVs are attached. **[Complete the table and attach one CV per person.]**

Proposed role	Name	Qualifications	Years of experience (insurance/Takafu l)	Relevant projects (project, client, role, year)
Project Manager	[Full name]	[Degrees, certifications]	[Years]	[3 to 5 projects]
Solution Architect	[Full name]	[Degrees, certifications]	[Years]	[3 to 5 projects]
Business Analyst (Takaful)	[Full name]	[Degrees, certifications]	[Years]	[3 to 5 projects]
Technical Lead	[Full name]	[Degrees, certifications]	[Years]	[3 to 5 projects]
QA Lead	[Full name]	[Degrees, certifications]	[Years]	[3 to 5 projects]
DevOps & Security Engineer	[Full name]	[Degrees, certifications]	[Years]	[3 to 5 projects]

Table 50: Key personnel summary

Annex C – Software Components and Network Flows

Server sizing per environment is in Section 15. This annex lists the software stack, network flows and expected storage growth for the on-premise option.

Layer	Software	Licence / source
Operating system	Linux: RHEL 9 or Ubuntu Server 24.04 LTS (64-bit)	IIFT/IITH
Container runtime	Docker Engine 27+ with Compose (or Kubernetes 1.30+)	Open source
Reverse proxy	Nginx 1.26+ (or existing F5 / WAF)	Open source / existing
Application runtime	Node.js 22 LTS (inside container images)	Open source
Database	PostgreSQL 16+ with streaming replication; pgBackRest or equivalent for backup/PITR	Open source
Object storage (optional)	MinIO (S3-compatible) or NFS share	Open source / existing
Monitoring	Prometheus, Grafana, Loki (or existing IITH tooling)	Open source / existing
Antivirus	ClamAV for document uploads (required in production)	Open source

Table 51: Software components

From	To	Port / protocol	Purpose
Internet / bank network	WAF / load balancer	443 (HTTPS)	Portal and e-sign
IIFT LAN	Internal reverse proxy	443 (HTTPS)	Back-office
Load balancer / proxy	Application VMs	8080 (HTTP, internal only) or 443	Application traffic
Application VMs	PostgreSQL	5432 (TLS)	Database
Application VMs	Active Directory	636 (LDAPS)	Back-office SSO
Application VMs	SMTP relay	587 (STARTTLS)	E-mail
Application VMs	Core / FIN / AML / SMS	443 (HTTPS) or 22 (SFTP)	Integrations
Monitoring server	Application VMs	9100 / 9464 (metrics)	Monitoring
PostgreSQL primary	Standby / DR replica	5432 (TLS)	Replication

Table 52: Network flows

Data	Year 1	Year 5	Notes
Structured data (PostgreSQL)	< 1 GB	< 5 GB	Policies, participants, payments, audit
Documents	10–20 GB	50–100 GB	Uploads and system-produced PDFs
Logs (online)	About 20 GB	About 20 GB	Rolling retention; archived to backup
Backups	About 150 GB	About 600 GB	35 days online plus monthly copies

Table 53: Storage growth estimate

Annex D – Glossary

Term	Meaning
Takaful	Islamic insurance based on mutual co-operation and shared responsibility, in which participants contribute to a common fund to help one another against defined risks.
Takaful operator	The company (here IIFT) that manages the Takaful fund on behalf of participants.
Participant	The person or entity that joins a Takaful plan and makes contributions (the Takaful equivalent of a policyholder).
Contribution	The amount paid by the participant for Takaful cover (the Takaful equivalent of a premium).
Tabarru'	The portion of the contribution donated to the participants' risk fund to help fellow participants who suffer a covered loss.

Term	Meaning
Wakalah	An agency contract under which the Takaful operator manages the fund for a fee (wakalah fee).
Participants' risk fund	The fund made up of tabarru' contributions from which claims are paid.
Nominee / beneficiary / executor	Persons designated to receive Takaful benefits or administer them on the participant's behalf.
Mortgage Takaful	Decreasing-term Takaful protecting a financing facility (hire purchase or property).
Banca / bancassurance	Distribution of Takaful products through bank partners and bank officers.
Main agent / sub-agent	Hierarchical agency relationship in which a main agent supervises sub-agents.
e-Policy / e-Receipt	Electronic policy document and receipt issued to the participant.
Grace period	Period (seven days) within which payment for an issued policy must be submitted before the agency's issuance is blocked.
Maker-checker	Control in which a transaction prepared by one user (maker) must be approved by another (checker) before it takes effect.
Quality check (QC)	Review of an application by IIFT before contract issuance.
End-of-day (EOD)	Daily batch that produces issuance reports, interfaces postings to FIN and reconciles with BRR.
Transactional outbox	Integration pattern in which outbound messages are stored in the same database transaction as the business change and delivered asynchronously, guaranteeing no loss.
Dead-letter	Holding area for messages that could not be delivered after the maximum number of retries.
Blind index	Keyed hash of an encrypted value that allows exact-match search without decrypting data.
Modular monolith	Single deployable application internally divided into independent modules with clear boundaries.
Hypercare	Period of intensified support immediately after go-live.
Point-in-time recovery	Ability to restore the database to any moment using base backups and transaction logs.